

Inflation for Classical Pair-Source Networks: Termination and Quantitative Obstructions

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Abstract

The inflation technique of Wolfe, Spekkens and Fritz tests whether an observed distribution is compatible with a causal structure that has latent common causes. Navascués and Wolfe proved that for correlation scenarios their hierarchy of finite inflation tests is asymptotically complete. They asked whether, for a fixed scenario with fixed observed alphabets, one finite order always suffices, and named the triangle as a possible case of asymptotic-only convergence. We answer the question for every pair-source scenario, in which each latent source is shared by exactly two binary observed variables, so that a scenario is a finite simple graph. A finite order characterizes compatibility if and only if every connected component of the graph is a double-star, a tree of diameter at most three; in that case order two suffices. The same classification holds for the ancestral-independence and recursively expressible strengthenings of the hierarchy, which coincide on every pair-source scenario. For every other graph and every order t there is a strictly positive rational law that passes the strongest order- t test and is incompatible. The obstructions are a parity witness at every order on every cycle and a witness at every order on the five-observer path whose incompatibility is a bilocal inequality; the terminating side is a reconstruction of a model from any order-two witness. For the triangle and the square we measure the failure: laws accepted at order t can lie at total-variation distance of order $1/t$ from the compatible set, while a convex-order argument bounds the distance by a constant times $t^{-1/2}$, so the worst-case convergence exponent lies between $1/2$ and 1 . On the square the two hierarchies differ by at most a factor $3/2$ in the order. For the triangle we also give an explicit nontermination family with a sharp rejecting-order exponent along one direction of approach to the compatible set, the distance asymptotics of that family, an explicit rejecting order under a promised distance, and a $2^{\Theta(B)}$ law for the worst rejecting order over rational inputs of B bits. The triangle nontermination theorem, its fan inequalities, the convex-order bound and the Finner inequality for arbitrary latent spaces are formalized in Lean.

1 Introduction

A correlation scenario has independent latent sources and observed variables that respond to the sources reaching them. The compatible laws of a fixed scenario form a nonconvex set, and finite latent-cardinality bounds give it a finite semialgebraic description, but an explicit characterization is unavailable even for the classical triangle with binary observations [9, 20]. Wolfe, Spekkens and Fritz introduced inflation as a test of compatibility [23]: at order t one copies each source t times and asks for a joint law on the copied observations with prescribed marginals. Navascués and Wolfe proved that their version of the hierarchy is asymptotically complete and asked whether a

fixed scenario with fixed observed alphabets admits one finite order that decides compatibility [15]. They named the triangle as a scenario where the answer could be no.

We answer the question for every pair-source scenario, meaning every scenario in which each source is shared by exactly two observed variables. Such a scenario is a finite simple graph G : vertices are observed variables and edges are sources. The triangle, the square and longer cycles, paths and stars are all pair-source scenarios.

Theorem A: which pair-source scenarios terminate. Call a tree of diameter at most three a double-star; it is an edge with any number of leaves at each end, and it includes single edges and stars.

Theorem 1.1. *Let G be a finite simple graph without isolated vertices, with binary observed variables. Some finite order of the Navascués–Wolfe hierarchy characterizes compatibility with G if and only if every connected component of G is a double-star, and then order two suffices. The same holds for the ancestral-independence and the recursively expressible strengthenings of the hierarchy from [23], which coincide with each other on every pair-source scenario. When some component is not a double-star, for every t there is a strictly positive rational law that passes every order- t test of all three hierarchies and is not compatible.*

The terminating half is a reconstruction (Theorem 5.1): on a double-star, source-disjoint observed blocks are independent under any order-two witness, and conditioning the witness on a leaf pattern reads off a pair of response tables for the central edge. It applies to arbitrary finite observed alphabets, with the order growing with the largest leaf alphabet. Navascués and Wolfe proved that star scenarios terminate at order two [15]; the reconstruction shows that the same mechanism reaches diameter three and no further. The obstructing half has two parts. Every graph that is not a double-star forest contains an induced cycle or an induced path on five vertices (Lemma 4.3), and a witness for an induced subgraph transports to the whole graph (Lemma 3.11). On every cycle we build a parity witness at every order (Theorem 6.8). On the five-vertex path we build a witness at every order whose incompatibility is a bilocal-type inequality across the two inner sources (Theorem 7.5). Both constructions are Fourier densities on auxiliary signs, chosen so that every moment that an inflation test prescribes takes its target value while the density stays positive. The equality of the ancestral-independence and recursively expressible hierarchies is a lemma about inflation graphs in which every latent node is a root and every observed node a sink (Lemma 3.9).

Theorem B: how far accepted laws can be from compatibility. For a hierarchy H and a scenario G let $H_t^H(G)$ be the largest total-variation distance from the compatible set among laws accepted at order t . By asymptotic completeness this tends to zero. For the triangle and the square we bracket the rate.

Theorem 1.2. *For every $t \geq 1$,*

$$\begin{aligned} \frac{5}{768t} &\leq H_t^{\text{exp}}(\square) = H_t^{\text{AI}}(\square) \leq H_t^{\text{NW}}(\square) \leq \min\left\{1, \frac{\sqrt{15}}{2\sqrt{t}}\right\}, \\ \frac{27}{5120t} &\leq H_t^{\text{exp}}(\triangle) = H_t^{\text{AI}}(\triangle) \leq H_t^{\text{NW}}(\triangle) \leq \frac{\sqrt{7}}{2\sqrt{t}}. \end{aligned}$$

The lower bounds come from a corrected Fourier density (Lemma 6.10) that keeps the parity law at correlation strength $q = \Theta(1/t)$ inside the order- t tests, together with a quantitative rigidity statement for parity-perfect compatible laws. The upper bound is a convex-order argument (Theorem 6.16): under any order- n witness, reading the copied table at independent uniform indices produces a compatible law, and the family of such laws is dominated in convex order by the empirical law of n samples from the target. It gives $H_n^{\text{NW}}(G) \leq \sqrt{K-1}/(2\sqrt{n})$ for K joint outcomes in any correlation scenario, without any factor depending on the number of sources. The worst-case exponent on the square and on the triangle therefore lies in $[1/2, 1]$; we do not determine it. On the square the two hierarchies are also close in order: $\mathcal{I}_{\lfloor 3t/2 \rfloor}^{\text{NW}}(\square) \subseteq \mathcal{I}_t^{\text{AI}}(\square) \subseteq \mathcal{I}_t^{\text{NW}}(\square)$ (Theorem 6.18).

Theorem C: the triangle along an explicit family. Before the classification we obtained the triangle case by a different construction, a defect cube, which gives finer information along one direction of approach to the compatible set. For every integer $t \geq 1$ put

$$\varepsilon_t = \frac{1}{2t^3}, \quad r_t = (1 - \varepsilon_t)^{t-1}, \quad P_t = \varepsilon_t \delta_{000} + (1 - \varepsilon_t) \text{Bern}(r_t)^{\otimes 3},$$

where $\text{Bern}(r)$ takes the value 1 with probability r .

Theorem 1.3. *For every $t \geq 1$, P_t satisfies the complete order- t Navascués–Wolfe inflation test for the triangle, together with every ancestral-independence and recursively expressible prescription, and is not compatible: $P_t(000)^2 - P_{t,A}(0)P_{t,B}(0)P_{t,C}(0) \geq \varepsilon_t^2/2 > 0$, in violation of the Finner inequality.*

For $0 < \varepsilon < 1/8$ put $P_\varepsilon = \varepsilon \delta_{000} + (1 - \varepsilon) \text{Bern}(1 - \frac{1}{2}\varepsilon^{2/3})^{\otimes 3}$; these laws converge to the compatible point δ_{111} . Their first rejecting order satisfies

$$\frac{1}{2} \leq \liminf_{\varepsilon \downarrow 0} \varepsilon^{1/3} t_{\min}^H(P_\varepsilon) \leq \limsup_{\varepsilon \downarrow 0} \varepsilon^{1/3} t_{\min}^H(P_\varepsilon) \leq 4 - 2\sqrt{3}$$

for each of the three hierarchies (Proposition 10.1). The lower bound is the passing construction; the upper bound is a fan inequality that every order- t feasible law obeys, $tP(000) \leq P_A(0) + \binom{t}{2}P_B(0)P_C(0)$ (Theorem 9.1), which also gives an explicit rejecting order for any Finner-violating law. Theorem 11.1 shows that ε is, to leading order, $(1 - 2^{-3/2})^{-1}$ times the total-variation distance from P_ε to the compatible set, so along this family the rejecting order grows like the inverse cube root of the distance. This exponent belongs to the family, not to the scenario: Theorem 1.2 shows that the worst case over all accepted laws decays no faster than $1/t$, and Proposition 9.2 gives a family approaching compatibility whose rejecting order is 2. For rational binary-triangle laws with at most B bits the worst required order is $2^{\Theta(B)}$ (Proposition 13.1), and a promised distance δ gives the rejecting-order bound $\lfloor (K-1)/(4\delta^2) \rfloor + 1$ in any correlation scenario with K joint outcomes (Corollary 12.1).

What is not shown. The classification is for pair-source scenarios; scenarios with a source shared by three or more observed variables are Question 15.4. We establish the strict separations between the three hierarchies at particular orders on the square, and we do not claim them at every order.

Organization. Section 2 fixes the three triangle hierarchies. Section 3 extends them to pair-source scenarios and proves the root-sink, source-disjointness and transport lemmas. Section 4 states Theorem A and deduces it from Sections 5 to 7, which treat double-stars, cycles together with the quantitative bounds of Theorem B, and the five-vertex path. Sections 8 to 13 prove Theorem C and its companions: the defect cube, its incompatibility and survival, the fan inequalities, the exponent, the distance asymptotics, the general distance bound and the bit-length law. Sections 14 to 16 discuss prior work, open problems and reproducibility. Appendix C collects supporting results on the triangle and the square that the main theorems do not use.

Method and verification. The proofs are analytic. Exact rational certificates check the finite instances: the defect-cube tables at orders one to three, the fan inequality along P_ε , and the new cycle, square, triangle and path witnesses at small orders. Section 16 lists the certificates and states which results are formalized in Lean and which are not.

2 The classical triangle and finite inflation tests

2.1 The triangle

Three mutually independent latent variables X, Y, Z take values in arbitrary measurable spaces with laws μ_X, μ_Y, μ_Z . Three observed variables with finite alphabets are generated by stochastic response kernels,

$$A \sim \kappa_A(\cdot \mid X, Z), \quad B \sim \kappa_B(\cdot \mid X, Y), \quad C \sim \kappa_C(\cdot \mid Z, Y),$$

conditionally independently given the sources. Write $\mathcal{C}_\Delta$ for the set of joint laws of (A, B, C) obtained this way for binary alphabets; the notation extends to any fixed finite alphabets. Assign each node an independent random seed and include that seed in one of its incident sources; the other node sharing that source ignores it. This preserves the triangle graph and shows that stochastic and deterministic responses generate the same set of observed laws [23]; we use whichever description is convenient. By a theorem of Rosset, Gisin and Wolfe [20], latent alphabets may be taken finite of bounded size, so $\mathcal{C}_\Delta$ is compact and semialgebraic; we use compactness in Section 11 and semialgebraicity in Section 13.

2.2 Inflation at order t

Fix $t \geq 1$. The order- t inflation of the triangle has t independent copies $X_1, \dots, X_t$ of X , and likewise of Y and Z , and the $3t^2$ copied observations

$$A^{ij} = A(X_i, Z_j), \quad B^{ik} = B(X_i, Y_k), \quad C^{jk} = C(Z_j, Y_k), \quad i, j, k \in [t],$$

each with the response kernel of its original. The copied ancestors are $\{X_i, Z_j\}$, $\{X_i, Y_k\}$, $\{Z_j, Y_k\}$. A *copied triangle* is $\Delta_{ijk} = \{A^{ij}, B^{ik}, C^{jk}\}$; the *diagonal triangles* are $\Delta_{\ell\ell\ell}$, $\ell \in [t]$. If $P \in \mathcal{C}_\Delta$ is realized by some model, then running the inflated model produces a joint law Γ_t on all copied observations with the following properties, which are therefore necessary for compatibility: independent permutations of the three index families leave Γ_t invariant, every copied triangle has law P , and the diagonal triangles, which depend on disjoint sets of source copies, are independent. The finite tests below ask for a joint law with some of these properties.

Definition 2.1 (Navascués–Wolfe feasible set). $P \in \mathcal{I}_t^{\text{NW}}$ if there is a probability law Γ_t on $\{0, 1\}^{3t^2}$ such that

- (i) Γ_t is invariant under independent permutations of the X -, Z - and Y -copy indices, that is under the group S_t^3 acting on the labels (i, j, k) ; and
- (ii) the diagonal law is a tensor power: $\text{Law}_{\Gamma_t}((A^{\ell\ell}, B^{\ell\ell}, C^{\ell\ell})_{\ell=1}^t) = P^{\otimes t}$.

This is the full-table symmetric formulation of [15] (equations (16)–(19) in the published numbering; the arXiv version numbers them (17)–(20)), whose degree- g diagonal conditions for $g \leq t$ follow from (ii) by marginalization. Navascués and Wolfe index the copies cyclically as $A(U_1^i, U_2^j)$, $B(U_2^i, U_3^j)$, $C(U_3^i, U_1^j)$; our labelling A^{ij} , B^{ik} , C^{jk} names the same variables with the shared index written in the same position, and the symmetry group is the same. They also note that further relations hold in a genuine inflation, such as $\text{Law}(A^{12}, B^{21}, C^{12}) = P_A \otimes P_B \otimes P_C$ at order two (written here in our indexing), and do not impose them in the hierarchy; those relations are instances of the ancestral-independence prescriptions of Definition 2.2 below, which our witness also satisfies. Membership in the Navascués–Wolfe feasible set is a linear feasibility problem in the 2^{3t^2} entries of Γ_t . Navascués and Wolfe prove that $\mathcal{I}_t^{\text{NW}} \downarrow \mathcal{C}_\Delta$ as $t \rightarrow \infty$, with a quantitative rate recalled in Section 12.

Definition 2.2 (Injectable sets and the AI feasible set). A set of copied observations is *injectable* if erasing copy indices maps it injectively onto a subset of $\{A, B, C\}$ and the shared latent indices agree; for the triangle these are the subsets of copied triangles. Two sets are *ancestrally independent* if their copied latent ancestors are disjoint. $P \in \mathcal{I}_t^{\text{AI}}$ if $P \in \mathcal{I}_t^{\text{NW}}$ with a witness Γ_t under which every injectable set has the corresponding marginal of P and every union of pairwise ancestrally independent injectable sets has the product of the corresponding marginals.

Definition 2.3 (Recursively expressible feasible set). Following [23, Definition 7], expressible sets are generated from injectable sets by two rules: if X' and Y' are d -separated by Z' in the inflated causal graph and the laws on $X' \cup Z'$ and $Y' \cup Z'$ are prescribed, then $p(x', y', z') = p(x', z')p(y', z')/p(z')$ (and 0 when $p(z') = 0$) is prescribed on $X' \cup Y' \cup Z'$; and marginals of prescribed sets are prescribed. $P \in \mathcal{I}_t^{\text{exp}}$ if $P \in \mathcal{I}_t^{\text{AI}}$ with a witness satisfying every prescription so generated.

For fixed P , each prescription fixes a marginal of Γ_t , so the strengthened tests are also linear feasibility problems. Restricting a witness to fewer copy indices preserves all three sets of conditions. Thus each hierarchy decreases with t , and at a fixed order the feasible sets are nested,

$$\mathcal{I}_t^{\text{exp}} \subseteq \mathcal{I}_t^{\text{AI}} \subseteq \mathcal{I}_t^{\text{NW}}, \quad (1)$$

and a test is stronger when its feasible set is smaller. All three contain $\mathcal{C}_\Delta$. We prove our witness lies in the smallest set, so the conclusion holds for all three; the Navascués–Wolfe set is the one whose asymptotic completeness is published and the one named in the question we answer.

For $H \in \{\text{NW}, \text{AI}, \text{exp}\}$ and $P \notin \mathcal{C}_\Delta$, write

$$t_{\min}^H(P) = \min\{t \geq 1 : P \notin \mathcal{I}_t^H\}.$$

Asymptotic completeness of the NW hierarchy and (1) make this a finite integer. We use

$$d_{\text{TV}}(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)|, \quad d_{\text{TV}}(P, \mathcal{C}) = \inf_{Q \in \mathcal{C}} d_{\text{TV}}(P, Q),$$

and write $d_2(P, \mathcal{C}) = \inf_{Q \in \mathcal{C}} \|P - Q\|_2$ for Euclidean distance.

2.3 The Finner inequality

Every $P \in \mathcal{C}_\Delta$ satisfies

$$P(000)^2 \leq P_A(0) P_B(0) P_C(0), \quad (2)$$

where P_A, P_B, P_C are the one-variable marginals. This is the event form of Finner's inequality [6], which Renou et al. [19] applied to the triangle network; we include a short proof in Section 8.2 because we need it for arbitrary stochastic responses and arbitrary latent alphabets.

3 Pair-source scenarios and the three finite tests

The triangle is one member of a family of scenarios indexed by finite graphs. We fix that family, write the three finite tests of Section 2 for it, and prove the three lemmas used in Section 4: the two stronger tests coincide, blocks of observers with disjoint sources are independent from order two on, and witnesses transport along induced subgraphs.

3.1 Scenarios

Definition 3.1 (Pair-source scenario). Let $G = (V, E)$ be a finite simple graph without isolated vertices. The associated scenario has one latent source S_e for each edge $e \in E$, taking values in an arbitrary measurable space, and one observed variable O_v with a finite alphabet $\mathcal{O}_v$ for each vertex $v \in V$. The sources are mutually independent, and

$$O_v \sim \kappa_v(\cdot \mid (S_e)_{e \ni v}),$$

conditionally independently across v given the sources. Write $\mathcal{C}_G$ for the set of joint laws of $(O_v)_{v \in V}$ obtained in this way. Unless stated otherwise every $\mathcal{O}_v$ is $\{0, 1\}$, and we call these the binary observations.

Each source is read by exactly the two endpoints of its edge. The triangle of Section 2 is the scenario of the 3-cycle, $\mathcal{C}_\Delta = \mathcal{C}_{C_3}$; the graph P_n is the path on n vertices.

Sources shared by three or more observers, and latent variables with latent parents, are outside this definition; see Remark 4.5.

Assign each vertex v an independent random seed and append it to the source of one edge at v , to be ignored by the other endpoint. The graph is unchanged and the seed makes the response at v deterministic, so stochastic and deterministic responses generate the same set $\mathcal{C}_G$ [23]. The seed needs an edge to hide on, which is why the definition excludes isolated vertices.

3.2 Inflation at order t

Definition 3.2 (Order- t inflation). Fix $t \geq 1$ and write $E(v) = \{e \in E : v \in e\}$ for the edges at v and $E(W) = \bigcup_{v \in W} E(v)$ for a vertex set W . The order- t inflation of G has independent copies $S_e^1, \dots, S_e^t$ of each source, and for every vertex v and every index tuple $i \in [t]^{E(v)}$ one copied observation

$$O_v^i \sim \kappa_v(\cdot \mid (S_e^{i_e})_{e \in E(v)}),$$

all conditionally independent given the source copies. Writing r also for the constant tuple with value r , the *row* $r \in [t]$ is the family $(O_v^r)_{v \in V}$, and the t rows form the diagonal.

A vertex of degree d has t^d copies, so the inflation carries $\sum_v t^{\deg v}$ observed variables. The group $\prod_{e \in E} S_t$ acts on the copy labels: $\sigma = (\sigma_e)_{e \in E}$ sends the label i to $\sigma(i)$ with $\sigma(i)_e = \sigma_e(i_e)$, and hence sends O_v^i to $O_v^{\sigma(i)}$. Each σ maps rows to rows when all its components are equal, and otherwise mixes them.

Call a probability law on $\prod_{v \in V} (\mathcal{O}_v)^{[t]^{E(v)}}$ a *table* of order t . Each of the three tests asks for a table with prescribed marginals.

Definition 3.3 (Navascués–Wolfe feasible set). $P \in \mathcal{I}_t^{\text{NW}}(G)$ if there is a table Γ_t of order t that is invariant under $\prod_{e \in E} S_t$ and whose diagonal law is a tensor power,

$$\text{Law}_{\Gamma_t}((O_v^r)_{v \in V} : r = 1, \dots, t) = P^{\otimes t}.$$

Definition 3.4 (Injectable sets and the AI feasible set). A set S of copied observations is *injectable* if the map π that erases copy indices is injective on S and any two members of S whose vertices share an edge carry the same index on that edge [23, Definition 4]. Equivalently, S is injectable if and only if there are a vertex set $W \subseteq V$ and an assignment $\iota : E(W) \rightarrow [t]$ with

$$S = \{O_v^{\iota|_{E(v)}} : v \in W\},$$

in which case $\pi(S) = W$. Two sets of copied observations are *ancestrally independent* if the sets of source copies they read are disjoint. $P \in \mathcal{I}_t^{\text{AI}}(G)$ if $P \in \mathcal{I}_t^{\text{NW}}(G)$ with a witness Γ_t under which every injectable set S has marginal $P_{\pi(S)}$ and every union of pairwise ancestrally independent injectable sets has the product of the corresponding marginals.

For the equivalence in Definition 3.4, given S injectable, let $W = \pi(S)$ and let $\iota(e)$ be the common index carried on e by the members of S at the endpoints of e that lie in W , which is well defined by the agreement requirement; conversely every set of the displayed form is injectable. For the triangle this reproduces the description in Appendix A: the injectable sets are the subsets of the copied triangles.

Definition 3.5 (Recursively expressible feasible set). Expressible sets of copied observations are generated from injectable sets by the two rules of [23, Definition 7], read in the inflated causal graph of Definition 3.2: if X and Y are d -separated by Z and the laws on $X \cup Z$ and on $Y \cup Z$ are prescribed, then

$$p(x, y, z) = \frac{p(x, z) p(y, z)}{p(z)} \quad \text{where } p(z) > 0, \quad p(x, y, z) = 0 \quad \text{where } p(z) = 0,$$

is prescribed on $X \cup Y \cup Z$; and every marginal of a prescribed set is prescribed. $P \in \mathcal{I}_t^{\text{exp}}(G)$ if $P \in \mathcal{I}_t^{\text{AI}}(G)$ with a witness satisfying every prescription so generated.

Proposition 3.6 (Soundness). *For every pair-source scenario G and every $t \geq 1$,*

$$\mathcal{C}_G \subseteq \mathcal{I}_t^{\text{exp}}(G) \subseteq \mathcal{I}_t^{\text{AI}}(G) \subseteq \mathcal{I}_t^{\text{NW}}(G).$$

Proof. The two inclusions on the right hold because each test adds prescriptions to the previous one. For the first, realize P by a model, run the model on the inflated source family of Definition 3.2, and let Γ_t be the law of the copied observations. Permuting the copies of one source permutes independent identically distributed variables, so Γ_t is invariant. Distinct rows read disjoint source copies and each row has law P , so the diagonal law is $P^{\otimes t}$. An injectable

set S reads, for each $v \in \pi(S)$, the copies indexed by a single global assignment ι , so the joint law of S is that of $(O_v)_{v \in \pi(S)}$ in the original model, namely $P_{\pi(S)}$. Ancestrally independent sets read disjoint source copies, hence are independent. The inflated model is a causal model for the inflated graph, so all conditional independences read off that graph hold, and the expressible prescriptions follow by induction on their generation. $\square$

3.3 The two stronger tests coincide

In the inflated graph every source copy is a root and every copied observation is a sink, which makes d -separation explicit.

Lemma 3.7 (Trail criterion). *Let X, Y, Z be disjoint sets of copied observations in the order- t inflation of a pair-source scenario. Every trail between a member of X and a member of Y alternates copied observations and source copies, each source copy on it being a fork and each internal copied observation a collider. Such a trail is active given Z if and only if all its internal copied observations lie in Z . Consequently $X \perp_d Y \mid Z$ if and only if no trail from X to Y has all its internal copied observations in Z .*

Proof. Every edge of the inflated graph joins a source copy to a copied observation, so the nodes of a trail alternate between the two kinds. A source copy has no parents, so both of its neighbours on the trail are its children and it is a fork; being a fork not in Z , since Z consists of copied observations, it never blocks. An internal copied observation has no children, so both of its neighbours on the trail are its parents and it is a collider; a collider blocks unless it or one of its descendants lies in Z , and it has no descendants, so it blocks exactly when it is not in Z . A trail is active given Z when no node on it blocks, which is the stated condition. The last sentence is the definition of d -separation. $\square$

The shortest trails are the two-step ones, $x - S_e^m - y$, which have no internal observation and are therefore always active: two copied observations reading a common source copy are never d -separated.

Definition 3.8 (Shared-parent graph and AI sets). For a set W of copied observations, the *shared-parent graph* on W joins two members when they read a common source copy. Call W an *AI set* if it is a union of pairwise ancestrally independent injectable sets.

Lemma 3.9 (Root-sink lemma). *Let G be a pair-source scenario and $t \geq 1$.*

- (i) *A set W of copied observations is an AI set if and only if every connected component of its shared-parent graph is injectable.*
- (ii) *A table satisfies every expressible prescription of Definition 3.5 if and only if it satisfies the injectable and ancestral-independence prescriptions of Definition 3.4.*

In particular $\mathcal{I}_t^{\text{exp}}(G) = \mathcal{I}_t^{\text{AI}}(G)$ for every t .

Proof. (i) If every component is injectable, then components with a common source copy would be joined, so distinct components are ancestrally independent and W is their union. Conversely, let $W = W_1 \cup \dots \cup W_m$ with the W_r injectable and pairwise ancestrally independent. Every edge of the shared-parent graph joins two members reading a common source copy, hence two members of the same W_r . So each component lies in one W_r , and a subset of an injectable set is injectable.

(ii) Necessity: injectable sets are expressible by the first rule, and if X and Y are ancestrally independent then by Lemma 3.7 no trail from X to Y can avoid internal copied observations,

and one with an internal copied observation is blocked by $\emptyset$; hence $X \perp_d Y \mid \emptyset$ and the join rule with $Z = \emptyset$ prescribes $p(x)p(y)$. Iterating gives the products of Definition 3.4.

Sufficiency: let Γ satisfy the prescriptions of Definition 3.4. We prove by induction on the generation of the expressible family that every generated set W is an AI set and that the law prescribed for it agrees with the marginal Γ_W . By (i) and the AI prescriptions, the marginal of an AI set is the product of $P_{\pi(K)}$ over the components K of its shared-parent graph.

For an injectable set the claim is the first AI prescription. For a marginal, the components of the shared-parent graph on a subset $W' \subseteq W$ are contained in those on W , hence injectable, so W' is again an AI set, and the marginal of the prescription is the prescription of the marginal.

Consider the join rule with $X \perp_d Y \mid Z$, where $X \cup Z$ and $Y \cup Z$ are already generated. Let K be a connected component of the shared-parent graph on $X \cup Y \cup Z$ and suppose K meets both X and Y . Among all paths inside K from a member of X to a member of Y choose one of minimal length. No internal vertex of it lies in X , since the subpath from that vertex to the Y -end would be shorter, and for the same reason none lies in Y ; so every internal vertex lies in Z . Inserting between consecutive vertices a source copy they share turns this path into a trail from X to Y whose internal copied observations all lie in Z , which is active by Lemma 3.7 and contradicts $X \perp_d Y \mid Z$. Hence every component of the shared-parent graph on $X \cup Y \cup Z$ lies in $X \cup Z$ or in $Y \cup Z$. Such a component is also a component of the shared-parent graph on the smaller set, since it is connected there and no member of the smaller set outside it is adjacent to it. As $X \cup Z$ and $Y \cup Z$ are AI sets, all these components are injectable, so $X \cup Y \cup Z$ is an AI set by (i).

It remains to check the prescribed law. Let $\mathcal{K}$ be the set of components of the shared-parent graph on $X \cup Y \cup Z$, split as $\mathcal{X} \sqcup \mathcal{Y} \sqcup \mathcal{Z}$ according to whether a component meets X , meets Y , or is contained in Z ; by the previous paragraph no component meets both X and Y , so this is a partition. Write $\alpha = \prod_{K \in \mathcal{X}} P_{\pi(K)}$, $\beta = \prod_{K \in \mathcal{Y}} P_{\pi(K)}$ and $\mu = \prod_{K \in \mathcal{Z}} P_{\pi(K)}$, each evaluated at the coordinates of its component. The components of the shared-parent graph on $X \cup Z$ are the members of $\mathcal{X} \cup \mathcal{Z}$ together with the components of $K \cap Z$ for $K \in \mathcal{Y}$; write ν_Y for the product of $P_{\pi(K)}$ over the latter, and ν_X for the analogous product coming from $\mathcal{X}$ inside $Y \cup Z$. Then

$$\Gamma_{X \cup Z} = \alpha \mu \nu_Y, \quad \Gamma_{Y \cup Z} = \beta \mu \nu_X, \quad \Gamma_Z = \mu \nu_X \nu_Y,$$

the last because the components of the shared-parent graph on Z are those of $K \cap Z$ for $K \in \mathcal{X} \cup \mathcal{Y}$ together with $\mathcal{Z}$. At a point where $\Gamma_Z > 0$ the factors μ , ν_X and ν_Y are positive and cancel,

$$\frac{\Gamma_{X \cup Z} \Gamma_{Y \cup Z}}{\Gamma_Z} = \alpha \beta \mu = \Gamma_{X \cup Y \cup Z},$$

which is the prescribed value. Where $\Gamma_Z = 0$ the rule prescribes 0, and $\Gamma_{X \cup Y \cup Z} \leq \Gamma_Z = 0$ there, so the two agree on zero fibres as well. This completes the induction. $\square$

Lemma 3.9 identifies the two stronger tests but says nothing about the Navascués–Wolfe test, and imposes no causal model on the table as a whole.

3.4 Blocks with disjoint sources

Lemma 3.10 (Source-disjoint independence). *Let $t \geq 2$ and $P \in \mathcal{I}_t^{\text{NW}}(G)$. If $U, W \subseteq V$ satisfy $E(U) \cap E(W) = \emptyset$, then*

$$P_{U \cup W} = P_U \otimes P_W.$$

Proof. Let Γ be a witness. Let σ be the element of $\prod_e S_t$ that transposes the copy indices 1 and 2 on every edge of $E(W)$ and fixes all other indices. For $u \in U$ every edge at u lies outside $E(W)$, so σ fixes O_u^1 ; for $w \in W$ every edge at w lies in $E(W)$, so σ sends O_w^1 to O_w^2 . Invariance of Γ gives

$$\text{Law}((O_u^1)_{u \in U}, (O_w^1)_{w \in W}) = \text{Law}((O_u^1)_{u \in U}, (O_w^2)_{w \in W}).$$

The left side is the marginal of row 1, namely $P_{U \cup W}$. The right side reads U -coordinates from row 1 and W -coordinates from row 2, and the two rows are independent with law P each, so it is $P_U \otimes P_W$. $\square$

Two vertices joined by an edge have that edge in both incidence sets, so the hypothesis says exactly that U and W are disjoint and no edge joins them. Iterating the lemma with U a single block and W the union of the others gives mutual independence of any family of pairwise source-disjoint blocks. In particular the observed law of a disconnected graph factors over its components from order two on.

3.5 Transport along induced subgraphs

Lemma 3.11 (Induced-subgraph transport). *Let H be an induced subgraph of G in which no vertex is isolated, and let $\mathcal{J} \in \{\mathcal{I}^{\text{NW}}, \mathcal{I}^{\text{AI}}, \mathcal{I}^{\text{exp}}\}$.*

- (i) *If $Q \in \mathcal{C}_G$ then $Q_{V(H)} \in \mathcal{C}_H$.*
- (ii) *If $P_H \in \mathcal{J}_t(H)$ then $P := P_H \otimes \text{Bern}(1/2)^{\otimes(V \setminus V(H))} \in \mathcal{J}_t(G)$.*
- (iii) *$d_{\text{TV}}(P, \mathcal{C}_G) \geq d_{\text{TV}}(P_H, \mathcal{C}_H)$. In particular, if $P_H \notin \mathcal{C}_H$ then $P \notin \mathcal{C}_G$.*

Proof. (i) Take a G -model for Q and keep the observations at $V(H)$. Since H is induced, every edge of G with both endpoints in $V(H)$ is an edge of H . An edge of G with exactly one endpoint v in $V(H)$ has its source read, among the retained observations, only by v ; append that source to the source of one edge of H at v , which exists because v is not isolated in H , and let the other endpoint of that edge ignore the appended component. The result is an H -model with the same observed law on $V(H)$.

(ii) Let Γ^H be a witness for P_H at order t and write $E_H(v)$ for the set of edges of H at v . Define a table Γ for G as follows: draw the copied observations $(O_v^j)_{v \in V(H), j \in [t]^{E_H(v)}}$ from Γ^H , draw independent fair bits ξ_w^k for $w \notin V(H)$ and $k \in [t]^{E(w)}$, and set

$$O_v^i := O_v^{i|E_H(v)} \quad (v \in V(H)), \quad O_w^k := \xi_w^k \quad (w \notin V(H)),$$

so that a copied observation at an H -vertex ignores the indices of the edges leaving H .

Symmetry: a permutation of the copy indices of an edge of H acts on the first family as the corresponding element of the symmetry group of the H -inflation and fixes the fair bits; a permutation of the copy indices of any other edge leaves every O_v^i with $v \in V(H)$ unchanged and permutes the fair bits among themselves, which preserves their law. **Diagonal law:** row r of the G -inflation restricted to $V(H)$ is row r of the H -inflation, and the outside entries of the t rows are $t|V \setminus V(H)|$ distinct fair bits, so the diagonal law is $P_H^{\otimes t} \otimes \text{Bern}(1/2)^{\otimes t(V \setminus V(H))} = P^{\otimes t}$.

Injectable prescriptions: let S be injectable in the G -inflation with $W = \pi(S)$. Its members at H -vertices, with the outside indices dropped, form an injectable set of the H -inflation, because index erasure is still injective and the indices on the edges of H still agree; so their joint law under Γ is $(P_H)_{W \cap V(H)}$. Its members at outside vertices are distinct fair bits, independent of everything else, and $P_W = (P_H)_{W \cap V(H)} \otimes \text{Bern}(1/2)^{\otimes(W \setminus V(H))}$, as required. Ancestrally

independent families: source copies of edges of H are source copies in G as well, so ancestrally independent injectable sets of the G -inflation have ancestrally independent H -parts, whose laws multiply under Γ^H , while the fair bits are independent of them and of each other. Hence Γ satisfies the conditions of Definition 3.4 whenever Γ^H does. The expressible case follows from the AI case by Lemma 3.9, applied in H and in G .

(iii) For $Q \in \mathcal{C}_G$, part (i) gives $Q_{V(H)} \in \mathcal{C}_H$, and marginalization does not increase total variation, so $d_{\text{TV}}(P, Q) \geq d_{\text{TV}}(P_{V(H)}, Q_{V(H)}) = d_{\text{TV}}(P_H, Q_{V(H)}) \geq d_{\text{TV}}(P_H, \mathcal{C}_H)$. Taking the infimum over Q gives the claim. $\square$

Part (ii) preserves strict positivity and rationality of the target, since the appended factor is the fair-bit law.

4 The termination classification

Definition 4.1 (Double-star). A *double-star* is a graph with a distinguished edge $\{b, c\}$, the *central edge*, such that every other vertex is adjacent to exactly one of b and c and to nothing else. Equivalently, a double-star is a tree of diameter at most three. Single edges and stars are double-stars.

Theorem 4.2 (Termination classification). *Let G be a finite simple graph without isolated vertices, with binary observations. The following are equivalent.*

- (a) $\mathcal{I}_t^{\text{NW}}(G) = \mathcal{C}_G$ for some finite t .
- (b) $\mathcal{I}_t^{\text{AI}}(G) = \mathcal{C}_G$ for some finite t .
- (c) $\mathcal{I}_t^{\text{exp}}(G) = \mathcal{C}_G$ for some finite t .
- (d) Every connected component of G is a double-star.

When they hold,

$$\mathcal{I}_2^{\text{NW}}(G) = \mathcal{I}_2^{\text{AI}}(G) = \mathcal{I}_2^{\text{exp}}(G) = \mathcal{C}_G.$$

When they fail, for every $t \geq 1$ there is a strictly positive rational law

$$P \in \mathcal{I}_t^{\text{exp}}(G) \setminus \mathcal{C}_G \subseteq \mathcal{I}_t^{\text{AI}}(G) \setminus \mathcal{C}_G \subseteq \mathcal{I}_t^{\text{NW}}(G) \setminus \mathcal{C}_G.$$

The graph and the alphabets stay fixed as t grows; only the law P varies. The proof combines Theorem 5.1 with Theorems 6.8 and 7.5 through the exhaustion lemma below.

Lemma 4.3 (Exhaustion). *Let K be a finite connected graph that has at least one edge and is not a double-star. Then K has an induced subgraph isomorphic to a cycle C_m with $m \geq 3$ or to the path P_5 .*

Proof. Suppose first that K contains a cycle, and let C be one of minimum length. A chord of C would split it into two shorter cycles, so C has no chords and is induced. Its length is at least 3.

Suppose next that K is a tree. We claim that a tree of diameter at most three is a double-star. If the diameter is 1 then K is a single edge. If it is 2 then K is a star. Let the diameter be 3 and fix a geodesic $u_0 u_1 u_2 u_3$. Let w be a vertex other than u_1 and u_2 , and consider the unique path from w to u_1 . If it avoids u_2 , then it also avoids u_3 , since the path from u_3 to u_1 passes through u_2 ; so concatenating with $u_1 u_2 u_3$ gives a path and $d(w, u_3) = d(w, u_1) + 2 \leq 3$, whence $d(w, u_1) = 1$. If it contains u_2 , then the path from w to u_2 avoids u_1 and hence u_0 , and the same argument with u_0 gives $d(w, u_2) = 1$. Every vertex is therefore adjacent to u_1 or u_2 , and K is a double-star with central edge $u_1 u_2$.

So a tree that is not a double-star has diameter at least 4. Take a geodesic of length at least 4 and let $v_0, \dots, v_4$ be five consecutive vertices on it. Any edge of K among them other than the four geodesic edges would create a cycle in a tree, so the induced subgraph on $\{v_0, \dots, v_4\}$ is P_5 . $\square$

Proof of Theorem 4.2. Assume (d). Each component is a double-star with binary observations, so Theorem 5.1 and Corollary 5.4 give $\mathcal{I}_2^{\text{NW}}(G) = \mathcal{C}_G$. With Proposition 3.6,

$$\mathcal{C}_G \subseteq \mathcal{I}_2^{\text{exp}}(G) \subseteq \mathcal{I}_2^{\text{AI}}(G) \subseteq \mathcal{I}_2^{\text{NW}}(G) = \mathcal{C}_G,$$

so all four sets coincide and (a), (b), (c) hold at $t = 2$.

Assume (d) fails, and fix $t \geq 1$. Some component K of G is not a double-star, and K has an edge because G has no isolated vertices. By Lemma 4.3, K has an induced subgraph H isomorphic to C_m for some $m \geq 3$ or to P_5 ; since K is induced in G , so is H , and H has no isolated vertices. Theorem 6.8 in the first case and Theorem 7.5 in the second supply a strictly positive rational law

$$P_H \in \mathcal{I}_t^{\text{exp}}(H) \setminus \mathcal{C}_H.$$

Let $P = P_H \otimes \text{Bern}(1/2)^{\otimes(V \setminus V(H))}$. By Lemma 3.11(ii), $P \in \mathcal{I}_t^{\text{exp}}(G)$, and by Lemma 3.11(iii), $P \notin \mathcal{C}_G$. It is strictly positive and rational because P_H is. By Proposition 3.6 the same law lies in $\mathcal{I}_t^{\text{AI}}(G)$ and in $\mathcal{I}_t^{\text{NW}}(G)$, so none of the three tests equals $\mathcal{C}_G$ at order t . As t was arbitrary, (a), (b) and (c) all fail.

The two paragraphs give (d) $\Rightarrow$ (a), (b), (c) and the negation of (d) $\Rightarrow$ the negation of each of (a), (b), (c), which is the stated equivalence, together with the two displayed conclusions. $\square$

For $m = 3$ the witness of Theorem 6.8 is a second nonterminating family for the triangle, alongside the defect construction of Theorem 8.2.

Remark 4.4 (Larger observed alphabets). Fix finite alphabets $\mathcal{O}_v$ with $|\mathcal{O}_v| \geq 2$. The classification is unchanged, with a different order in the terminating case. If some component is not a double-star, regard the binary law P of Theorem 4.2 as a law on $\prod_v \mathcal{O}_v$ supported on $\{0, 1\}^V$. A compatible realization of such a law has responses that leave $\{0, 1\}$ with probability zero, hence is a binary realization, so the law remains incompatible; and its witness, read in the larger alphabets, satisfies the same prescriptions. The transferred law is supported on the binary sub-alphabet, so the strict positivity in Theorem 4.2 is a statement about binary observations. If every component is a double-star, Theorem 5.1 gives termination at

$$T = \max(\{2\} \cup \{|\mathcal{O}_v| : v \text{ a leaf}\}),$$

the leaves being taken with respect to a choice of central edge in each component; the bound does not involve the alphabets at the central vertices.

Remark 4.5 (Scenarios that are not pair-source). Let three observers read one common source. That source can carry a sample of the whole observed triple, so every law on the three outcomes is compatible and the order-one test already characterizes the compatible set, while the observed adjacency graph of this scenario is the triangle, which does not terminate at any order. The adjacency graph alone therefore carries no classification once source hyperedges are allowed.

Scope. Theorem 4.2 decides which binary pair-source scenarios admit a finite characterizing order, and for the terminating ones it gives the order. It does not decide the worst-case distance $\sup\{d_{\text{TV}}(P, \mathcal{C}_G) : P \in \mathcal{I}_t^{\text{NW}}(G)\}$ for a nonterminating G , which is Question 15.1 for the triangle. Nontermination together with the convergence $\mathcal{I}_t^{\text{NW}}(G) \downarrow \mathcal{C}_G$ of [15] gives infinitely many strict inclusions $\mathcal{I}_{t+1}^{\text{NW}}(G) \subsetneq \mathcal{I}_t^{\text{NW}}(G)$, but not strictness at every order. It says nothing about quantum or no-signalling resources.

5 Finite reconstruction on double-stars

Fix a double-star G in the sense of Definition 4.1: two central vertices b, c joined by the central edge $y = \{b, c\}$, left leaves $\ell_1, \dots, \ell_p$ joined to b by edges $e_1, \dots, e_p$, and right leaves $r_1, \dots, r_q$ joined to c by edges $f_1, \dots, f_q$. Observed alphabets $\mathcal{O}_v$ are finite and arbitrary. Put

$$T = \max(\{2\} \cup \{|\mathcal{O}_{\ell_s}|\}_{s \leq p} \cup \{|\mathcal{O}_{r_j}|\}_{j \leq q}),$$

In the order- T inflation the leaves have one index each, and the centres have one index per incident edge. Write

$$L_s^m = O_{\ell_s}^m, \quad R_j^m = O_{r_j}^m, \quad B^{\mathbf{i}, n} = O_b^{(\mathbf{i}, n)}, \quad C^{n, \mathbf{k}} = O_c^{(n, \mathbf{k})},$$

where $m, n \in [T]$, $\mathbf{i} \in [T]^p$ records the index on each e_s , and $\mathbf{k} \in [T]^q$ the index on each f_j . Row m consists of L_s^m , R_j^m , $B^{m, m}$ and $C^{m, m}$, with $\mathbf{i}$ and $\mathbf{k}$ constant equal to m .

For a law P on $\prod_v \mathcal{O}_v$ write $P(\beta, \gamma, \xi, \zeta)$ for the probability that $O_b = \beta$, $O_c = \gamma$, $O_{\ell_s} = \xi_s$ for all s and $O_{r_j} = \zeta_j$ for all j .

Theorem 5.1 (Double-star reconstruction). *For every double-star G with finite observed alphabets,*

$$\mathcal{I}_T^{\text{NW}}(G) = \mathcal{C}_G.$$

Proof. The inclusion $\mathcal{C}_G \subseteq \mathcal{I}_T^{\text{NW}}(G)$ is Proposition 3.6. Let $P \in \mathcal{I}_T^{\text{NW}}(G)$ and let Γ be a witness at order T . We recover a model for P in four steps.

(i) *The leaves and all their copies are independent.* Distinct leaves have disjoint incidence sets, each consisting of a single edge, so Lemma 3.10, available because $T \geq 2$, applies to one leaf against the union of the others and, iterated, gives

$$P_{\text{leaf}}(\xi, \zeta) := P(O_{\ell_s} = \xi_s \ \forall s, \ O_{r_j} = \zeta_j \ \forall j) = \prod_s P_{\ell_s}(\xi_s) \prod_j P_{r_j}(\zeta_j). \quad (3)$$

Under Γ the T rows are independent with law P each, and the leaf entries of a single row are independent by (3). Every copied leaf observation occurs in exactly one row, so the whole family $\{L_s^m, R_j^m\}$ is independent under Γ , with $L_s^m \sim P_{\ell_s}$ and $R_j^m \sim P_{r_j}$.

(ii) *A positive conditioning event.* For each left leaf choose $x_s^1, \dots, x_s^T \in \mathcal{O}_{\ell_s}$ so that every symbol of $\text{supp } P_{\ell_s}$ occurs in the list, which is possible because $T \geq |\mathcal{O}_{\ell_s}|$; fill the remaining positions with repetitions of a supported symbol. Choose $z_j^1, \dots, z_j^T$ for each right leaf in the same way. Let

$$E = \{L_s^m = x_s^m \ \forall s, m\} \cap \{R_j^m = z_j^m \ \forall j, m\}.$$

By (i),

$$\Gamma(E) = \prod_{s,m} P_{\ell_s}(x_s^m) \prod_{j,m} P_{r_j}(z_j^m) > 0. \quad (4)$$

(iii) *The conditional marginal of the two centres.* Fix $\mathbf{i} \in [T]^p$ and $\mathbf{k} \in [T]^q$. Choose permutations $\sigma_s \in S_T$ of the copy indices of e_s with $\sigma_s(1) = i_s$, permutations $\tau_j \in S_T$ of those of f_j with $\tau_j(1) = k_j$, and the identity on the central edge y . The resulting element of the symmetry group carries row m to

$$\left(B^{(\sigma_s(m))_s, m}, C^{m, (\tau_j(m))_j}, (L_s^{\sigma_s(m)})_s, (R_j^{\tau_j(m)})_j \right),$$

which for $m = 1$ is $(B^{\mathbf{i}, 1}, C^{1, \mathbf{k}}, (L_s^{i_s})_s, (R_j^{k_j})_j)$. By invariance the joint law of these T images equals the joint law of the T rows, namely $P^{\otimes T}$. As m runs over $[T]$ the index $\sigma_s(m)$ runs over $[T]$, so every copied leaf observation occurs exactly once among the images and the event E is determined by them. Marginalizing the images of rows $2, \dots, T$ over their two centre entries and using (3),

$$\Gamma(B^{\mathbf{i}, 1} = \beta, C^{1, \mathbf{k}} = \gamma, E) = P(\beta, \gamma, x^{\mathbf{i}}, z^{\mathbf{k}}) \prod_s \prod_{m \neq i_s} P_{\ell_s}(x_s^m) \prod_j \prod_{m \neq k_j} P_{r_j}(z_j^m),$$

where $x^{\mathbf{i}} = (x_s^{i_s})_s$ and $z^{\mathbf{k}} = (z_j^{k_j})_j$. Dividing by (4) and using (3) once more,

$$\Gamma(B^{\mathbf{i}, 1} = \beta, C^{1, \mathbf{k}} = \gamma \mid E) = P(O_b = \beta, O_c = \gamma \mid O_{\ell_s} = x_s^{i_s} \forall s, O_{r_j} = z_j^{k_j} \forall j), \quad (5)$$

the conditioning event on the right having positive probability whenever every $x_s^{i_s}$ and $z_j^{k_j}$ is supported. This identity is the finite-order step, a statement about the witness derived from symmetry and the diagonal law alone.

(iv) *Two response tables and fresh sources.* For $\xi \in \prod_s \text{supp } P_{\ell_s}$ pick positions $\mathbf{i}(\xi)$ with $x_s^{i_s(\xi)} = \xi_s$, and for $\zeta \in \prod_j \text{supp } P_{r_j}$ pick $\mathbf{k}(\zeta)$ with $z_j^{k_j(\zeta)} = \zeta_j$. Under the conditional law $\Gamma(\cdot \mid E)$ define the random maps

$$F(\xi) = B^{\mathbf{i}(\xi), 1}, \quad G(\zeta) = C^{1, \mathbf{k}(\zeta)},$$

with arbitrary fixed values on unsupported arguments. Both are functions of the same table, so the pair (F, G) has a joint law μ , and (5) reads

$$\mu(F(\xi) = \beta, G(\zeta) = \gamma) = P(O_b = \beta, O_c = \gamma \mid \xi, \zeta) \quad (6)$$

for all supported ξ, ζ .

Now take fresh independent sources: let the central source be $Y^* = (F, G) \sim \mu$, let the source of e_s be $X_s^* \sim P_{\ell_s}$ and the source of f_j be $Z_j^* \sim P_{r_j}$, all mutually independent. Respond by

$$O_{\ell_s} = X_s^*, \quad O_{r_j} = Z_j^*, \quad O_b = F(X_1^*, \dots, X_p^*), \quad O_c = G(Z_1^*, \dots, Z_q^*).$$

Each response reads only the sources of its incident edges: b reads $X_1^*, \dots, X_p^*$ and Y^* , and c reads Y^* and $Z_1^*, \dots, Z_q^*$. This is a model for the scenario G , and by (6) and (3) its observed law is

$$\prod_s P_{\ell_s}(\xi_s) \prod_j P_{r_j}(\zeta_j) \mu(F(\xi) = \beta, G(\zeta) = \gamma) = P_{\text{leaf}}(\xi, \zeta) P(\beta, \gamma \mid \xi, \zeta) = P(\beta, \gamma, \xi, \zeta).$$

Unsupported leaf values are never sampled, so the arbitrary choices there do not matter. Hence $P \in \mathcal{C}_G$. $\square$

The construction samples new independent sources; it does not assume that conditioning on E preserves any independence among the latent variables of a hypothetical model behind Γ . The only fact used about Γ is (5). A central source with $\prod_s |\mathcal{O}_b|^{\|\text{supp } P_{\ell_s}\|} \cdot \prod_j |\mathcal{O}_c|^{\|\text{supp } P_{r_j}\|}$ values suffices, since that is the number of pairs of response tables.

Corollary 5.2 (Binary observations). *For a double-star with binary observations, $\mathcal{I}_2^{\text{NW}}(G) = \mathcal{C}_G$.*

Proof. All leaf alphabets have two elements, so $T = 2$. $\square$

Corollary 5.3 (Four-observer path). *Consider $A(X)$, $B(X, Y)$, $C(Y, Z)$, $D(Z)$, the path P_4 with A and D at the ends. Then $\mathcal{I}_T^{\text{NW}}(P_4) = \mathcal{C}_{P_4}$ with $T = \max\{2, |\mathcal{O}_A|, |\mathcal{O}_D|\}$. For binary endpoints $T = 2$, whatever the finite alphabets of B and C .*

Proof. P_4 is the double-star with central edge $\{B, C\}$, one left leaf A and one right leaf D ; apply Theorem 5.1, whose bound involves only the leaf alphabets. $\square$

This is the four-on-line scenario of [15, Figure 8]. Navascués and Wolfe list it with the triangle among the scenarios whose compatible set is cut out by inequality constraints, show that order two does not suffice for the triangle, and leave the four-on-line case open. The corollary settles it at order two for binary endpoints.

Corollary 5.4 (Disjoint unions). *Let G be a disjoint union of double-stars $G_1, \dots, G_n$ with orders $T_1, \dots, T_n$ as above, and let $T = \max\{2, T_1, \dots, T_n\}$. Then $\mathcal{I}_T^{\text{NW}}(G) = \mathcal{C}_G$.*

Proof. Let $P \in \mathcal{I}_T^{\text{NW}}(G)$ with witness Γ . Distinct components are source-disjoint, so Lemma 3.10, iterated, gives $P = \bigotimes_a P_{V(G_a)}$. The restriction of Γ to the copied observations at the vertices of G_a is a table for the order- T inflation of G_a : it is invariant under permutations of the copy indices of the edges of G_a , and its diagonal law is the corresponding marginal of $P^{\otimes T}$, namely $(P_{V(G_a)})^{\otimes T}$. Hence $P_{V(G_a)} \in \mathcal{I}_T^{\text{NW}}(G_a) \subseteq \mathcal{I}_{T_a}^{\text{NW}}(G_a)$, the inclusion because discarding the copy indices outside $[T_a]$ leaves a witness at order T_a . Theorem 5.1 now gives $P_{V(G_a)} \in \mathcal{C}_{G_a}$. Realize each component by its own model on its own independent source family; the joint observed law is the product, which is P . $\square$

A star is the case $q = 0$, with c one of its leaves. There the bound of Theorem 5.1 is not sharp: order two suffices with arbitrary finite alphabets [15, Section 4.1], by taking independent sources equal to the leaf values and letting the centre respond with its conditional law given them. Theorem 5.1 adds the central edge, where the two centres share a source and the leaf values on the two sides must be assembled into one pair of response tables.

6 Cycles

Section 3 writes the three finite tests for an arbitrary pair-source scenario. For every cycle and every order we construct a strictly positive law that the strongest of them accepts and that no model produces, then measure how far the accepted laws can sit from compatibility.

Throughout, the m -cycle C_m , $m \geq 3$, has observed variables $O_0, \dots, O_{m-1}$ and sources $S_0, \dots, S_{m-1}$, indices read modulo m , with O_v a child of S_{v-1} and S_v ; the copied observations of Definition 3.2 are written O_v^{ij} , children of $S_{v-1,i}$ and $S_{v,j}$. The triangle is C_3 , with $\mathcal{C}_\Delta = \mathcal{C}_{C_3}$, read as $A = O_0$, $B = O_1$, $C = O_2$ and $X = S_0$, $Y = S_1$, $Z = S_2$; the square is C_4 , written $\square$, read as $A(X, W)$, $B(X, Y)$, $C(Y, Z)$, $D(Z, W)$ with $X = S_0$, $Y = S_1$, $Z = S_2$, $W = S_3$.

6.1 Parity witnesses on every cycle

Lemma 6.1 (Auxiliary parity density). *For $N \geq 1$ and $q \geq 0$ let*

$$H_{N,q}(s) = 2^{-N} \sum_{\substack{S \subseteq [N] \\ |S| \text{ even}}} (-q)^{|S|/2} \prod_{w \in S} s_w, \quad s \in \{-1, 1\}^N. \quad (7)$$

Then $H_{N,q}$ is normalized, its characters are

$$\mathbb{E}_{H_{N,q}} \left[\prod_{w \in S} s_w \right] = \begin{cases} 0, & |S| \text{ odd}, \\ (-q)^{|S|/2}, & |S| \text{ even}, \end{cases} \quad (8)$$

and if $N^2 q \leq 1/4$ then $H_{N,q}(s) \geq \frac{2}{3} 2^{-N} > 0$ for every s .

Proof. Normalization and (8) are orthogonality of the characters of $\{-1, 1\}^N$ under the uniform measure. For positivity, $\binom{N}{2j} \leq N^{2j}$, so

$$|2^N H_{N,q}(s) - 1| \leq \sum_{j \geq 1} \binom{N}{2j} q^j \leq \sum_{j \geq 1} (N^2 q)^j \leq \sum_{j \geq 1} 4^{-j} = \frac{1}{3}. \quad \square$$

For $F \subseteq \{0, \dots, m-1\}$ let $\partial F = \{w : |F \cap \{w, w+1\}| = 1\}$ be the set of edges of C_m incident to an odd number of vertices of F . Every vertex meets two edges, so $|\partial F|$ is even.

Definition 6.2 (Cycle parity family). For $m \geq 3$ and $q \geq 0$ let $P_{m,q}$ be the signed measure on $\{-1, 1\}^m$ with density $2^{-m} \sum_F (-q)^{|\partial F|/2} \prod_{v \in F} o_v$ against the counting measure, so that it has total mass 1 and

$$\mathbb{E}_{P_{m,q}} \left[\prod_{v \in F} O_v \right] = (-q)^{|\partial F|/2} \quad \text{for every } F. \quad (9)$$

If $m^2 q \leq 1/4$ then $P_{m,q}$ is a probability law, being the law of $(\sigma_{v-1} \sigma_v)_{v=0}^{m-1}$ for $\sigma \sim H_{m,q}$. The exact ranges in the two cases we use are wider.

Taking F the whole vertex set gives $\mathbb{E}[\prod_v O_v] = 1$, so wherever $P_{m,q}$ is a probability law the product of all m signs is 1 almost surely; taking F a contiguous arc gives mean $-q$. For $m = 3$ the family is the parity-even law with observed means x, y, z ,

$$\Pi(x, y, z)(\alpha, \beta, \gamma) = \frac{1}{4}(1 + x\alpha + y\beta + z\gamma) \quad \text{on } \alpha\beta\gamma = 1, \quad P_{3,q} = \Pi(-q, -q, -q), \quad (10)$$

which is a probability law exactly for $0 \leq q \leq 1/3$; and for $m = 4$ it is the square parity family $P_q := P_{4,q}$ with atoms

$$\frac{1}{8}(1-6q+q^2) \text{ at } 0000, \quad \frac{1}{8}(1-q^2) \text{ at } 0011, 0110, 1001, 1100, \quad \frac{1}{8}(1+q)^2 \text{ at } 0101, 1010, 1111, \quad (11)$$

zero on odd parity, in the convention false = 0 $\leftrightarrow$ +1. It is a law exactly for $0 \leq q \leq 3 - 2\sqrt{2}$.

Fix $t \geq 1$, put $N = mt$ and draw auxiliary signs $s = (s_{v,i})_{v \in \{0, \dots, m-1\}, i \in [t]}$ from $H_{N,q}$. Define every copied observation by

$$O_v^{ij} = s_{v-1,i} s_{v,j}. \quad (12)$$

For a set F of copied observations let ∂F denote the set of copied sources occurring an odd number of times in $\prod_{O \in F} O$; each copied observation contributes two incidences, so $|\partial F|$ is even, and (8) gives

$$\mathbb{E} \left[\prod_{O \in F} O \right] = (-q)^{|\partial F|/2}. \quad (13)$$

Lemma 6.3 (Boundaries of prescribed characters). *Let $F_1, \dots, F_r$ be pairwise ancestrally independent injectable sets of copied observations of C_m and let $U_i \subseteq F_i$. Then each ∂U_i has even cardinality and contains at most one copy of each source, so it is empty or meets at least two source families; the sets $\partial U_1, \dots, \partial U_r$ are pairwise disjoint; and $\partial(U_1 \cup \dots \cup U_r) = \bigsqcup_i \partial U_i$, which again is empty or meets at least two source families. Moreover ∂U_i is the image, under erasure of copy indices, of the boundary of the corresponding original observed set.*

Proof. An injectable set contains at most one copied observation over each original vertex and its ancestry contains at most one copy of each original source, so ∂U_i contains at most one coordinate from each family; being of even cardinality it is empty or has at least two elements in distinct families. Erasing copy indices carries the incidence count of U_i to that of its image, which gives the last claim. Ancestrally independent sets have disjoint ancestries, hence disjoint coordinate sets and disjoint boundaries, and the boundary of a union of sets with disjoint ancestries is the disjoint union of the boundaries. If that union is nonempty then some ∂U_i is nonempty and already meets two families. $\square$

Lemma 6.4 (The parity witness passes every prescription). *Let $m \geq 3$, $t \geq 1$ and $0 < q \leq 1/(4m^2t^2)$. The law of (12) is a strictly positive witness placing $P_{m,q}$ in $\mathcal{I}_t^{\text{AI}}(C_m) = \mathcal{I}_t^{\text{exp}}(C_m)$.*

Proof. Positivity of $H_{N,q}$ holds since $N^2q = m^2t^2q \leq 1/4$. The density (7) is symmetric in the N coordinates, so the witness is invariant under independent permutations of the copy indices of each source. By Lemma 6.3 an injectable set F has ∂F equal to the boundary of its image, so (13) and (9) give it the prescribed marginal of $P_{m,q}$. For pairwise ancestrally independent injectable $F_1, \dots, F_r$ and any $U_i \subseteq F_i$, Lemma 6.3 and (13) give

$$\mathbb{E} \left[\prod_i \prod_{O \in U_i} O \right] = (-q)^{\frac{1}{2} \sum_i |\partial U_i|} = \prod_i \mathbb{E} \left[\prod_{O \in U_i} O \right],$$

and characters determine a law on a finite sign cube, so the blocks are independent with the prescribed marginals. The diagonal rows are pairwise ancestrally independent injectable copied cycles, so the diagonal law is $P_{m,q}^{\otimes t}$. Lemma 3.9 supplies the recursive prescriptions. $\square$

The witness is a law on the copied observations, not the output of an inflated model: under $H_{N,q}$ the auxiliary signs are correlated across families.

Lemma 6.5 (Exact parity rigidity). *Let $R \in \mathcal{C}_\Delta$ with $ABC = 1$ almost surely. Then there are $\sigma, \tau, v \in [-1, 1]$ with $(\mathbb{E}A, \mathbb{E}B, \mathbb{E}C) = (\sigma\tau, \sigma v, \tau v)$. In particular $\mathbb{E}A \cdot \mathbb{E}B \cdot \mathbb{E}C \geq 0$.*

Proof. Absorb the response seeds into incident sources, so that $A = A(X, Z)$, $B = B(X, Y)$, $C = C(Z, Y)$ are deterministic signs. For μ_Y -almost every y the identity $A(X, Z)B(X, y)C(Z, y) = 1$ holds for $\mu_X \otimes \mu_Z$ -almost every (X, Z) ; fix such a y_0 and put $S(X) = B(X, y_0)$, $T(Z) = C(Z, y_0)$, so that $A = ST$ almost surely. Substituting in $ABC = 1$ gives $(S(X)B(X, Y))(T(Z)C(Z, Y)) = 1$ almost surely. Write $B' = SB$ and $C' = TC$. Conditionally on Y these are independent signs, and they agree almost surely, so each equals a function $U(Y)$ almost surely. Hence $A = ST$, $B = SU$, $C = TU$ almost surely, and independence of the three sources gives the means with $\sigma = \mathbb{E}S$, $\tau = \mathbb{E}T$, $v = \mathbb{E}U$. Their product is $(\sigma\tau v)^2$. $\square$

Rigidity of the parity-perfect (parity token-counting) triangle correlations is prior work: the exact statement is due to Boreiri, Girardin, Ulu, Lipka-Bartosik, Brunner and Sekatski [2], and Boreiri, Ulu, Brunner and Sekatski [3] give a noise-robust form (a compatible law with parity

error ϵ is within 3ϵ of a parity-perfect compatible law); Lemma 6.6 is a moment form of that robustness. Renou and Beigi [17, 18] prove rigidity of token-counting distributions on every ring scenario with bipartite sources; their statements concern integer token counts rather than parity, so they do not contain the lemma below, but they are the prior rigidity results on cycles. We give the two proofs because we need the statements for arbitrary latent alphabets and stochastic responses, and because the constant of Lemma 6.6 enters the distance bounds below.

Lemma 6.6 (Quantitative parity rigidity). *Let $R \in \mathcal{C}_\Delta$ and $W = \mathbb{E}_R[ABC]$. Then*

$$\max\{\mathbb{E}_R A, \mathbb{E}_R B, \mathbb{E}_R C\} \geq -4(1 - W).$$

Proof. Let $a(x, z)$, $b(x, y)$, $c(z, y)$ be the conditional means of A , B , C given the sources; they take values in $[-1, 1]$, and since the three responses use independent private randomness, $W = \mathbb{E}[abc]$ and $\mathbb{E}_R A = \mathbb{E}[a]$, with the expectations over the independent sources X, Y, Z . We use two inequalities: for $p, r \in [-1, 1]$,

$$|p - r| \leq 1 - pr, \quad (14)$$

since for $p \geq r$ the difference $(1 - pr) - (p - r) = (1 - p)(1 + r)$ is nonnegative, and for $u, w \in [-1, 1]$,

$$|u + w| \geq 2uw, \quad (15)$$

since for $uw \leq 0$ the right side is nonpositive and for u, w of the same sign $|u| + |w| - 2|u||w| = |u|(1 - |w|) + |w|(1 - |u|) \geq 0$.

Choose y_0 with $\mathbb{E}_{X,Z}[1 - a(X, Z)b(X, y_0)c(Z, y_0)] \leq 1 - W$; it exists because the average over Y of the left side is $1 - W$. Put $S(X) = b(X, y_0)$ and $T(Z) = c(Z, y_0)$. By (14) with $p = a$, $r = ST$,

$$\mathbb{E}[a - ST] \leq \mathbb{E}[1 - aST] \leq 1 - W, \quad \text{so} \quad |\mathbb{E}[a] - \mathbb{E}[S] \mathbb{E}[T]| \leq 1 - W,$$

using the independence of X and Z . Next let $u(y) = \mathbb{E}_X[S(X)b(X, y)]$ and $w(y) = \mathbb{E}_Z[T(Z)c(Z, y)]$, both in $[-1, 1]$, and $v(y) = \text{sgn}(u(y) + w(y))$. By (15),

$$\begin{aligned} \mathbb{E}_Y[1 - vu] + \mathbb{E}_Y[1 - vw] &= \mathbb{E}_Y[2 - |u + w|] \leq 2 - 2\mathbb{E}_Y[uw] \\ &= 2 - 2\mathbb{E}[STbc] \leq 2 - 2(W - \mathbb{E}[a - ST]) \leq 4(1 - W), \end{aligned}$$

where $\mathbb{E}_Y[uw] = \mathbb{E}[S(X)T(Z)b(X, Y)c(Z, Y)]$ by independence of the sources and $\mathbb{E}[STbc] \geq \mathbb{E}[abc] - \mathbb{E}[ST - a]$ because $|bc| \leq 1$. Hence $\mathbb{E}_Y[1 - vu] \leq 4(1 - W)$ and $\mathbb{E}_Y[1 - vw] \leq 4(1 - W)$. By (14) with $p = b(X, Y)$ and $r = S(X)v(Y)$,

$$\mathbb{E}[b - Sv] \leq \mathbb{E}[1 - bSv] = \mathbb{E}_Y[1 - vu] \leq 4(1 - W), \quad \text{so} \quad |\mathbb{E}[b] - \mathbb{E}[S] \mathbb{E}[v]| \leq 4(1 - W),$$

and likewise $|\mathbb{E}[c] - \mathbb{E}[T] \mathbb{E}[v]| \leq 4(1 - W)$. Write $\sigma = \mathbb{E}[S]$, $\tau = \mathbb{E}[T]$, $\nu = \mathbb{E}[v]$. If all three means were below $-4(1 - W)$, then $\sigma\tau \leq \mathbb{E}[a] + (1 - W) < 0$, $\sigma\nu < 0$ and $\tau\nu < 0$, whose product $(\sigma\tau\nu)^2$ is nonnegative, a contradiction. $\square$

For $W = 1$ the lemma says that some mean is nonnegative, which is Lemma 6.5 again. A compatible law at total-variation distance δ from a parity-perfect law has $1 - W \leq 2\delta$, so the lemma also replaces the repair step of [3]: we construct no repaired model, and the bound holds for arbitrary latent alphabets and stochastic responses.

Lemma 6.7 (Distance of the cycle family). *Let $m \geq 3$ and let $q > 0$ be such that $P_{m,q}$ is a probability law. Then $d_{\text{TV}}(P_{m,q}, \mathcal{C}_{C_m}) \geq q/10$; in particular $P_{m,q} \notin \mathcal{C}_{C_m}$.*

Proof. Partition $\{0, \dots, m-1\}$ into three nonempty contiguous arcs $\Lambda_1, \Lambda_2, \Lambda_3$ and let $\varphi(o) = (V_1, V_2, V_3)$ with $V_r = \prod_{v \in \Lambda_r} o_v$. If $R \in \mathcal{C}_{C_m}$, then $\varphi_* R \in \mathcal{C}_\Delta$: the three sources on the arc boundaries are independent, each arc product is a function of the two boundary sources adjacent to its arc together with the sources internal to that arc, and those internal sources can be absorbed as private randomness of the arc. Each $\partial \Lambda_r$ has two elements, and $\partial(\Lambda_1 \cup \Lambda_2) = \partial \Lambda_3$, so (9) gives

$$\varphi_* P_{m,q} = \Pi(-q, -q, -q).$$

Let $R \in \mathcal{C}_{C_m}$ and $\delta = d_{\text{TV}}(P_{m,q}, R)$. Total variation does not increase under φ , so $Q = \varphi_* R \in \mathcal{C}_\Delta$ satisfies $d_{\text{TV}}(\Pi(-q, -q, -q), Q) \leq \delta$. Since $|\mathbb{E}_P f - \mathbb{E}_Q f| \leq 2d_{\text{TV}}(P, Q)$ for $|f| \leq 1$, the parity $W = \mathbb{E}_Q[V_1 V_2 V_3]$ satisfies $1 - W \leq 2\delta$ and each mean $\mathbb{E}_Q V_r$ is at most $-q + 2\delta$. Lemma 6.6 gives $-q + 2\delta \geq -4(1 - W) \geq -8\delta$, that is $\delta \geq q/10$. $\square$

Theorem 6.8 (Cycles at every order). *Let $m \geq 3$ and $t \geq 1$, put*

$$q = \frac{1}{4m^2 t^2}, \quad \eta = \frac{q}{32m},$$

and let $\tilde{P}_{m,q} = T_\eta^{\otimes m} P_{m,q}$, where T_η flips a sign with probability η . Then $\tilde{P}_{m,q}$ is a strictly positive rational law with

$$\tilde{P}_{m,q} \in \mathcal{I}_t^{\text{exp}}(C_m) = \mathcal{I}_t^{\text{AI}}(C_m), \quad d_{\text{TV}}(\tilde{P}_{m,q}, \mathcal{C}_{C_m}) \geq \frac{11q}{160} = \frac{11}{640 m^2 t^2} > 0.$$

Consequently no finite order of any of the three hierarchies characterizes $\mathcal{C}_{C_m}$.

Proof. Apply T_η independently to every copied observation of the witness of Lemma 6.4. The resulting law is strictly positive, still invariant under the copy-index permutations, and its characters are those of (13) multiplied by $(1 - 2\eta)^{|F|}$. The same factor multiplies the target characters, and $|F|$ adds over disjoint blocks, so every injectable marginal and every ancestrally independent product of Lemma 6.4 persists with $\tilde{P}_{m,q}$ in place of $P_{m,q}$, as does the diagonal law. Lemma 3.9 again supplies the recursive prescriptions.

For the distance, $d_{\text{TV}}(\tilde{P}_{m,q}, P_{m,q}) \leq m\eta = q/32$ by a union bound over the flipped coordinates, and Lemma 6.7 gives $d_{\text{TV}}(\tilde{P}_{m,q}, \mathcal{C}_{C_m}) \geq q/10 - q/32 = 11q/160$. Since t was arbitrary and $\tilde{P}_{m,q} \notin \mathcal{C}_{C_m}$, the feasible set at order t is strictly larger than $\mathcal{C}_{C_m}$ for every t . $\square$

For $m = 3$ this gives a worst-case distance of order t^{-2} for the triangle, in place of the order t^{-3} that the family of Section 8 supplies for Question 15.1.

6.2 A corrected density and a linear lower bound

The density (7) fixes every auxiliary character, including characters that no prescription refers to. Changing those characters restores positivity at a larger q .

Lemma 6.9 (Phase telescoping). *For real $\theta_1, \dots, \theta_m$,*

$$1 - \cos\left(\sum_{g=1}^m \theta_g\right) \leq m \sum_{g=1}^m (1 - \cos \theta_g).$$

Proof. $1 - \cos \phi = \frac{1}{2}|1 - e^{i\phi}|^2$, and $1 - \prod_g e^{i\theta_g} = \sum_g (\prod_{g' < g} e^{i\theta_{g'}})(1 - e^{i\theta_g})$, so $|1 - \prod_g e^{i\theta_g}| \leq \sum_g |1 - e^{i\theta_g}|$. Cauchy-Schwarz gives $(\sum_g a_g)^2 \leq m \sum_g a_g^2$. $\square$

Lemma 6.10 (Corrected density). *Fix $m \in \{3, 4\}$, $t \geq 1$ and $0 < q \leq 1/(16t)$, and set $c = 4$ for $m = 3$ and $c = 5$ for $m = 4$. On m families of t signs $s_{g,i}$ put*

$$f_g = \prod_{i=1}^t (1 + i\sqrt{q} s_{g,i}), \quad R = (1 + q)^{t/2}, \quad W(s) = \operatorname{Re} \prod_{g=1}^m f_g + c \sum_{g=1}^m (1 - \operatorname{Re} f_g). \quad (16)$$

Then W is a real polynomial in q with rational coefficients, $W \geq 3/5$ for $m = 3$ and $W \geq 1/3$ for $m = 4$, the law $\rho = 2^{-mt}W$ is a probability law, and its characters are

$$\mathbb{E}_\rho \left[\prod_{w \in S} s_w \right] = \begin{cases} 0, & |S| \text{ odd}, \\ (1 - c)(-q)^{|S|/2}, & S \neq \emptyset \text{ even, inside one family}, \\ (-q)^{|S|/2}, & |S| \text{ even, meeting at least two families.} \end{cases} \quad (17)$$

Proof. Taking real parts keeps only even powers of $i\sqrt{q}$, so W is a real polynomial in q . Since $|1 + i\sqrt{q}s| = (1 + q)^{1/2}$ we may write $f_g = R e^{i\theta_g}$, and then

$$\begin{aligned} W &= R^m \cos \left(\sum_g \theta_g \right) + c \sum_g (1 - R \cos \theta_g) \\ &= R^m - cm(R - 1) - R^m \left(1 - \cos \sum_g \theta_g \right) + cR \sum_g (1 - \cos \theta_g). \end{aligned}$$

Lemma 6.9 gives

$$W \geq R^m - cm(R - 1) + (cR - mR^m) \sum_g (1 - \cos \theta_g).$$

From $tq \leq 1/16$ we get $R^2 = (1 + q)^t \leq \sum_{j \geq 0} (tq)^j \leq 16/15$ and $R - 1 = (R^2 - 1)/(R + 1) \leq 1/30$. For $m = 3$, $c = 4$: $3R^3 \leq 4R$ because $3R^2 \leq 16/5 \leq 4$, and $W \geq 1 - 12/30 = 3/5$. For $m = 4$, $c = 5$: $4R^4 \leq 5R$ because $4R^3 \leq 4(16/15)^{3/2} < 5$, and $W \geq 1 - 20/30 = 1/3$.

For the characters, work under the uniform law on the sign cube, where the families are independent and $\mathbb{E}f_g = 1$. For S with parts S_g in the families,

$$\mathbb{E} \left[\prod_g f_g \prod_{w \in S} s_w \right] = \prod_g \mathbb{E} \left[f_g \prod_{w \in S_g} s_w \right] = (i\sqrt{q})^{|S|},$$

whose real part is 0 for odd $|S|$ and $(-q)^{|S|/2}$ for even $|S|$. Also $\mathbb{E}[(1 - \operatorname{Re} f_g) \prod_{w \in S} s_w]$ vanishes unless S is a nonempty subset of family g , in which case it equals $-\operatorname{Re}(i\sqrt{q})^{|S|}$. Adding the contributions gives (17); the case $S = \emptyset$ gives $\mathbb{E}_\rho 1 = 1$. $\square$

Theorem 6.11 (The square at $q = \Theta(1/t)$). *For every $t \geq 1$ and every $0 < q \leq 1/(16t)$, the square parity law P_q of (11) lies in $\mathcal{I}_t^{\text{AI}}(\square) = \mathcal{I}_t^{\text{exp}}(\square)$.*

Proof. Take $m = 4$ and $c = 5$ in Lemma 6.10, name the four families x, y, z, w after the sources X, Y, Z, W , and define every copied observation by

$$A^{il} = x_i w_l, \quad B^{ij} = x_i y_j, \quad C^{jk} = y_j z_k, \quad D^{kl} = z_k w_l,$$

that is by (12) for $m = 4$. The density (16) is symmetric within each family, so the law is invariant under independent permutations of the copy indices of each source. By Lemma 6.3 every boundary occurring in an AI prescription, and every boundary of a subset of one of its blocks, is empty or meets at least two families; by (17) such characters take the value $(-q)^{|\partial \cdot|/2}$, unchanged from (13). The computation of Lemma 6.4 therefore applies verbatim: every injectable set has the marginal of $P_{4,q} = P_q$, ancestrally independent injectable blocks are independent, and the diagonal law is $P_q^{\otimes t}$. The moments that Lemma 6.10 changed are the nonempty even characters inside a single family, and no prescription refers to one. Lemma 3.9 gives the recursive prescriptions. $\square$

Theorem 6.12 (The triangle at $q = \Theta(1/t)$). *For every $t \geq 1$ and every $0 < q \leq 1/(16t)$, the parity-even law $\Pi(-q, -q, -q)$ of (10) lies in $\mathcal{I}_t^{\text{AI}}(\Delta) = \mathcal{I}_t^{\text{exp}}(\Delta)$.*

Proof. Take $m = 3$ and $c = 4$ in Lemma 6.10, name the three families x, y, z after the sources X, Y, Z , and define

$$A^{ij} = x_i z_j, \quad B^{ik} = x_i y_k, \quad C^{jk} = z_j y_k,$$

which is (12) for $m = 3$ in the index convention of Section 2. Positivity and normalization are Lemma 6.10 with $W \geq 3/5$. Symmetry within each family is again immediate. By Lemma 6.3 every character prescribed by an injectable marginal, by an ancestrally independent product, or by the diagonal law has boundary empty or meeting at least two families, so (17) returns $(-q)^{|\partial \cdot|/2}$ and the argument of Lemma 6.4 applies unchanged. The original copied triangle has $\mathbb{E}A = \mathbb{E}[x_i z_j] = -q$ and likewise $\mathbb{E}B = \mathbb{E}C = -q$, while $ABC = x_i z_j x_i y_k z_j y_k = 1$, so its law is $\Pi(-q, -q, -q) = P_{3,q}$. Lemma 3.9 gives the recursive prescriptions. $\square$

The exact certificates described in Section 16 check the square and triangle witnesses of the two theorems, with and without the flips of Corollaries 6.14 and 6.15, at orders one and two for the square and one to three for the triangle: positivity of every atom, symmetry, the full diagonal law and every ancestral-independence prescription.

6.3 Distance and the worst-case profile

For $H \in \{\text{NW}, \text{AI}, \text{exp}\}$ and a pair-source scenario G set

$$H_t^H(G) = \sup_{P \in \mathcal{I}_t^H(G)} d_{\text{TV}}(P, \mathcal{C}_G), \quad (18)$$

the worst-case total-variation distance from compatibility of a law the order- t test accepts, with d_{TV} half the ℓ^1 distance as in Section 2.

Lemma 6.13 (Max-moment inequality for the square). *Every $P \in \mathcal{C}_{\square}$ in sign notation satisfies*

$$\max\{\mathbb{E}A, \mathbb{E}B, \mathbb{E}[AB]\} + 4P(ABCD = -1) \geq 0.$$

Proof. Absorb the response seeds into incident sources, so $A = A(X, W)$, $B = B(X, Y)$, $C = C(Y, Z)$, $D = D(Z, W)$ are deterministic signs and X, Y, Z, W are independent. Fix z and write $g_z(Y) = C(Y, z)$, $h_z(W) = D(z, W)$ and

$$\delta_z = \Pr(A(X, W)B(X, Y) \neq g_z(Y)h_z(W)),$$

so that $\mathbb{E}_Z \delta_Z = P(ABCD = -1) =: \delta$. Put

$$u_z(x) = \mathbb{E}_W[A(x, W)h_z(W)], \quad v_z(x) = \mathbb{E}_Y[B(x, Y)g_z(Y)].$$

Since Y and W are independent, $\mathbb{E}[ABg_zh_z \mid X = x] = u_z(x)v_z(x)$, and as ABg_zh_z is a sign, $\mathbb{E}_X[1 - u_zv_z] = 2\delta_z$.

Let $f_z(x) = \text{sgn}(u_z(x) + v_z(x))$, with $\text{sgn } 0 = 1$, and compare A and B with $A'_z = f_z(X)h_z(W)$ and $B'_z = f_z(X)g_z(Y)$. Then

$$e_z := \Pr(A \neq A'_z) + \Pr(B \neq B'_z) = \mathbb{E}_X \left[1 - \frac{1}{2}|u_z + v_z| \right] \leq \mathbb{E}_X[1 - u_zv_z] = 2\delta_z,$$

using $|u + v| \geq 2uv$ on $[-1, 1]^2$. Independence of X, Y, W gives

$$(\mathbb{E}A'_z)(\mathbb{E}B'_z)(\mathbb{E}[A'_zB'_z]) = (\mathbb{E}f_z)^2(\mathbb{E}g_z)^2(\mathbb{E}h_z)^2 \geq 0,$$

so at least one of the three comparison moments is nonnegative. Each differs from the corresponding moment of P by at most $2e_z$, since $|\mathbb{E}U - \mathbb{E}V| \leq 2\Pr(U \neq V)$ for signs and $\Pr(AB \neq A'_zB'_z) \leq \Pr(A \neq A'_z) + \Pr(B \neq B'_z)$. Hence

$$\max\{\mathbb{E}A, \mathbb{E}B, \mathbb{E}[AB]\} \geq -2e_z \geq -4\delta_z.$$

The left side does not depend on z ; averaging over z finishes the proof. $\square$

Corollary 6.14 (Full-support survivors on the square). $d_{\text{TV}}(P_q, \mathcal{C}_{\square}) \geq q/6$ for $0 < q \leq 3 - 2\sqrt{2}$. With flip probability $\eta = q/64$ applied to each of the four bits and to every copied observation, $\tilde{P}_q = T_{\eta}^{\otimes 4}P_q$ is strictly positive, lies in $\mathcal{I}_t^{\text{AI}}(\square) = \mathcal{I}_t^{\text{exp}}(\square)$ for $q \leq 1/(16t)$, and satisfies $d_{\text{TV}}(\tilde{P}_q, \mathcal{C}_{\square}) \geq 5q/48$. At $q_t = 1/(16t)$,

$$H_t^{\text{exp}}(\square) = H_t^{\text{AI}}(\square) \geq \frac{5}{768t}.$$

Proof. If $R \in \mathcal{C}_{\square}$ and $d = d_{\text{TV}}(P_q, R)$, each of $\mathbb{E}_R A, \mathbb{E}_R B, \mathbb{E}_R[AB]$ is at most $-q + 2d$ and the parity error of R is at most d , since P_q has none. Lemma 6.13 gives $0 \leq -q + 2d + 4d$, so $d \geq q/6$. The noise argument of Theorem 6.8 applies to the witness of Theorem 6.11, and $d_{\text{TV}}(\tilde{P}_q, P_q) \leq 4\eta = q/16$, so $d_{\text{TV}}(\tilde{P}_q, \mathcal{C}_{\square}) \geq q/6 - q/16 = 5q/48$. Substituting $q_t = 1/(16t)$ gives $5/(768t)$, and Lemma 3.9 identifies the two suprema. $\square$

Corollary 6.15 (Full-support survivors on the triangle). $d_{\text{TV}}(\Pi(-q, -q, -q), \mathcal{C}_{\triangle}) \geq q/10$ for $0 < q \leq 1/3$. With flip probability $\eta = q/192$ on each of the three bits and on every copied observation, the noisy target is strictly positive, lies in $\mathcal{I}_t^{\text{AI}}(\triangle) = \mathcal{I}_t^{\text{exp}}(\triangle)$ for $q \leq 1/(16t)$, and is at distance at least $27q/320$ from $\mathcal{C}_{\triangle}$. At $q_t = 1/(16t)$,

$$H_t^{\text{exp}}(\triangle) = H_t^{\text{AI}}(\triangle) \geq \frac{27}{5120t}.$$

Proof. The distance bound is Lemma 6.7 for $m = 3$, whose proof uses only Lemma 6.6 on the triangle itself. The three flips cost at most $3\eta = q/64$ in total variation, and $q/10 - q/64 = 27q/320$. Feasibility is Theorem 6.12 with the noise argument of Theorem 6.8. Substituting $q_t = 1/(16t)$ gives $27/(5120t)$. $\square$

6.4 A convex-order upper bound

Theorem 6.16 (Convex order along the diagonal). *Let G be a correlation scenario, that is, finitely many independent latent sources and observed variables with finite alphabets, each a stochastic function of the sources it reads, with K joint outcomes; the order- n inflation and the set $\mathcal{I}_n^{\text{NW}}(G)$ are defined as in Section 3 with sources in place of edges. Let $P \in \mathcal{I}_n^{\text{NW}}(G)$ with witness Γ . For a deterministic table ω of copied observations let q_ω be the observed law obtained by sampling one copy index uniformly and independently for each source and reading ω at those indices. Then $q_\omega \in \mathcal{C}_G$ for every ω , and for every convex Φ on the observed simplex*

$$\mathbb{E}_\Gamma \Phi(q_\omega) \leq \mathbb{E} \Phi(\hat{P}_n), \quad (19)$$

where $\hat{P}_n$ is the empirical law of n independent draws from P . In particular

$$\mathbb{E}_\Gamma \|q_\omega - P\|_2^2 \leq \frac{1 - \|P\|_2^2}{n}, \quad H_n^{\text{NW}}(G) \leq \min\left\{1, \frac{\sqrt{K-1}}{2\sqrt{n}}\right\}. \quad (20)$$

Proof. Fix ω . Sampling an index $I_e \in [n]$ uniformly and independently for each source e and letting each observed variable v output $\omega_v((I_e)_{e \ni v})$, where $e \ni v$ ranges over the sources v reads, is a model of G with independent sources I_e and deterministic responses, so $q_\omega \in \mathcal{C}_G$.

Draw one uniform permutation π_e of $[n]$ for each source, independently of each other and of $\omega \sim \Gamma$, and set

$$T_\ell = \left(\omega_v((\pi_e(\ell))_{e \ni v}) \right)_v, \quad \ell \in [n], \quad \hat{P} = \frac{1}{n} \sum_{\ell=1}^n \delta_{T_\ell}.$$

For each fixed choice of permutations, the symmetry of Γ carries $(T_1, \dots, T_n)$ to the diagonal rows, so $(T_1, \dots, T_n) \sim P^{\otimes n}$; this remains true after averaging over the permutations, so $\hat{P}$ has the law of $\hat{P}_n$. Conditionally on ω and on a fixed ℓ , the indices $(\pi_e(\ell))_e$ are independent and uniform on $[n]$, so $\mathbb{E}[\delta_{T_\ell} \mid \omega] = q_\omega$ and hence $\mathbb{E}[\hat{P} \mid \omega] = q_\omega$. Conditional Jensen gives $\Phi(q_\omega) \leq \mathbb{E}[\Phi(\hat{P}) \mid \omega]$, and averaging over ω gives (19).

Take $\Phi(\mu) = \|\mu - P\|_2^2$. Then $\mathbb{E}\|\hat{P}_n - P\|_2^2 = \frac{1}{n} \sum_x P(x)(1 - P(x)) = (1 - \|P\|_2^2)/n$, which is the first part of (20). Some ω attains at most the mean, and its q_ω is compatible, so

$$d_{\text{TV}}(P, \mathcal{C}_G) \leq \frac{1}{2} \|P - q_\omega\|_1 \leq \frac{\sqrt{K}}{2} \|P - q_\omega\|_2 \leq \frac{\sqrt{K}}{2} \sqrt{\frac{1 - \|P\|_2^2}{n}} \leq \frac{\sqrt{K-1}}{2\sqrt{n}},$$

using $\|P\|_2^2 \geq 1/K$. Total variation never exceeds 1. $\square$

The sampling mechanism is that of the finite de Finetti estimate of [15, Section 4.2 and eq. (A7)], which bounds the same quantity by $L(1 - \|P\|_2^2)/n$ for L sources through the collision probability of two index samples. Using the full diagonal law through a conditional expectation, rather than its first two degrees, removes the factor L . Section 12 draws the rejecting-order consequence.

Corollary 6.17 (Brackets for the square and the triangle). *For every $t \geq 1$,*

$$\begin{aligned} \frac{5}{768t} &\leq H_t^{\text{exp}}(\square) = H_t^{\text{AI}}(\square) \leq H_t^{\text{NW}}(\square) \leq \min\left\{1, \frac{\sqrt{15}}{2\sqrt{t}}\right\}, \\ \frac{27}{5120t} &\leq H_t^{\text{exp}}(\triangle) = H_t^{\text{AI}}(\triangle) \leq H_t^{\text{NW}}(\triangle) \leq \frac{\sqrt{7}}{2\sqrt{t}}. \end{aligned}$$

Proof. Combine Corollaries 6.14 and 6.15 with Theorem 6.16 at $K = 16$ and $K = 8$, and with Proposition 3.6. $\square$

The two exponents do not meet. Whether $H_t^{\text{NW}}(\square) = O(1/t)$ is open, and so is the corresponding question for the triangle; this is the form Question 15.1 takes once the lower bounds above replace the family bound.

6.5 Order conversion on the square

Theorem 6.18 (NW and AI differ by a bounded change of order). *For every $t \geq 2$,*

$$\mathcal{I}_{\lfloor 3t/2 \rfloor}^{\text{NW}}(\square) \subseteq \mathcal{I}_t^{\text{AI}}(\square) = \mathcal{I}_t^{\text{exp}}(\square) \subseteq \mathcal{I}_t^{\text{NW}}(\square).$$

Proof. The right inclusion is Proposition 3.6 and the equality is Lemma 3.9. For the left inclusion let Γ be an order- n Navascués–Wolfe witness for P with $n = \lfloor 3t/2 \rfloor$ and let Γ' be its restriction to the copy indices $1, \dots, t$ of each source. Restriction preserves symmetry and the diagonal law of degree t , so $\Gamma' \in \mathcal{I}_t^{\text{NW}}$; we check the AI prescriptions.

Fix an AI set of Γ' and decompose it into the connected components of its shared-parent graph, which are injectable and pairwise ancestrally independent. A component's ancestry uses two, three or four source types; the two-type components are single copied observations and correspond to edges of the four-cycle $X - Y - Z - W - X$ of source types. Any two subsets of a four-element set of sizes at least three, or of sizes three and two, intersect, so a component using at least three source types conflicts with every other component and needs a diagonal row of its own. Let b be the number of these.

The remaining components form a multigraph on the source-type four-cycle, with multiplicities n_A, n_B, n_C, n_D on the edges A, B, C, D . Opposite edges are disjoint, so the colours $\max(n_A, n_C)$ for the pair $\{A, C\}$ and $\max(n_B, n_D)$ for the pair $\{B, D\}$ suffice, and their sum equals the maximum source-type degree Δ : if $n_A \geq n_C$ and $n_B \geq n_D$ the sum is $n_A + n_B = \deg X$, and $\deg X$ dominates the other three degrees. So $b + \Delta$ rows suffice.

Choose a source type of residual degree Δ , say X , together with its two neighbours Y and W on the cycle. A component using at least three source types omits at most one, so it uses at least two of X, Y, W ; each of the Δ two-type components incident to X uses exactly two of them. The components have pairwise disjoint copied ancestries, and the three families X, Y, W supply $3t$ copies in Γ' , so $2b + 2\Delta \leq 3t$ and therefore $b + \Delta \leq \lfloor 3t/2 \rfloor = n$.

Assign each component to the diagonal row of its colour. Components sharing a row use disjoint source types, and distinct components use distinct copies within each family, so independent permutations of the copy indices of the four sources place all components on their assigned rows simultaneously. Under Γ the rows are independent with law P . Within one row the components have disjoint original source types, so Lemma 3.10, available because $n \geq 2$, makes P factor across them. Hence the joint law of the placed components is the product of their P -marginals. Undoing the permutations, the AI set of Γ' has exactly the prescribed product. $\square$

The factor $3/2$ is the cost of placing all components on diagonal rows at once; ancestrally disjoint collections of $\lfloor 3t/2 \rfloor$ injectable components whose source-type conflict graph is complete show that this method cannot do better. Hence $H_{\lfloor 3t/2 \rfloor}^{\text{NW}}(\square) \leq H_t^{\text{AI}}(\square) \leq H_t^{\text{NW}}(\square)$, so the two hierarchies have the same polynomial decay exponent if either has one.

6.6 Low-order thresholds

On the parity family we settle the first ten orders exactly. The reduction below turns feasibility for a parity-perfect target into a finite linear program in the numbers of negative auxiliary signs; we state it for the square, where it also covers the Navascués–Wolfe test.

Lemma 6.19 (Count-moment reduction). *Let $0 \leq q \leq 3 - 2\sqrt{2}$ and $t \geq 1$. Then $P_q \in \mathcal{I}_t^{\text{AI}}(\square)$ if and only if there are $w_a \geq 0$, $a = (a_X, a_Y, a_Z, a_W) \in \{0, \dots, t\}^4$, with $\sum_a w_a = 1$ and*

$$\sum_a w_a \prod_g K_{k_g}^{(t)}(a_g) = (-q)^{S/2}, \quad S = \sum_g k_g, \quad (21)$$

for exactly those degree tuples $k \in \{0, \dots, t\}^4$ with S even and $2 \max_g k_g \leq S$, where

$$K_k^{(t)}(a) = \binom{t}{k}^{-1} \sum_j (-1)^j \binom{a}{j} \binom{t-a}{k-j}$$

is the normalized Krawtchouk polynomial.

Proof sketch. Every copied square is injectable, so under any witness it has law P_q and hence parity +1 almost surely. Parity forcing across the copy indices then gives sign potentials: from $A^{il} B^{ij} C^{jk} D^{kl} = 1$ for all indices, the product $B^{ij} C^{jk}$ depends only on (i, k) , so writing M^{ik} for it and $x_i = M^{i1}$, $y_j = C^{j1}$ gives $B^{ij} = x_i y_j$ and, by the same argument on the other pairs, every supported array is of the form $A^{il} = x_i w_l$, $B^{ij} = x_i y_j$, $C^{jk} = y_j z_k$, $D^{kl} = z_k w_l$. The potential is unique up to the simultaneous flip of all four families, so averaging over the two gauges and over independent permutations of the copy indices reduces a witness to a distribution on the four counts a_g of negative signs. Under that symmetrization the average of a character with k_g distinct indices in family g is $\prod_g K_{k_g}^{(t)}(a_g)$, by the definition of the Krawtchouk polynomial as the mean of a k -element sign product drawn without replacement.

The degrees that the AI prescriptions constrain are exactly those in the statement. Necessity: by Lemma 6.3 a nonempty prescribed boundary meets at least two source families, so no single family can carry more than half of the total degree, and the total degree is even. Sufficiency: a balanced degree tuple can be realized by ancestrally disjoint paths in the source four-cycle, matching opposite-type terminals through intermediate copies and then adjacent terminals directly, so the corresponding character is prescribed. The prescribed value is $(-q)^{S/2}$ by (9). $\square$

Lemma 6.20 (The Navascués–Wolfe form). *Under the hypotheses of Lemma 6.19, $P_q \in \mathcal{I}_t^{\text{NW}}(\square)$ if and only if (21) holds for exactly the degree tuples in*

$$D_{\text{NW}}(t) = \{b_1 + \dots + b_t : b_r \in \{0, 1\}^4 \text{ of even weight}\} \subseteq D_{\text{AI}}(t),$$

where $D_{\text{AI}}(t)$ is the degree set of Lemma 6.19. The same reductions hold for the triangle family $\Pi(-q, -q, -q)$ with three families of signs and the potentials $A^{ij} = x_i z_j$, $B^{ik} = x_i y_k$, $C^{jk} = z_j y_k$.

Proof sketch. Every copied square is carried to the first diagonal row by independent copy-index permutations, so under a symmetric witness with diagonal law $P_q^{\otimes t}$ it has law P_q and parity +1; the potential representation and the symmetrization then proceed as in Lemma 6.19. The diagonal law prescribes exactly the characters $\prod_r \prod_{v \in F_r} O_v^{(r)}$ with one vertex set F_r per diagonal row; in potential form row r contributes the indicator of ∂F_r to the degree tuple, and the

boundaries of vertex sets of a connected graph are exactly its even-weight edge sets. Conversely a distribution on counts with the prescribed moments yields a symmetric witness with the diagonal law by the construction in the proof of Lemma 6.19. For the triangle, $A^{ij}B^{ik}C^{jk} = 1$ at $j = 1$ gives $B^{ik} = A^{i1}C^{1k}$, so $x_i = A^{i1}$ and $y_k = C^{1k}$ serve as potentials for B , and $z_j = A^{1j}x_1$ completes the representation, unique up to a global flip. $\square$

Lemma 6.21 (Monotonicity along the parity direction). *Let $H \in \{\text{NW}, \text{AI}, \text{exp}\}$, $G \in \{\triangle, \square\}$, and $0 \leq q' \leq q$ in the range where the parity target is a law. If the parity target with parameter q lies in $I_t^H(G)$, so does the one with parameter q' .*

Proof. By Lemmas 6.19 and 6.20 we may take a witness to be a law on sign potentials. Multiply every potential sign by an independent ± 1 variable of mean $\rho = \sqrt{q'/q}$. A character whose boundary has S elements is multiplied by ρ^S , so its value $(-q)^{S/2}$ becomes $(-q')^{S/2}$, while symmetry and the block structure of every prescription are unchanged. Lemma 3.9 covers the recursively expressible test. $\square$

By Lemma 6.21 the set of parameters accepted at a given order is an interval $[0, q_t^H]$, and we bracket q_t^H by a rational primal witness of Lemmas 6.19 and 6.20 at a point below it and rational dual certificates above it.

Proposition 6.22 (The parity thresholds at orders up to ten). *For the square and the triangle, the thresholds $q_t^{\text{AI}} = q_t^{\text{exp}}$ and q_t^{NW} satisfy the bounds of Table 1 for $1 \leq t \leq 10$, each lower bound being a rational witness and each upper bound a dual certificate valid on the whole interval up to the endpoint of the family. In particular:*

- (i) $q_t^{\text{AI}} < q_t^{\text{NW}}$ at every even $t \leq 10$, so $\mathcal{I}_t^{\text{AI}} \subsetneq \mathcal{I}_t^{\text{NW}}$ there on both scenarios;
- (ii) the two brackets coincide at every odd $t \leq 9$;
- (iii) $q_{t+1}^{\text{NW}} < q_t^{\text{AI}}$ for every $t \leq 9$ except $t = 3$, where q_3^{NW} and q_4^{NW} share the same bracket;
- (iv) on the square, q_2^{AI} is the zero r of $g(q) = q^3 - 33q^2 + 27q - 3$ and $P_q \in \mathcal{I}_2^{\text{NW}}(\square)$ for every $q \leq 0.17156287$; on the triangle, $q_2^{\text{AI}} = 2 - \sqrt{3}$, $q_3^{\text{NW}} = q_4^{\text{NW}} = q_3^{\text{AI}} = 1/5$ and $q_5^{\text{AI}} = q_5^{\text{NW}} = 1/7$.

Consequently $\mathcal{I}_3^{\text{NW}}(\square) \subsetneq \mathcal{I}_2^{\text{AI}}(\square) = \mathcal{I}_2^{\text{exp}}(\square) \subsetneq \mathcal{I}_2^{\text{NW}}(\square)$.

Proof. All statements are exact certificates, described in Section 16: for each order and hierarchy a nonnegative rational count witness at the lower endpoint, checked against every prescribed moment of Lemmas 6.19 and 6.20, and a chain of rational dual functionals, each nonpositive on all count configurations, whose prescribed expectations are polynomials in q certified positive by Sturm sequences on intervals covering everything above the upper endpoint. Lemma 6.21 turns the point witnesses into the intervals. The exact values arise where the dual polynomial factors: at square order two the dual is the functional D with $\mathbb{E}D = (1+q)g(q)$, verified coefficient by coefficient; on the triangle the duals are $-(q+1)(q^2-4q+1)$ at order two and $(5q-1)(q+1)^2$, $(5q-1)(q+1)^3$ at orders three and four, and the order-five dual vanishes at $1/7$. The degree sets $D_{\text{AI}}(t)$ and $D_{\text{NW}}(t)$ are enumerated by brute force over all ancestral-independence families for $t \leq 3$ and, for every $t \leq 10$, every tuple of $D_{\text{AI}}(t)$ is realized by an explicit family that the verifier checks against the definitions. The chain follows from the order-two and order-three brackets on the square, with the equality from Lemma 3.9. $\square$

The table shows two regularities that the theorems do not explain. The products tq_t^H increase towards constants near 0.4 on the square and 0.9 on the triangle, an order of magnitude above

t	q_t^{AI}	square q_t^{NW}	q_t^{AI}	triangle q_t^{NW}
1	$3 - 2\sqrt{2}$	$3 - 2\sqrt{2}$	$1/3$	$1/3$
2	0.1324743290	≥ 0.1715628752	$2 - \sqrt{3}$	$1/3$
3	0.0950228950	0.0950228950	$1/5$	$1/5$
4	0.0835279580	0.0950228950	0.1730047060	$1/5$
5	0.0657380020	0.0657380020	$1/7$	$1/7$
6	0.0603176600	0.0645356170	0.1283741760	0.1381856900
7	0.0502565110	0.0502565110	0.1092774840	0.1092774840
8	0.0470271490	0.0494541280	0.1011416960	0.1070388380
9	0.0406784540	0.0406784540	0.0888129680	0.0888129680
10	0.0384981990	0.0400943600	0.0833179940	0.0870307060

Table 1: Certified brackets for the parity thresholds. A decimal entry x means $x \leq q_t^H \leq x+4 \cdot 10^{-9}$; the exact entries are the roots of the dual polynomials; the entry ≥ 0.1715628752 is a lower bound only, the family ending at $3 - 2\sqrt{2} \approx 0.1715728753$.

the 1/16 of Theorems 6.11 and 6.12; and the two hierarchies coincide along this direction at every odd order up to nine while separating at every even order, with a ratio $q_t^{\text{NW}}/q_t^{\text{AI}} - 1$ decreasing like $0.4/t$. Neither pattern is proved beyond the certified orders; uncertified numerics at triangle order fifteen show a relative gap of 10^{-4} between the two hierarchies, which may be the end of the coincidence or conditioning error.

7 The five-observer path

A path is a tree, and trees carry no parity obstruction of the kind used in Section 6. The scenario, in the language of Section 3, is

$$A(X), \quad B(X, L), \quad C(L, R), \quad D(R, Z), \quad E(Z),$$

with X, L, R, Z mutually independent. We write $\mathcal{C}_{P_5}$ for its compatible set, record the endpoint observations A, E as bits x, z and the middle observations B, C, D as signs. The order- t inflation of Definition 3.2 has copies $X_a, L_i, R_j, Z_e, a, i, j, e \in [t]$, and the $3t^2 + 2t$ copied observations

$$A^a, \quad B^{ai}, \quad C^{ij}, \quad D^{je}, \quad E^e.$$

7.1 The target

Definition 7.1 (Path target). For $0 < h < 1$ let

$$P_h(x, b, c, d, z) = \frac{1}{32} \left[1 + bcd \frac{1+h}{4} (1 + (-1)^{x+z}) \right], \quad x, z \in \{0, 1\}, \quad b, c, d \in \{-1, 1\}. \quad (22)$$

Lemma 7.2 (Properties of P_h). P_h is a probability law, every atom is at least $(1-h)/64$, the pair (A, E) consists of independent fair bits, every proper nonempty moment of (B, C, D) conditional on (A, E) vanishes, and

$$f_{xz} := \mathbb{E}[BCD \mid A = x, E = z] = \frac{1+h}{4} (1 + (-1)^{x+z}) = \begin{cases} (1+h)/2, & x = z, \\ 0, & x \neq z. \end{cases} \quad (23)$$

Proof. The bracket in (22) lies between $1 - (1+h)/2 = (1-h)/2$ and $1 + (1+h)/2$, so every atom lies in $[(1-h)/64, (3+h)/64]$ and in particular is positive. Summing bcd over $\{-1, 1\}^3$ gives zero, so the atoms sum to $32 \cdot \frac{1}{32} = 1$ and the (A, E) marginal is uniform on four cells. Given (x, z) , the conditional density of (b, c, d) is $\frac{1}{8}[1 + bcd f_{xz}]$, whose only nonvanishing nontrivial character is bcd , with value f_{xz} ; that is (23) and the vanishing of the proper moments. $\square$

7.2 A bilocal obstruction

Lemma 7.3 (Bilocal inequality on the path). *Let $R \in \mathcal{C}_{P_5}$ with $R(A = x, E = z) > 0$ for all four (x, z) , put $f_{xz} = \mathbb{E}_R[BCD \mid A = x, E = z]$ and*

$$I = \frac{1}{4} \sum_{x,z} f_{xz}, \quad J = \frac{1}{4} \sum_{x,z} (-1)^{x+z} f_{xz}.$$

Then $\sqrt{|I|} + \sqrt{|J|} \leq 1$.

Proof. Absorb the response seeds into incident sources. Conditioning on $\{A = x\}$ reweights the law of X alone, and conditioning on $\{E = z\}$ reweights the law of Z alone, so under the conditional law X, L, R, Z are still mutually independent, and L, R keep their original laws. Writing

$$b_x(\ell) = \mathbb{E}[B \mid L = \ell, A = x], \quad d_z(\rho) = \mathbb{E}[D \mid R = \rho, E = z], \\ c(\ell, \rho) = \mathbb{E}[C \mid L = \ell, R = \rho],$$

all with values in $[-1, 1]$, conditional independence of the three responses given the sources gives

$$f_{xz} = \mathbb{E}_{L,R}[b_x(L) c(L, R) d_z(R)].$$

Put $\beta_{\pm} = \frac{1}{2}(b_0 \pm b_1)$ and $\delta_{\pm} = \frac{1}{2}(d_0 \pm d_1)$. Then

$$I = \mathbb{E}_{L,R}[\beta_+(L) c(L, R) \delta_+(R)], \quad J = \mathbb{E}_{L,R}[\beta_-(L) c(L, R) \delta_-(R)],$$

so, using $|c| \leq 1$ and the independence of L and R ,

$$|I| \leq p_+ r_+, \quad |J| \leq p_- r_-, \quad p_{\pm} = \mathbb{E}_L |\beta_{\pm}|, \quad r_{\pm} = \mathbb{E}_R |\delta_{\pm}|.$$

For real u, v with $|u|, |v| \leq 1$, $|\frac{u+v}{2}| + |\frac{u-v}{2}| = \max(|u|, |v|) \leq 1$, so $p_+ + p_- \leq 1$ and $r_+ + r_- \leq 1$. Cauchy–Schwarz gives

$$\sqrt{|I|} + \sqrt{|J|} \leq \sqrt{p_+ r_+} + \sqrt{p_- r_-} \leq \sqrt{p_+ + p_-} \sqrt{r_+ + r_-} \leq 1. \quad \square$$

Inequalities of this shape originate with Branciard, Rosset, Gisin and Pironio [4] for the bilocal scenario with measurement inputs; Lemma 7.3 is the specialization to a single fixed setting, one binary central output and two binary endpoint observations, proved here for arbitrary latent alphabets and stochastic responses.

Corollary 7.4 (Quantitative incompatibility). *For every $0 < h < 1$,*

$$P_h \notin \mathcal{C}_{P_5} \quad \text{and} \quad d_{\text{TV}}(P_h, \mathcal{C}_{P_5}) \geq \frac{h}{96}.$$

Proof. By (23), $I = J = (1 + h)/4$ for P_h , so $\sqrt{I} + \sqrt{J} = \sqrt{1 + h} > 1$ and Lemma 7.3 rules out compatibility.

Let $R \in \mathcal{C}_{P_5}$ and $d = d_{\text{TV}}(P_h, R)$. If $d \geq 1/8$ then $d > 1/96 > h/96$, so assume $d < 1/8$. Each endpoint cell satisfies $|R(x, z) - \frac{1}{4}| \leq d$, hence $R(x, z) \geq \frac{1}{4} - d > \frac{1}{8} > 0$ and Lemma 7.3 applies to R . Write $N_R(x, z) = \mathbb{E}_R[BCD \mathbf{1}\{A = x, E = z\}]$ and N_P for the same quantity under P_h , so $f_{xz}^R = N_R(x, z)/R(x, z)$ and likewise for P_h . The eight atoms of one endpoint cell contribute at most $2d$ to $\sum |R - P_h|$, so $|N_R - N_P| \leq 2d$, while $|R(x, z) - P_h(x, z)| \leq d$. Since $|N_P| \leq P_h(x, z) = \frac{1}{4}$,

$$|f_{xz}^R - f_{xz}^P| \leq \frac{|N_R - N_P|}{R(x, z)} + \frac{|N_P| |P_h(x, z) - R(x, z)|}{R(x, z) P_h(x, z)} \leq \frac{2d + d}{R(x, z)} \leq 24d.$$

Averaging, $|I_R - I_P| \leq 24d$ and $|J_R - J_P| \leq 24d$. If $24d < h/4$ then $I_R, J_R > \frac{1+h}{4} - \frac{h}{4} = \frac{1}{4}$, so $\sqrt{|I_R|} + \sqrt{|J_R|} > 1$, contradicting Lemma 7.3. Hence $24d \geq h/4$. $\square$

7.3 The witness at every order

Theorem 7.5 (The path at every order). *For every $t \geq 1$ put $\gamma = 1/(4t)$ and $h_t = \gamma^2 = 1/(16t^2)$. Then*

$$P_{h_t} \in \mathcal{I}_t^{\text{exp}}(P_5) = \mathcal{I}_t^{\text{AI}}(P_5), \quad P_{h_t} \notin \mathcal{C}_{P_5},$$

with a strictly positive rational witness. Consequently $t_{\min}^H(P_{h_t}) > t$ for $H \in \{\text{NW}, \text{AI}, \text{exp}\}$, no finite order of any of the three hierarchies characterizes $\mathcal{C}_{P_5}$, and

$$d_{\text{TV}}(P_{h_t}, \mathcal{C}_{P_5}) \geq \frac{1}{1536 t^2}.$$

Proof. The auxiliary law. On signs $u = (u_1, \dots, u_t)$ and $v = (v_1, \dots, v_t)$ set

$$H_t(u, v) = 2^{-2t} \operatorname{Re} \left[\prod_{i=1}^t (1 + i\gamma u_i) \prod_{j=1}^t (1 - i\gamma v_j) \right]. \quad (24)$$

Writing $\sigma = (u_1, \dots, u_t, -v_1, \dots, -v_t) \in \{-1, 1\}^{2t}$, the bracket is $\prod_{k=1}^{2t} (1 + i\gamma \sigma_k)$, whose real part keeps the even characters with coefficient $(i\gamma)^{|S|} = (-h_t)^{|S|/2}$, so

$$H_t = 2^{-2t} \sum_{\substack{S \subseteq [2t] \\ |S| \text{ even}}} (-h_t)^{|S|/2} \prod_{k \in S} \sigma_k.$$

This is H_{2t, h_t} of Lemma 6.1 read in the coordinates σ , and $(2t)^2 h_t = 1/4$, so H_t is a probability law with

$$H_t(u, v) \geq \frac{2}{3} 2^{-2t} > 0. \quad (25)$$

All its values are rational.

The witness. Draw $(u, v) \sim H_t$; independently draw fair bits X_a and Z_e , fair signs r_i and s_j , and fair signs ν_{ij} , all mutually independent, $a, i, j, e \in [t]$. With $u^0 = 1$ and $u^1 = u$, define every copied observation by

$$A^a = X_a, \quad B^{ai} = r_i u_i^{X_a}, \quad C^{ij} = \begin{cases} r_i s_j, & u_i = v_j, \\ r_i s_j \nu_{ij}, & u_i \neq v_j, \end{cases} \quad D^{je} = s_j v_j^{Z_e}, \quad E^e = Z_e. \quad (26)$$

Each copied observation is a function of the data attached to its copied parents together with its own private sign: r_i and u_i belong to L_i , and s_j and v_j to R_j . Call the resulting law Γ_t ; by (25) it is strictly positive and rational. As in Section 8.1, Γ_t is a law on the copied observations and not the output of an inflated model: under H_t the families u and v are correlated.

Formal weights. Give each u_i the complex weight $(1 + i\gamma u_i)/2$ and each v_j the weight $(1 - i\gamma v_j)/2$, and keep the actual probabilities of X, Z, r, s, ν . Write $\mathbb{E}_*$ for the resulting formal expectation, a product functional in which all $2t + 2t + t^2 + 2t$ coordinate families are independent, with $\mathbb{E}_* u_i = i\gamma$ and $\mathbb{E}_* v_j = -i\gamma$. Since the weights of the auxiliary coordinates multiply to a complex number whose real part is H_t , and the other coordinates carry real weights,

$$\mathbb{E}_{\Gamma_t}[F] = \operatorname{Re} \mathbb{E}_*[F] \quad \text{for every real-valued } F. \quad (27)$$

Every copied path has law P_h . Fix (a, i, j, e) and condition on $X_a = x, Z_e = z$, which are fair and independent and reproduce the (A, E) marginal of P_h . The masks r_i and s_j are fair signs independent of everything else and each occurs in exactly one of B^{ai}, D^{je} and in C^{ij} , so every product of a proper nonempty subset of $\{B^{ai}, C^{ij}, D^{je}\}$ carries r_i or s_j to an odd power and has formal expectation zero. In the triple product the masks cancel:

$$B^{ai} C^{ij} D^{je} = u_i^x v_j^z \cdot (\mathbf{1}\{u_i = v_j\} + \nu_{ij} \mathbf{1}\{u_i \neq v_j\}),$$

and $\mathbb{E}_* \nu_{ij} = 0$, so $\mathbb{E}_*[B^{ai} C^{ij} D^{je}] = \mathbb{E}_*[u_i^x v_j^z \mathbf{1}\{u_i = v_j\}]$. The formal weight of $(u_i, v_j) = (+1, +1)$ is $\frac{1+i\gamma}{2} \cdot \frac{1-i\gamma}{2} = \frac{1+h}{4}$, and the weight of $(-1, -1)$ is the same real number, so

$$\mathbb{E}_*[u_i^x v_j^z \mathbf{1}\{u_i = v_j\}] = \frac{1+h}{4} (1 + (-1)^{x+z}) = f_{xz},$$

which is real. Every character of the copied path is thus one of: a character of the fair bits X_a, Z_e alone; a character carrying a proper nonempty subset of $\{B^{ai}, C^{ij}, D^{je}\}$, with formal expectation zero; or a character carrying all three, with formal expectation $\frac{1}{4} \sum_{x,z} \chi(x, z) f_{xz}$. All of these are real and all agree with the corresponding characters of P_h in Lemma 7.2, so by (27)

$$\operatorname{Law}_{\Gamma_t}(A^a, B^{ai}, C^{ij}, D^{je}, E^e) = P_h,$$

and $\mathbb{E}_*[G]$ is real for every function G of a copied path.

Injectable marginals. An injectable set of the path contains at most one copied observation over each of A, B, C, D, E and its copy indices agree on shared sources, so it lies in a copied path (a, i, j, e) with consistent indices, after completing unused indices arbitrarily. Its marginal is therefore the corresponding marginal of P_h , and $\mathbb{E}_*[G]$ is real for every function G of it.

Ancestral products and the diagonal law. Let $F_1, \dots, F_r$ be pairwise ancestrally independent injectable sets. Distinct copied sources carry distinct auxiliary coordinates and distinct masks, and distinct copied observations carry distinct private signs, so the F_k depend on disjoint coordinate families and $\mathbb{E}_*[\prod_k G_k] = \prod_k \mathbb{E}_*[G_k]$ for functions G_k of F_k . Each factor is real by the previous paragraph, hence so is the product, and (27) gives

$$\mathbb{E}_{\Gamma_t}\left[\prod_k G_k\right] = \prod_k \mathbb{E}_*[G_k] = \prod_k \mathbb{E}_{\Gamma_t}[G_k].$$

Taking the G_k to be indicators of atoms shows the blocks are independent with the prescribed marginals. The diagonal rows $(A^\ell, B^{\ell\ell}, C^{\ell\ell}, D^{\ell\ell}, E^\ell)$, $\ell \in [t]$, are pairwise ancestrally independent injectable copied paths, so the diagonal law is $P_h^{\otimes t}$.

Symmetry. The density (24) is symmetric in $u_1, \dots, u_t$ and in $v_1, \dots, v_t$ separately, the families (X_a) , (Z_e) , (r_i) , (s_j) and (ν_{ij}) are i.i.d., and (26) is equivariant, so Γ_t is invariant under independent permutations of the four copy-index families. Together with the previous two paragraphs this places P_{h_t} in $\mathcal{I}_t^{\text{AI}}(P_5)$, and Lemma 3.9 gives $\mathcal{I}_t^{\text{exp}}(P_5) = \mathcal{I}_t^{\text{AI}}(P_5)$.

Conclusion. Corollary 7.4 gives $P_{h_t} \notin \mathcal{C}_{P_5}$ and $d_{\text{TV}}(P_{h_t}, \mathcal{C}_{P_5}) \geq h_t/96 = 1/(1536t^2)$. Membership at order t forces $t_{\min}^H(P_{h_t}) > t$ for each of the three hierarchies, by Proposition 3.6. $\square$

Along this family the first rejecting order grows without bound while the target stays uniformly inside the simplex: since $h_t \leq 1/16$, Lemma 7.2 gives

$$P_{h_t}(x, b, c, d, z) \geq \frac{1 - h_t}{64} \geq \frac{15}{1024}$$

for every t and every atom. The obstruction therefore does not come from vanishing probabilities. The distance to $\mathcal{C}_{P_5}$ shrinks at rate t^{-2} , faster than the rate $t^{-1/2}$ that Theorem 6.16 forces; whether the path carries survivors at distance of order t^{-1} , as the square and the triangle do, is not settled here.

8 The explicit nontermination family

For $0 < \varepsilon < 1$ and $0 \leq r \leq 1$ define the three-bit law

$$Q(\varepsilon, r) = \varepsilon \delta_{000} + (1 - \varepsilon) \text{Bern}(r)^{\otimes 3}, \quad (28)$$

a point mass at 000 with weight ε mixed with three independent bits that equal 1 with probability r . Throughout, $\text{Bern}(r)$ denotes the law on $\{0, 1\}$ with $\text{Bern}(r)(1) = r$.

Theorem 8.1 (Membership at every finite order). *Let $t \geq 1$, $0 < \varepsilon < 1$ and $0 \leq r \leq (1 - \varepsilon)^{t-1}$. Then*

$$Q(\varepsilon, r) \in \mathcal{I}_t^{\text{exp}} \subseteq \mathcal{I}_t^{\text{AI}} \subseteq \mathcal{I}_t^{\text{NW}}.$$

Theorem 8.2 (No finite characterizing order). *For every integer $t \geq 1$, with $\varepsilon_t = 1/(2t^3)$ and $r_t = (1 - \varepsilon_t)^{t-1}$, the law $P_t = Q(\varepsilon_t, r_t)$ satisfies*

$$P_t \in \mathcal{I}_t^{\text{exp}} \subseteq \mathcal{I}_t^{\text{AI}} \subseteq \mathcal{I}_t^{\text{NW}} \quad \text{and} \quad P_t(000)^2 - P_{t,A}(0)P_{t,B}(0)P_{t,C}(0) \geq \frac{\varepsilon_t^2}{2} > 0,$$

so $P_t \notin \mathcal{C}_{\Delta}$. Hence none of the three finite-order tests characterizes $\mathcal{C}_{\Delta}$ at any finite order.

Theorem 8.2 includes $t = 1$, where $P_1 = \frac{1}{2}(\delta_{000} + \delta_{111})$, the perfectly correlated three-bit law, which is the standard example of a triangle-incompatible distribution and passes the trivial order-one test.

Corollary 8.3 (Larger alphabets). *For the triangle with observed alphabets of any sizes $n_A, n_B, n_C \geq 2$, no finite order of the Navascués–Wolfe hierarchy characterizes the compatible set.*

Proof. Regard P_t as a law on the larger alphabets, supported on $\{0, 1\}^3$. A compatible realization of a law supported on $\{0, 1\}^3$ has response kernels that output values outside $\{0, 1\}$ with probability zero, so it is a binary realization; hence P_t remains incompatible. The order- t witness Γ_t of Theorem 8.2, viewed as a law on the larger copied alphabets, still has the required symmetry and diagonal law. So $P_t \in \mathcal{I}_t^{\text{NW}} \setminus \mathcal{C}$ for the larger scenario as well. $\square$

8.1 The defect cube

Fix $t \geq 1$, $0 < \varepsilon < 1$ and $0 \leq r \leq (1 - \varepsilon)^{t-1}$. The witness is a joint law Γ_t on the $3t^2$ copied observations built from independent bits indexed by the cells of the cube $[t]^3$.

Roots. Let $D_{ijk} \sim \text{Bern}(\varepsilon)$, $(i, j, k) \in [t]^3$, be mutually independent *defect* bits, and let $N_v \sim \text{Bern}(s)$, one for each copied observation v , be mutually independent *private* bits, independent of the defects, with

$$s = 1 - \frac{r}{(1 - \varepsilon)^{t-1}} \in [0, 1]. \quad (29)$$

The endpoint $r = 0$ uses $s = 1$; the endpoint $r = (1 - \varepsilon)^{t-1}$ uses $s = 0$, in which case the private bits play no role.

Outputs. Define

$$\begin{aligned} A^{ij} &= \mathbf{1}\{N_{A^{ij}} = 0 \text{ and } D_{ijk} = 0 \ \forall k\}, \\ B^{ik} &= \mathbf{1}\{N_{B^{ik}} = 0 \text{ and } D_{ijk} = 0 \ \forall j\}, \\ C^{jk} &= \mathbf{1}\{N_{C^{jk}} = 0 \text{ and } D_{ijk} = 0 \ \forall i\}. \end{aligned} \quad (30)$$

So A^{ij} reads the line of cells $\Lambda(A^{ij}) = \{(i, j, k) : k \in [t]\}$, B^{ik} the line $\Lambda(B^{ik}) = \{(i, j, k) : j \in [t]\}$, and C^{jk} the line $\Lambda(C^{jk}) = \{(i, j, k) : i \in [t]\}$; each output equals 1 exactly when its private bit and every defect on its line are 0. Let Γ_t be the joint law of the outputs. For $s = 0$ its entries are

$$\Gamma_t(w) = \sum_{d \in \{0,1\}^{[t]^3} : F_t(d)=w} \varepsilon^{|d|} (1 - \varepsilon)^{t^3 - |d|}, \quad (31)$$

For rational ε , these entries are rational. Here F_t is the map (30) from defect configurations to observed assignments. The entries are nonnegative and sum to $(\varepsilon + 1 - \varepsilon)^{t^3} = 1$.

Γ_t does not come from running an inflated triangle model; it is a probability law on the copied observations, which is what a finite inflation test asks for.

Lemma 8.4 (Disjoint ancestry gives disjoint inputs). *If two sets of copied observations have disjoint copied latent ancestry in the order- t inflation, then their defect lines and their private bits are disjoint, and the two sets of outputs are independent under Γ_t .*

Proof. Two lines through the cube intersect only if they share the index that corresponds to a common source type: $\Lambda(A^{ij}) \cap \Lambda(B^{i'k}) \neq \emptyset$ forces $i = i'$, and then X_i is a common ancestor; similarly $\Lambda(A^{ij}) \cap \Lambda(C^{j'k}) \neq \emptyset$ forces $j = j'$ and $\Lambda(B^{ik}) \cap \Lambda(C^{jk'}) \neq \emptyset$ forces $k = k'$; two lines of the same type are disjoint unless the observations coincide. Taking unions, disjoint ancestry implies disjoint defect supports. Distinct observations have distinct private bits. Outputs that are functions of disjoint families of independent bits are independent. $\square$

Lemma 8.5 (Law of a copied triangle). *For every $(i, j, k) \in [t]^3$, $\text{Law}_{\Gamma_t}(A^{ij}, B^{ik}, C^{jk}) = Q(\varepsilon, r)$.*

Proof. The three lines of Δ_{ijk} meet pairwise exactly at the cell (i, j, k) . Condition on D_{ijk} . If $D_{ijk} = 1$, all three outputs are 0. If $D_{ijk} = 0$, each output depends on its own private bit and on the $t - 1$ remaining cells of its line; these three families are pairwise disjoint, so the outputs are conditionally independent, each equal to 1 with probability $(1 - s)(1 - \varepsilon)^{t-1} = r$ by (29). Averaging over D_{ijk} gives (28). $\square$

Lemma 8.6 (Symmetry). Γ_t is invariant under independent permutations of the X -, Z - and Y -copy indices.

Proof. A permutation σ of the X -indices sends $D_{ijk} \mapsto D_{\sigma(i)jk}$, $N_{A^{ij}} \mapsto N_{A^{\sigma(i)j}}$, and so on; it permutes the two families of i.i.d. bits and carries (30) to itself. The same holds for the other two index families. $\square$

8.2 Genuine incompatibility

Lemma 8.7 (Finner inequality for the triangle). Every $P \in \mathcal{C}_\Delta$ satisfies the inequality $P(000)^2 \leq P_A(0)P_B(0)P_C(0)$.

Proof. Let the sources have laws μ_X, μ_Y, μ_Z on arbitrary measurable spaces and let

$$f(x, z) = \Pr(A = 0 \mid x, z), \quad g(x, y) = \Pr(B = 0 \mid x, y), \quad h(z, y) = \Pr(C = 0 \mid z, y)$$

be the response probabilities, measurable functions with values in $[0, 1]$. Conditional independence of the responses given the sources gives $P(000) = \int f(x, z)g(x, y)h(z, y) d\mu_X d\mu_Y d\mu_Z$. Put $u(x)^2 = \int g(x, y)^2 d\mu_Y(y)$ and $v(z)^2 = \int h(z, y)^2 d\mu_Y(y)$. Cauchy–Schwarz in y , then Cauchy–Schwarz in the independent pair (x, z) , then $f^2 \leq f$, $g^2 \leq g$, $h^2 \leq h$ (all three functions lie in $[0, 1]$) give

$$\begin{aligned} P(000) &\leq \int f(x, z)u(x)v(z) d\mu_X d\mu_Z \\ &\leq \left(\int f^2 d\mu_X d\mu_Z \right)^{1/2} \left(\int u^2 d\mu_X \int v^2 d\mu_Z \right)^{1/2} \leq \sqrt{P_A(0)P_B(0)P_C(0)}. \end{aligned}$$

Tonelli’s theorem justifies every interchange, since all integrands are nonnegative and bounded. The argument uses neither finiteness of the latent alphabets nor determinism of the responses. $\square$

The inequality is the triangle case of Finner’s generalized Hölder inequality [6]; Renou et al. [19] derived it for the triangle network in this event form and proved it for quantum sources and for no-signalling boxes combined by local wirings.

Lemma 8.8 (Violation by the witness). Let $t \geq 1$, $0 < \varepsilon < t^{-3}$ and $P = Q(\varepsilon, (1 - \varepsilon)^{t-1})$. Then $P(000)^2 > P_A(0)P_B(0)P_C(0)$; in particular $P \notin \mathcal{C}_\Delta$. For $\varepsilon = \varepsilon_t = 1/(2t^3)$,

$$P_t(000)^2 - P_{t,A}(0)P_{t,B}(0)P_{t,C}(0) \geq \varepsilon_t^2 - t^3 \varepsilon_t^3 = \frac{\varepsilon_t^2}{2}.$$

Proof. With $r = (1 - \varepsilon)^{t-1}$, $P(000) = \varepsilon + (1 - \varepsilon)(1 - r)^3 \geq \varepsilon$ and, by symmetry of the three coordinates, $P_A(0) = P_B(0) = P_C(0) = 1 - (1 - \varepsilon)r = 1 - (1 - \varepsilon)^t \leq t\varepsilon$ by Bernoulli’s inequality. Hence $P(000)^2 \geq \varepsilon^2$ and $P_A(0)P_B(0)P_C(0) \leq t^3 \varepsilon^3$, and $\varepsilon^2 > t^3 \varepsilon^3$ exactly when $\varepsilon < t^{-3}$. For $\varepsilon_t = 1/(2t^3)$ the difference is at least $\varepsilon_t^2(1 - t^3 \varepsilon_t) = \varepsilon_t^2/2$. $\square$

At $t = 1$ the witness is $P_1 = \frac{1}{2}(\delta_{000} + \delta_{111})$ with $P_1(000)^2 = 1/4$ and $P_{1,A}(0)^3 = 1/8$; the margin $1/8$ is $\varepsilon_1^2/2$.

8.3 Survival of every finite order

The defect law Γ_t witnesses $Q(\varepsilon, r) \in \mathcal{I}_t^{\text{exp}}$, which is Theorem 8.1. The three feasible sets require successively more of Γ_t .

8.3.1 The Navascués–Wolfe conditions

Symmetry is Lemma 8.6. For the diagonal law, let R_ℓ be the union of the three lines of the diagonal triangle $\Delta_{\ell\ell\ell}$:

$$R_\ell = \{(\ell, \ell, k) : k \in [t]\} \cup \{(\ell, j, \ell) : j \in [t]\} \cup \{(i, \ell, \ell) : i \in [t]\}. \quad (32)$$

Equivalently, R_ℓ is the set of cells with at least two coordinates equal to ℓ ; it has $3t - 2$ cells.

Lemma 8.9 (Diagonal supports are mutually disjoint). *If $\ell \neq m$ then $R_\ell \cap R_m = \emptyset$. Consequently the t diagonal triangles depend on mutually disjoint families of independent bits, are mutually independent under Γ_t , and their joint law is $Q(\varepsilon, r)^{\otimes t}$.*

Proof. A cell has three coordinates; it cannot have two of them equal to ℓ and two equal to $m \neq \ell$. The private bits of distinct diagonal triangles are distinct. Mutual independence of functions of disjoint families of independent variables follows, and Lemma 8.5 identifies each factor. $\square$

The lemma proves the tensor-power diagonal law, so $Q(\varepsilon, r) \in \mathcal{I}_t^{\text{NW}}$.

8.3.2 The ancestral-independence conditions

Every injectable set is a subset of some copied triangle Δ_{ijk} (Definition 2.2), so its prescribed marginal is a projection of $Q(\varepsilon, r)$ and is matched by Lemma 8.5. A union of pairwise ancestrally independent injectable sets is, by Lemma 8.4 applied to the union, a family of outputs depending on pairwise disjoint families of independent bits, hence mutually independent, with the prescribed product law. This proves $Q(\varepsilon, r) \in \mathcal{I}_t^{\text{AI}}$. In particular the higher-degree relations that Navascués and Wolfe mention but do not impose, such as $\text{Law}(A^{12}, B^{21}, C^{12}) = P_A \otimes P_B \otimes P_C$, hold for Γ_t : the three observations A^{12}, B^{21}, C^{12} have ancestor sets $\{X_1, Z_2\}$, $\{X_2, Y_1\}$ and $\{Z_1, Y_2\}$. Their lines $(1, 2, \cdot)$, $(2, \cdot, 1)$ and $(\cdot, 1, 2)$ are pairwise disjoint.

8.3.3 The recursively expressible conditions

The AI conditions do not by themselves give the conditional-product prescriptions of Definition 2.3, which use d -separations in the genuine inflated graph. Every such separation also holds for the defect law.

Lemma 8.10 (Latent-projection containment). *Let H_t be the directed acyclic graph with the defect and private bits as root vertices and the copied observations as their children according to (30). The latent projection of H_t onto the observed vertices is a subgraph of the latent projection of the order- t inflated triangle onto the same vertices.*

Proof. Both projections have only the observed vertices, all of which are sinks, and a bidirected edge between two observed vertices exactly when they share a latent parent. In H_t a private bit has one child and creates no edge, and a defect D_{ijk} has children A^{ij}, B^{ik}, C^{jk} , creating the three edges $A^{ij} \leftrightarrow B^{ik}$, $A^{ij} \leftrightarrow C^{jk}$, $B^{ik} \leftrightarrow C^{jk}$. In the genuine inflation these three pairs share X_i, Z_j and Y_k respectively, so the same three edges are present. $\square$

Lemma 8.11 (Recursive prescriptions hold). *Every marginal prescribed by the recursion of Definition 2.3 equals the corresponding marginal of Γ_t .*

Proof. Γ_t is Markov with respect to H_t , so every conditional independence among observed sets that is implied by d -separation in H_t holds under Γ_t . For these two-layer graphs, an active path between observed vertices alternates between latent roots and observed colliders; every internal observed vertex must belong to the conditioning set, since observed vertices have no descendants. Equivalently, in the bidirected projection an active path has all its internal vertices in the conditioning set. Deleting bidirected edges cannot create such a path. Lemma 8.10 therefore transfers every observed d -separation of the genuine inflation to H_t . Now induct on the recursion. Injectable base sets are matched by Lemma 8.5. At a conditional-product step with $X' \perp_d Y' \mid Z'$, the transferred conditional independence gives $\Gamma_t(x', y', z') = \Gamma_t(x', z')\Gamma_t(y', z')/\Gamma_t(z')$ whenever $\Gamma_t(z') > 0$, while for $\Gamma_t(z') = 0$ both numerator marginals and the joint atom vanish, matching the zero branch. Marginalization of a matched set is matched. Copy-index equalities among prescribed marginals follow from Lemma 8.6. $\square$

This proves $Q(\varepsilon, r) \in \mathcal{I}_t^{\text{exp}}$ and completes the proof of Theorem 8.1. Theorem 8.2 follows by combining Theorem 8.1 at $r = r_t = (1 - \varepsilon_t)^{t-1}$ with Lemma 8.8.

9 Finite-order fan inequalities

The inequalities of this section bound, for every law feasible at order t , the probability of the all-zero atom by the one-variable zero marginals.

Theorem 9.1 (Fan inequalities). *Let P be a law on three bits with $a = P_A(0)$, $b = P_B(0)$, $c = P_C(0)$, $z = P(000)$. If $P \in \mathcal{I}_t^{\text{NW}}$ then*

$$tz \leq a + \binom{t}{2}bc \quad \text{and} \quad t(z^2 - abc) \leq az - abc. \quad (33)$$

Consequently, if $z^2 > abc$ (a Finner violation), then

$$t_{\min}^{\text{NW}}(P) \leq \left\lfloor \frac{z \min\{a, b, c\} - abc}{z^2 - abc} \right\rfloor + 1, \quad (34)$$

and the same bound holds for $t_{\min}^{\text{AI}}$ and $t_{\min}^{\text{exp}}$.

This gives an explicit rejecting order in terms of four probabilities. Relabelling the local outcomes gives the corresponding bound for a Finner violation at any atom.

Proposition 9.2 (No uniformly divergent distance lower bound). *For $0 < p < 1$ let $R_p = (1 - p)\delta_{111} + p\delta_{000}$. Then $R_p \notin \mathcal{C}_{\Delta}$, $R_p \rightarrow \delta_{111}$ as $p \downarrow 0$, and $t_{\min}^H(R_p) = 2$ for all three hierarchies and all p . Hence no inequality $t_{\min}^H(P) \geq c d_{\text{TV}}(P, \mathcal{C}_{\Delta})^{-\alpha}$ with $c, \alpha > 0$ holds for all incompatible P ; the growth in Proposition 10.1 is a property of the direction of approach to the boundary.*

Proof of Theorem 9.1. For $t = 1$, the two inequalities reduce to $z \leq a$ and $z^2 \leq az$, which hold since $0 \leq z \leq a$. Now let Γ be an order- t feasible law for P , $t \geq 2$, and consider the $2t + 1$ copied observations A^{11}, B^{1k}, C^{1k} , $k \in [t]$: the fan of copied triangles $\Delta_{11k} = \{A^{11}, B^{1k}, C^{1k}\}$ sharing the vertex A^{11} . Define the indicators $A = \mathbf{1}[A^{11} = 0]$, $B_k = \mathbf{1}[B^{1k} = 0]$, $C_k = \mathbf{1}[C^{1k} = 0]$ and $T_k = AB_kC_k$. Three identities follow from Definition 2.1: $\mathbb{E}A = a$ and $\mathbb{E}T_k = z$, because a permutation of the Y -indices carries Δ_{11k} to the diagonal triangle Δ_{111} , whose law is P ; and for $k \neq \ell$,

$$\mathbb{E}[B_k C_{\ell}] = bc, \quad (35)$$

because the independent permutations $X_1 \mapsto 1$, $Z_1 \mapsto 2$, $Y_k \mapsto 1$, $Y_\ell \mapsto 2$ (extended arbitrarily to $[t]$) carry $B^{1k}, C^{1\ell}$ to B^{11}, C^{22} , which lie in the two diagonal triangles $\Delta_{111}, \Delta_{222}$ whose joint law is $P \otimes P$ by the degree-two diagonal condition. The argument uses no product prescription beyond the diagonal law, and (35) is not available for $k = \ell$.

First inequality. For every deterministic assignment of the fan,

$$\sum_k AB_k C_k \leq A + \frac{1}{2} \sum_{k \neq \ell} B_k C_\ell. \quad (36)$$

If $A = 0$ the left side vanishes. If $A = 1$, put $S = \sum_k B_k C_k$; then $\sum_k B_k \geq S$ and $\sum_k C_k \geq S$, so the right side is $1 + \frac{1}{2}[(\sum_k B_k)(\sum_k C_k) - S] \geq 1 + \frac{1}{2}(S^2 - S)$, and the right side minus the left side is at least $\frac{1}{2}(S - 1)(S - 2) \geq 0$ for every integer $S \geq 0$. Taking expectations under Γ and using the three identities gives $tz \leq a + \binom{t}{2}bc$.

Second inequality. Let $U = \sum_k T_k$. Since $U = AU$, Cauchy–Schwarz gives $(\mathbb{E}U)^2 \leq \mathbb{E}A \cdot \mathbb{E}U^2$, and $U^2 = \sum_k T_k + \sum_{k \neq \ell} T_k T_\ell$ with $T_k T_\ell \leq B_k C_\ell$. Hence $(tz)^2 \leq a[tz + t(t - 1)bc]$, which rearranges to $t(z^2 - abc) \leq az - abc$.

Rejecting order. If $z^2 > abc$, the second inequality fails as soon as $t > (az - abc)/(z^2 - abc)$; the fan can be rooted at whichever of A, B, C has the smallest zero marginal, giving (34). The ratio is at least 1 since numerator minus denominator equals $z(\min\{a, b, c\} - z) \geq 0$, so the bound is never below 2, consistent with order one never rejecting. The smaller feasible sets of (1) reject no later. $\square$

Inequality (36) is a pointwise marginal-polytope certificate in the sense of [23], on a $1 \times 1 \times t$ box of source copies; here it is available in closed form at every order.

Proof of Proposition 9.2. R_p has $a = b = c = z = p$, so $z^2 = p^2 > p^3 = abc$ and Lemma 8.7 gives incompatibility. Order one is passed by every law. At order two the first fan inequality reads $2p \leq p + p^2$, which fails for $p < 1$; so $t_{\min}^H(R_p) = 2$ for all three hierarchies, while $d_{\text{TV}}(R_p, \mathcal{C}_\Delta) \leq p \rightarrow 0$. $\square$

10 Sharp rejecting-order exponent

10.1 Finite rejection for each incompatible law

Navascués and Wolfe [15] prove that for every correlation scenario G and every $P \in \mathcal{I}_t^{\text{NW}}(G)$ there is a compatible law within Euclidean distance $O(t^{-1/2})$ of P , with an explicit constant depending on the number of source types; since the compatible set is closed [20], every incompatible P is rejected at some finite order:

$$P \notin \mathcal{C}_G \implies \exists t : P \notin \mathcal{I}_t^{\text{NW}}(G).$$

The quantitative form of this convergence is Theorem 6.16 and Corollary 12.1.

10.2 The family P_ε

Proposition 10.1 (One family with divergent rejecting order). *For $0 < \varepsilon < 1/8$ let $P_\varepsilon = Q(\varepsilon, 1 - \frac{1}{2}\varepsilon^{2/3})$, and write $\sigma = \frac{1}{2}\varepsilon^{2/3}$, $m = \varepsilon + (1 - \varepsilon)\sigma$ for the common one-variable zero*

marginal and $z = \varepsilon + (1 - \varepsilon)\sigma^3$ for $P_\varepsilon(000)$. Then $P_\varepsilon \notin \mathcal{C}_\Delta$, $P_\varepsilon \rightarrow \delta_{111} \in \mathcal{C}_\Delta$ as $\varepsilon \downarrow 0$, and for $H \in \{\text{exp, AI, NW}\}$ the first rejecting order $t_{\min}^H(P_\varepsilon) = \min\{t : P_\varepsilon \notin \mathcal{I}_t^H\}$ is finite and satisfies

$$\left\lfloor 1 + \frac{1}{2}\varepsilon^{-1/3} \right\rfloor + 1 \leq t_{\min}^H(P_\varepsilon) \leq \lfloor \tau_-(\varepsilon) \rfloor + 1 \quad (\varepsilon \leq 1/64),$$

$$\tau_- = \frac{z + m^2/2 - \sqrt{(z + m^2/2)^2 - 2m^3}}{m^2},$$

so that

$$\frac{1}{2} \leq \liminf_{\varepsilon \downarrow 0} \varepsilon^{1/3} t_{\min}^H(P_\varepsilon) \leq \limsup_{\varepsilon \downarrow 0} \varepsilon^{1/3} t_{\min}^H(P_\varepsilon) \leq 4 - 2\sqrt{3} \approx 0.536.$$

In particular $t_{\min}^H(P_\varepsilon) = \Theta(\varepsilon^{-1/3})$ for each of the three hierarchies.

The lower bound is the passing construction of Section 8, the upper bound the first fan inequality of Theorem 9.1; the exponents agree, the constants do not. Section 11 proves the distance statements for the same family.

10.3 Proof of the exponent bounds

Proof of Proposition 10.1. Put $\sigma = \frac{1}{2}\varepsilon^{2/3}$, so $P_\varepsilon = Q(\varepsilon, 1 - \sigma)$, $m = \varepsilon + (1 - \varepsilon)\sigma$ and $z = \varepsilon + (1 - \varepsilon)\sigma^3$. *Incompatibility:* $z \geq \varepsilon$ and $m \leq \varepsilon + \sigma = \varepsilon^{2/3}(\varepsilon^{1/3} + \frac{1}{2})$, so $m^3 < \varepsilon^2 \leq z^2$ because $\varepsilon^{1/3} < \frac{1}{2}$ when $\varepsilon < 1/8$; Lemma 8.7 applies. *Limit:* $\varepsilon \rightarrow 0$ and $1 - \sigma \rightarrow 1$, so $P_\varepsilon \rightarrow \delta_{111}$, which is compatible. *Lower bound:* if $t \leq 1 + \frac{1}{2}\varepsilon^{-1/3}$ then $(t - 1)\varepsilon \leq \sigma$, and Bernoulli's inequality gives $(1 - \varepsilon)^{t-1} \geq 1 - (t - 1)\varepsilon \geq 1 - \sigma$, so Theorem 8.1 applies with $r = 1 - \sigma$ and $P_\varepsilon \in \mathcal{I}_t^{\text{exp}}$; hence every rejecting order exceeds $\lfloor 1 + \frac{1}{2}\varepsilon^{-1/3} \rfloor$. Finiteness is asymptotic completeness together with (1).

Upper bound. Write $v_t = tz - m - \binom{t}{2}m^2$; by the first fan inequality with $a = b = c = m$, positivity of v_t rejects at order t . For $t = x\varepsilon^{-1/3} + O(1)$ one has $tz = x\varepsilon^{2/3} + O(\varepsilon)$, $m = \frac{1}{2}\varepsilon^{2/3} + O(\varepsilon)$ and $\binom{t}{2}m^2 = \frac{x^2}{8}\varepsilon^{2/3} + O(\varepsilon)$, so

$$v_t = \left(x - \frac{1}{2} - \frac{x^2}{8}\right)\varepsilon^{2/3} + O(\varepsilon),$$

whose leading coefficient is positive exactly for $4 - 2\sqrt{3} < x < 4 + 2\sqrt{3}$. Fixing any x slightly above $4 - 2\sqrt{3}$ gives rejection for all small ε , which is the limsup bound. For the exact finite statement, v_t is the quadratic $-\frac{m^2}{2}t^2 + (z + \frac{m^2}{2})t - m$ in t , with discriminant $D = (z + m^2/2)^2 - 2m^3$ and roots $\tau_\pm = (z + m^2/2 \pm \sqrt{D})/m^2$. For $\varepsilon \leq 1/64$ one has $m \leq \frac{3}{4}\varepsilon^{2/3}$ and $z \geq \varepsilon$, so $D \geq \varepsilon^2 - 2m^3 \geq \varepsilon^2(1 - \frac{27}{32}) = \frac{5}{32}\varepsilon^2$ and $\tau_+ - \tau_- = 2\sqrt{D}/m^2 \geq \frac{4\sqrt{10}}{9}\varepsilon^{-1/3} > 1$; since $v_1 = z - m < 0$ the integer $\lfloor \tau_- \rfloor + 1$ lies strictly between the roots, where $v_t > 0$. Hence $t_{\min}^{\text{NW}}(P_\varepsilon) \leq \lfloor \tau_- \rfloor + 1 = (4 - 2\sqrt{3})\varepsilon^{-1/3} + O(1)$. □

The constants $\frac{1}{2}$ and $4 - 2\sqrt{3}$ come from the passing construction and from the fan certificate; whether $\varepsilon^{1/3}t_{\min}^H(P_\varepsilon)$ converges, and to what, is open. Exact certificates at $\varepsilon = k^{-3}$ record where the two bounds meet (Section 16).

11 Distance to the compatible set

The parameter ε of the family P_ε is, to leading order, a fixed multiple of the distance from P_ε to the compatible set, in total variation and in Euclidean norm. With Proposition 10.1 the exponent becomes a statement about the geometry near the boundary of $\mathcal{C}_\Delta$.

Theorem 11.1 (Distance asymptotics). *For $P_\varepsilon = Q(\varepsilon, 1 - \frac{1}{2}\varepsilon^{2/3})$, $0 < \varepsilon < 1/8$,*

$$d_{\text{TV}}(P_\varepsilon, \mathcal{C}_\Delta) = (1 - 2^{-3/2})\varepsilon + O(\varepsilon^{4/3}), \quad d_2(P_\varepsilon, \mathcal{C}_\Delta) = \sqrt{8/7}(1 - 2^{-3/2})\varepsilon + O(\varepsilon^{4/3}).$$

Corollary 11.2 (Direction-specific boundary complexity). *For $H \in \{\text{NW}, \text{AI}, \text{exp}\}$,*

$$t_{\min}^H(P_\varepsilon) = \Theta(d_{\text{TV}}(P_\varepsilon, \mathcal{C}_\Delta)^{-1/3}) = \Theta(d_2(P_\varepsilon, \mathcal{C}_\Delta)^{-1/3}) \quad (\varepsilon \downarrow 0).$$

This is a statement about one direction of approach to the boundary; by Proposition 9.2 no such lower bound holds uniformly over all incompatible laws.

Proof of Theorem 11.1. Put $\sigma = \frac{1}{2}\varepsilon^{2/3}$, $m = \varepsilon + (1 - \varepsilon)\sigma$, $z = \varepsilon + (1 - \varepsilon)\sigma^3$, $c_0 = 2^{-3/2}$ and $d_0 = 1 - c_0$.

Lower bounds. The product law $\text{Bern}(1 - \sigma)^{\otimes 3}$ is compatible and within total variation ε of P_ε . Compactness of $\mathcal{C}_\Delta$ gives a nearest law in either norm [20]. A total-variation minimizer Q therefore has distance $\delta = O(\varepsilon)$. Since probabilities of events differ by at most δ ,

$$Q(000) \geq z - \delta, \quad Q_V(0) \leq m + \delta \quad (V = A, B, C).$$

Finner gives

$$z - \delta \leq (m + \delta)^{3/2} = c_0\varepsilon + O(\varepsilon^{4/3}),$$

so $\delta \geq d_0\varepsilon - O(\varepsilon^{4/3})$.

A Euclidean minimizer Q_2 also satisfies $\|P_\varepsilon - Q_2\|_2 = O(\varepsilon)$, by comparison with the product law. Cauchy–Schwarz bounds each one-variable marginal error by $2\|P_\varepsilon - Q_2\|_2$, since each marginal sums four atoms. Hence Finner gives $Q_2(000) \leq c_0\varepsilon + O(\varepsilon^{4/3})$. Set $D = P_\varepsilon(000) - Q_2(000)$; then $D \geq d_0\varepsilon - O(\varepsilon^{4/3}) > 0$ for small ε . The other seven coordinates of $Q_2 - P_\varepsilon$ sum to D , so

$$\|Q_2 - P_\varepsilon\|_2^2 \geq D^2 + \frac{D^2}{7}.$$

This proves the Euclidean lower bound.

A compatible approximation. Set $h = d_0/7$ and $u = \sigma + (c_0 + h)\varepsilon$. Each source carries a base bit with success probability $\sqrt{u}$ and a flag with success probability $h\varepsilon$, all six bits independent. A node outputs 0 if both incident base bits equal 1 or either incident flag equals 1. These probabilities lie in $[0, 1]$ for small ε , and the resulting law Q_ε belongs to $\mathcal{C}_\Delta$.

Without flags, the atom probabilities are $u^{3/2}$ at 000, $u - u^{3/2}$ at each one-zero atom, zero at each two-zero atom, and $1 - 3u + 2u^{3/2}$ at 111. A single flag forces the two nodes incident to its source to zero. Except on a base-output event of probability $O(u)$, it transfers mass from 111 to the two-zero atom associated with that source. Multiple flags have probability $O(\varepsilon^2)$. The errors from these exceptions are $O(\varepsilon u + \varepsilon^2) = O(\varepsilon^{5/3})$.

Using $u^{3/2} = c_0\varepsilon + O(\varepsilon^{4/3})$ gives the following atom probabilities, with an $O(\varepsilon^{4/3})$ remainder in each entry:

Atom	P_ε	Q_ε
000	ε	$c_0\varepsilon$
Each of 011, 101, 110	σ	$\sigma + h\varepsilon$
Each of 001, 010, 100	0	$h\varepsilon$
111	$1 - 3\sigma - \varepsilon$	$1 - 3\sigma - (c_0 + 6h)\varepsilon$

Since $1 - c_0 = 7h$, the deviation at 000 is $-d_0\varepsilon + O(\varepsilon^{4/3})$ and at each of the other seven atoms it is $h\varepsilon + O(\varepsilon^{4/3})$. Its total variation is $d_0\varepsilon + O(\varepsilon^{4/3})$ and its Euclidean norm is $\sqrt{8/7}d_0\varepsilon + O(\varepsilon^{4/3})$, proving both upper bounds. $\square$

Corollary 11.2 follows from Theorem 11.1 and Proposition 10.1. The constant $1 - 2^{-3/2}$ is the fraction of the rare atom $P_\varepsilon(000) \approx \varepsilon$ that no compatible law can retain once its zero marginals are pinned near $\frac{1}{2}\varepsilon^{2/3}$, by the Finner inequality.

12 Distance-promised rejecting orders

Theorem 8.2 says that the order needed cannot be fixed in advance for the scenario. For a single instance, Theorem 6.16 bounds it under a distance promise and Proposition 13.1 bounds it in the input size.

Corollary 12.1 (Distance-promised rejecting order). *Let G be a correlation scenario with K joint outcomes and let $\delta_2 = d_2(P, \mathcal{C}_G)$ and $\delta = d_{\text{TV}}(P, \mathcal{C}_G)$. If $P \in \mathcal{I}_n^{\text{NW}}(G)$ then*

$$\delta_2^2 \leq \frac{1 - \|P\|_2^2}{n}, \quad (37)$$

so for incompatible P

$$t_{\min}^{\text{NW}}(P) \leq \left\lfloor \frac{1 - \|P\|_2^2}{\delta_2^2} \right\rfloor + 1 \leq \left\lfloor \frac{K - 1}{4\delta^2} \right\rfloor + 1;$$

for the binary triangle, $t_{\min}^{\text{NW}}(P) \leq \lfloor 7/(4\delta^2) \rfloor + 1$.

Proof. The first inequality is (20), since some ω attains at most the mean and q_ω is compatible. Feasibility at order n then forces $n \leq (1 - \|P\|_2^2)/\delta_2^2$, and $\|P - Q\|_2^2 \geq \|P - Q\|_1^2/K = 4d_{\text{TV}}(P, Q)^2/K$ with $\|P\|_2^2 \geq 1/K$ gives the second form. The triangle has $K = 8$. $\square$

The proof uses no convexity of $\mathcal{C}_G$: we extract one compatible law from the witness. The bound gives a rejecting order under a promise rather than a decision procedure, since δ is not directly evaluable.

13 Rational input bit-length complexity

Proposition 13.1 (Required order in the input size). *Encode each of the eight rational atoms by its reduced numerator and positive denominator in ordinary binary, with delimiters of linear overhead. For a law P on three bits, let $\ell(P)$ be its total bit length. For each $H \in \{\text{NW}, \text{AI}, \text{exp}\}$,*

$$\max\{t_{\min}^H(P) : \ell(P) \leq B, P \notin \mathcal{C}_\Delta\} = 2^{\Theta(B)}.$$

Proof of Proposition 13.1. Lower bound. With $k = 2^j$, $\varepsilon = k^{-3}$ and $\sigma = 1/(2k^2)$, the eight atoms of P_ε are rationals with denominator $8k^9$ and numerators of $O(j)$ bits, so $\ell(P_\varepsilon) = O(j)$, while Proposition 10.1 gives $t_{\min}^H(P_\varepsilon) > k/2 = 2^{j-1}$. Choosing j proportional to B gives $2^{\Omega(B)}$.

Upper bound. By [20], $\mathcal{C}_\Delta$ is the image of a compact semialgebraic parameter space (six latent values per source and stochastic binary response tables) under a polynomial map, hence a closed semialgebraic subset of the simplex. The function $u(P) = d_2(P, \mathcal{C}_\Delta)^2$ is therefore semialgebraic, and its graph is described by a fixed quantifier-free formula $\Phi(P, u)$ in integer polynomials of fixed degrees, by real quantifier elimination [1]. For a rational P of bit length B , substituting P and clearing denominators turns the atoms of Φ into univariate integer polynomials in u of fixed degree and coefficient height $2^{O(B)}$. The set of u satisfying $\Phi(P, u)$ is the single point $u(P)$; discard identically zero specialized polynomials, whose comparisons have constant truth values. If none of the remaining polynomials vanished at $u(P)$, their signs would persist on a neighbourhood and the formula would define an interval. Thus some nonzero specialized polynomial f vanishes at $u(P)$. Remove any factor u^r from f to obtain $g(u) = a_0 + \dots + a_d u^d$ with $a_0 \neq 0$, the same nonzero root, and coefficient height at most $H = 2^{O(B)}$. For $0 < u < 1/(H+1)$,

$$\left| \sum_{i=1}^d a_i u^i \right| \leq H \sum_{i=1}^d u^i < \frac{Hu}{1-u} < 1 \leq |a_0|,$$

so $g(u) \neq 0$. Hence $u(P) \geq 2^{-O(B)}$ when P is incompatible, and Corollary 12.1 gives $t_{\min}^{\text{NW}}(P) \leq (1 - \|P\|_2^2)/u(P) + 1 = 2^{O(B)}$; the two smaller feasible sets reject no later. $\square$

With the binary triangle held fixed, a quantifier-free membership formula contains a bounded number of integer polynomials of bounded degree. Evaluating their signs at a rational input takes time polynomial in B . Quantifier elimination supplies such a formula in principle [1], although deriving and storing it may be impractical. Thus the exponential inflation-order bound coexists with a polynomial-time decision procedure for this fixed scenario. For an incompatible rational input, one can also compute the first rejecting order by enumerating exact rational feasibility problems.

14 Relation to prior work

Finite-order convergence. Navascués and Wolfe [15, Section 4.1] give a binary family that passes second-order triangle inflation while violating Finner, ask whether any finite order characterizes compatibility, and prove that star scenarios terminate at order two. Theorem 4.2 places the star result inside the double-star class and shows that every other pair-source scenario fails at every order. The terminating half reuses their mechanism, source-disjoint independence at order two, and adds the conditioning step that reads off the central responses; their four-on-line scenario is the first case beyond stars (Corollary 5.3). Henson, Lal and Pusey [11] classify causal structures by whether the observed conditional independences already cut out the classical compatible set; that criterion parts from Theorem 4.2 at the four-on-line scenario, which fails theirs and terminates at order two. Wolfe, Spekkens and Fritz define the recursively expressible sets and leave their general study to future work [23]; their example of a strict gap between expressible and ancestral-independence prescriptions has an observed node with observed children, outside the root-sink class, and Lemma 3.9 shows that no gap exists on that class. They also exhibit a polynomial optimization problem whose inflation relaxations fail to attain the exact

optimum at any finite order, in a scenario where every law is compatible. Our results concern the compatibility tests themselves.

Network inequalities. Finner’s inequality [6] supplies the incompatibility test for the defect cube. Renou et al. [19] prove its triangle-network form for quantum sources and for no-signalling boxes combined by local wirings, and give the classical Cauchy–Schwarz argument used here. The incompatibility of the five-path witness is a bilocal inequality across the two inner sources, of the kind introduced by Branciard, Rosset, Gisin and Pironio [4]; Lemma 7.3 is the classical square-root form with the two endpoint observers treated as settings. The incompatibility of the cycle witnesses is the rigidity of parity-perfect triangle correlations, due to Boreiri et al. [2], with the noise-robust form of [3] (Lemma 6.6 is a moment version that needs no repaired model); Renou and Beigi [17] prove token-counting rigidity on every ring scenario. Section 6 reproves the parity statements for arbitrary latent alphabets and stochastic responses.

The first fan inequality follows the marginal-polytope method of Wolfe, Spekkens and Fritz [23, Section IV.D]: prove a pointwise inequality on copied observations, then substitute the prescribed marginals. Fraser and Wolfe [8] derive triangle inequalities from the wagon-wheel and web inflations. Theorem 9.1 gives a closed formula indexed by the order, together with a second-moment bound. Da Silva, Pozas-Kerstjens and Parisio [5] study local models and conjectural boundaries for permutation-invariant binary-triangle distributions; the parity-even face used in Section 6 is one of those symmetric families. Pozas-Kerstjens, Gisin and Renou [16] prove network nonlocality along continuous families of distributions with inflation; the families here are classical and approach the compatible set rather than leave it.

Witness constructions. The Fourier densities of Sections 6 and 7 are signed combinations of characters on auxiliary signs made positive by a real-part or a correction term. Sign-potential representations of parity-perfect arrays and the Krawtchouk reduction of Lemma 6.19 are standard in coding theory [14]. The Shearer-type measure that keeps the rare-selector witnesses of Appendix C positive is the one from [21].

Quantitative convergence and effectivity. Theorem 6.16 strengthens the finite-sampling estimate in [15]: the diagonal of a witness is dominated in convex order by an empirical law, which gives $\sqrt{K-1}/(2\sqrt{n})$ with no factor depending on the number L of sources, against the factor L in their collision estimate. The other published rates of this family also carry the sources: Gitton’s convergence theorem for postselected inflation [10, Theorem 8] has a constant of order $\sqrt{12^{C-1} \sum_c d_c S_c (S_c - \frac{1}{2})}$ in the source multiplicities S_c , and Fraser’s combinatorial hierarchy [7] converges at rate $O(LC/K)$ in its own parameters. The convex-order domination of sampling without replacement by sampling with replacement goes back to Hoeffding [12, Theorem 4]. The bit-length upper bound combines the convex-order estimate with the finite latent-cardinality theorem of Rosset, Gisin and Wolfe [20], quantifier elimination, and an elementary bound on nonzero polynomial roots. Its lower bound uses rational members of P_ε .

Quantum inflation. Wolfe et al. [22] develop quantum inflation, and Ligthart, Gachechiladze and Gross [13] construct a provably convergent variant. The present results apply to classical inflation and the two strengthenings defined in Sections 2 and 3.

15 Open problems

Question 15.1 (Worst-case convergence exponent). For the square and the triangle, Corollary 6.17 gives $\Omega(1/t) \leq H_t^H \leq O(t^{-1/2})$ for each hierarchy H . Which exponent in $[1/2, 1]$ is sharp, and does it depend on the hierarchy?

The lower bound uses one parity direction with the worst-case profile of Lemma 6.13. Along the parity direction itself the exponent is one for both hierarchies: the certified thresholds of Proposition 6.22 give tq_t increasing towards about 0.42 on the square and 0.93 on the triangle through order ten, so any law accepted by the Navascués–Wolfe test at distance of order $t^{-1/2}$ from the compatible set, if one exists, lies in another direction. The same question is open for longer cycles, where Theorem 6.8 gives only $\Omega(m^{-2}t^{-2})$, and for the five-vertex path, where Theorem 7.5 gives $\Omega(t^{-2})$.

Question 15.2 (Exact leading constant along the defect family). Does $\varepsilon^{1/3}t_{\min}^H(P_\varepsilon)$ converge? If so, what is its limit, and do the three hierarchies share it?

The current bounds give the interval $[\frac{1}{2}, 4 - 2\sqrt{3}]$. The finite certificates determine the first rejecting order at $\varepsilon = k^{-3}$ for $k \in \{4, 6, 8, 10, 16\}$; the passing and fan bounds differ at the larger tested values.

Question 15.3 (Separation of the hierarchies). Theorem 6.18 shows that on the square the Navascués–Wolfe and ancestral-independence hierarchies differ by at most a factor $3/2$ in the order. Is the factor sharp, and does a bounded conversion hold on every pair-source scenario?

Along the parity direction the certified thresholds of Proposition 6.22 separate the two hierarchies at every even order up to ten and make them coincide at every odd order up to nine, on both the square and the triangle; we do not know whether the inclusion $\mathcal{I}_t^{\text{AI}} \subseteq \mathcal{I}_t^{\text{NW}}$ is strict at every order on any scenario, nor why the parity direction alternates.

Question 15.4 (Beyond pair-source scenarios). Which scenarios with a source shared by three or more observed variables admit a finite characterizing order?

The reconstruction of Theorem 5.1 uses that every source has two endpoints, and the exhaustion of Lemma 4.3 is a statement about graphs. Both would need replacements for hypergraphs. The nontermination side transfers to any scenario that contains a non-double-star pair-source scenario as an induced substructure in the sense of Lemma 3.11.

Question 15.5 (Explicit rejecting orders beyond Finner). Can one bound the rejecting order in directly evaluable quantities of an incompatible input law that satisfies every Finner inequality?

Theorem 9.1 covers Finner violations on the triangle and Lemma 6.13 covers parity violations on the square. The distance and bit-length bounds apply more broadly, but computing the distance or extracting the constants from quantifier elimination need not be practical.

Question 15.6 (Quantum and no-signalling analogues). For which quantum or no-signalling inflation hierarchies can one construct incompatible laws with an unbounded first rejecting order and determine its growth?

This question requires specifying both the hierarchy and the compatible resource model; the classical proofs here do not decide it.

16 Reproducibility

The proofs do not depend on computation. The accompanying artifact contains exact rational certificates and checkers for the following finite instances:

- structural certificates at orders $t = 1, 2, 3$: the root law, the line supports (30), and the target atoms of P_t as exact rationals; the checker verifies nonnegativity and normalization, covariance of the supports under every adjacent-transposition generator of S_t^3 , all t^3 copied-triangle marginals by exact enumeration over the relevant roots, mutual disjointness of the diagonal supports R_ℓ , all 8^t diagonal equations against $P_t^{\otimes t}$, and the exact Finner gap;
- a sparse full order-two inflation table with 166 nonzero coordinates out of 2^{12} and denominator 2^{32} , produced by a separate enumeration of all 256 defect configurations; the checker treats it as an untrusted linear-programming vector and verifies normalization, nonnegativity, all 12,288 generator-coordinate symmetry equations and all 64 diagonal equations;
- a negative control: the same table with one denominator unit moved from the all-one to the all-zero atom, which is normalized and nonnegative and is rejected by the diagonal equations;
- an exact rational order-two Farkas certificate rejecting $R_{1/2}$ from Proposition 9.2;
- fan certificates: for fourteen members P_ε , $\varepsilon = k^{-3}$, the exact atoms, the Bernoulli passing certificate at order $1 + k/2$, the exact fan value v_t at the first rejecting order of the fan inequality, and the second-moment rejecting order; an all-order certificate of the pointwise inequality (36) as a nonnegative polynomial identity in the count S , checked exhaustively over 174,760 assignments and 271,500 count orbits; the 2,660 index-permutation embeddings behind (35); and eight rejected negative controls. The checker uses explicit exceptions and passes identically under an optimized interpreter.
- classification certificates (`artifact/certificates/classification/`): for the cycles C_3, C_4, C_5 at orders one and two, the five-vertex path at orders one and two, the square parity witness of Theorem 6.11 at orders one and two, and the triangle witness of Theorem 6.12 at orders one to three, each with and without the flips that make the target strictly positive, the verifier reconstructs the global witness from the density formula and checks normalization and positivity of every atom, invariance under every copy-index permutation, the full diagonal law, every injectable marginal and every ancestral-independence product (all 18,054 families at path order two and all 26,671 at C_5 order two), the incompatibility of each target (parity rigidity on the cycles, the bilocal inequality on the path, a max-moment inequality on the square and on the triangle), and rejection of a negative control with one unit of mass moved; at triangle order three, where the table has 2^{27} atoms, the diagonal law and the 15,850 prescriptions are checked by exact Fourier characters rather than atom by atom; the minimum of the corrected densities of Lemma 6.10 is scanned exactly over all assignments up to $t = 5$ for the square and $t = 6$ for the triangle; for Proposition 6.22(iv) at order two, count-form primal witnesses at $q = 3/20$ (order two, Navascués–Wolfe) and $q = 1/10$ (order two, ancestral independence) expanded to full tables of 2^{16} atoms, the order-two dual with its polynomial identity $\mathbb{E}D = (1 + q)g(q)$ verified coefficient by coefficient over the 81 count configurations, and the order-three dual over 256 configurations at $q = 1/10$.
- parity-threshold certificates (`artifact/certificates/exponent/`): for the square and the triangle, for each hierarchy and each order $t \leq 10$, a rational count witness at the lower endpoint of Table 1, checked against every moment prescribed by Lemmas 6.19 and 6.20,

and a chain of rational dual functionals with Sturm certificates of positivity on intervals covering everything above the upper endpoint; the degree sets are enumerated by brute force for $t \leq 3$ and every ancestral-independence degree tuple is realized by an explicit family checked against the definitions for $t \leq 10$; the verifier runs 20,004 exact checks in under a minute. The count-moment reduction itself and Lemma 6.21 are proved in the text and assumed by the verifier.

The certified first rejecting orders of P_ε for $\varepsilon = k^{-3}$ are 4, 5, 6, 7, 10 at $k = 4, 6, 8, 10, 16$, where the passing and fan bounds meet; for $k = 32, 128, 1024, 65536$ the fan bounds are 19, 70, 551, 35122 against lower bounds on the first rejecting order of 18, 66, 514, 32770. The exact Finner gaps at $t = 1, 2, 3$ are $1/8$, $9273825/2^{32}$ and $327279338388988503553/1792720717930698493280256$, above the uniform bounds $\varepsilon_t^2/2 = 1/8, 1/512, 1/5832$. The certificate checkers use Python integers and exact fractions, without a linear-programming solver or floating-point comparisons. The hardened nontermination checker raises an explicit exception on every failed check and refuses to run under an optimized interpreter; the artifact keeps the original checker unchanged for provenance and includes mutation tests (truncated, empty, corrupted, wrong-order and unsymmetric inputs, and optimized-interpreter runs) with their logs. A manifest with SHA-256 identities and a fresh replay record with operating system, interpreter version, commands, wall times and output hashes accompany the artifact.

To replay the checks, use Python 3 from the directory containing `paper/` and `artifact/`:

```
python3 -B artifact/verifiers/run_replay.py
python3 -B artifact/verifiers/mutation_tests.py
python3 -B artifact/verifiers/mutation_tests_fan.py
python3 -B artifact/certificates/classification/verify_classification.py
python3 -B artifact/certificates/exponent/verify_exponent.py
```

The replay runner copies each certificate set into a temporary directory and writes new records under `artifact/replay_logs/`. Its independent distance check requires `mpmath`; the exact certificate checkers use the Python standard library. Replaying changes the log files, so the supplied manifest identifies the distributed snapshot, not regenerated logs. The finite checks verify the listed tables and certificates; the all-order and distance-asymptotic claims rest on the proofs above.

Formal verification. Theorem 8.2 and the results it rests on are formalized in Lean 4 with Mathlib: Lemma 8.7 for arbitrary probability spaces and measurable stochastic responses; the defect-cube law with its symmetry, diagonal law, injectable marginals and ancestral products (Lemmas 8.4, 8.5, 8.6 and 8.9, and Theorem 8.1 for $\mathcal{I}_t^{\text{AI}} \subseteq \mathcal{I}_t^{\text{NW}}$); the margin of Lemma 8.8; Appendix A; Theorem 9.1 with the rejecting-order bound; Proposition 9.2; the finite bounds of Proposition 10.1; the triangle case of Proposition 12.1; and the ℓ^2 form (20) of Theorem 6.16 for the triangle with the bound $\sqrt{7}/(2\sqrt{n})$ of Corollary 6.17. The main statement is `no.finite_characterizing_orderM`, whose compatible set allows arbitrary latent probability spaces; the earlier finite-latent version is kept alongside it. These modules contain no `sorry`, and every theorem depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`.

The pair-source results of Sections 3 to 7 are formalized over a graph-structure development whose definitions are proved to agree with the triangle module on the triangle. Proved there, with binary observed variables and finite latent alphabets throughout: the three tests with their nesting; soundness (Proposition 3.6); the trail criterion and the root-sink lemma (Lemmas 3.7

and 3.9); source-disjoint independence at order two (Lemma 3.10); induced-subgraph transport (Lemma 3.11); the exhaustion lemma (Lemma 4.3); the double-star reconstruction with its labelling step (Theorem 5.1 for binary alphabets, Corollary 5.4); the cycle target, its characters and exact parity rigidity (Lemmas 6.1 and 6.5); the quantitative rigidity of Lemma 6.6; the cycle witness at every order, its incompatibility and the distance bound (Lemmas 6.4 and 6.7); the positivity and moment table of the corrected density in the triangle case and the triangle witness at $q = 1/(16t)$ (Lemma 6.10 with $m = 3$, Theorem 6.12); the five-path target, the bilocal inequality, the distance bound and the witness at every order (Lemmas 7.2 and 7.3, Corollary 7.4, Theorem 7.5); local flips; and Theorem 4.2 itself in the binary case, assembled from these (`classification_NW`, `classification_AI`, `classification_exp`). The coverage map `lean/coverage/inflation-nontermination.json` lists the declarations against the statements. Not formalized: the square witness of Theorem 6.11 (recorded as a statement with `sorry` in a separate library that the audited development does not import), Lemma 6.13, Corollaries 6.14 and 6.15, the convex-order inequality for general convex functions, Theorem 6.18, Lemma 6.20, Lemma 6.21, Proposition 6.22, the larger-alphabet transfer of Remark 4.4, Theorem 11.1, Proposition 13.1, the limit form of Proposition 10.1, Corollary 8.3 and Appendix C. The Lean sources accompany the artifact.

A Injectable sets of the triangle

We justify the description of injectable sets used in Definition 2.2. In the order- t inflated triangle, a set S of copied observations is injectable when the map that erases copy indices is injective on S and, for any two members of S that are copies of variables sharing a source in the triangle, the copy index of that source agrees [23, Definition 4]. Injectivity allows at most one copy of each of A , B , C . If S contains A^{ij} and $B^{i'k}$ then $i = i'$; if it contains A^{ij} and $C^{j'k}$ then $j = j'$; if it contains B^{ik} and $C^{jk'}$ then $k = k'$. Hence $S \subseteq \Delta_{ijk}$ for suitable (i, j, k) , any missing index being chosen freely. Conversely every subset of a copied triangle is injectable. This proves the characterization for all t .

The degree- t diagonal condition (ii) of Definition 2.1 also implies the lower-degree conditions of [15]: the law of any $g \leq t$ diagonal triangles is a marginal of $P^{\otimes t}$, hence $P^{\otimes g}$, and by symmetry the same holds for any g copied triangles with pairwise distinct indices in all three families.

B Certificate formats

Structural certificate at order t . A JSON object with the root law (kind, $\varepsilon_t = 1/(2t^3)$ and $1 - \varepsilon_t$ as exact fractions), the value r_t , the list of t^3 roots D_{ijk} , for each of the $3t^2$ outputs the list of roots on its line, the eight target atoms of P_t as exact fractions, and the claimed counts of diagonal equations (8^t) and symmetry generators ($3(t - 1)$). The checker recomputes every claimed quantity from the definitions and compares exactly. The checker computes the copied-triangle marginals by enumerating the 2^{3t-2} assignments of the roots on the three lines through a cell, which makes $t = 3$ (with 2^7 assignments per triangle and $8^3 = 512$ diagonal equations) immediate.

Full order-two table. A JSON object listing the 12 observation names in a fixed order, the denominator $2^{32} = (2t^3)^{t^3}$ at $t = 2$, and the nonzero numerators $\Gamma_2(w)$ of (31) for 166 assignments $w \in \{0, 1\}^{12}$; unlisted assignments have numerator 0. The checker verifies $\sum_w \Gamma_2(w) = 1$,

nonnegativity, $\Gamma_2(w) = \Gamma_2(\pi w)$ for the three adjacent transpositions π and all 4096 assignments, and the 64 equations $\sum_{w: w_{\text{diag}}=(x,y)} \Gamma_2(w) = P_2(x)P_2(y)$ for the diagonal coordinates $(A^{11}, B^{11}, C^{11}, A^{22}, B^{22}, C^{22})$.

Negative control. The same object with the numerators of the all-one and all-zero assignments changed by -1 and $+1$. Normalization and nonnegativity hold; the diagonal equation at $(000, 000)$ fails. The checker is required to reject it.

Promised-rejection calibration. An exact Farkas certificate showing that the rational law $\frac{1}{2}(\delta_{000} + \delta_{111})$ is infeasible at order two with margin $1/2$, which checks rejection of $R_{1/2}$ from Proposition 9.2. This finite check does not certify the general convergence constants.

Fan certificates. A JSON list of records, one per k in the list 4, 6, 8, 10, 16, 32, 64, 128, 256, 512, 1024, 4096, 16384, 65536, with $\varepsilon = k^{-3}$, $\sigma = 1/(2k^2)$, the eight exact atoms of P_ε , the values m , z and the Finner gap $z^2 - m^3$, the passing order $1 + k/2$ with the exact Bernoulli certificate $(t-1)\varepsilon \leq \sigma$, the first order at which $v_t = tz - m - \binom{t}{2}m^2$ is positive together with the exact rational value of v_t there, and the first order at which the second-moment inequality fails. An all-order certificate records the identity $1 + \frac{1}{2}S(S-1) - S = \frac{1}{2}(S-1)(S-2)$ and its nonnegativity on integers by cases, so that (36) needs no order-by-order enumeration; the checker nevertheless enumerates all assignments of the fan for small t and all count orbits for larger t , and verifies that the index permutations used in (35) extend to permutations of $[t]$. Negative controls perturb the atoms, the parameters and the claimed orders and must be rejected.

C Further quantitative results

This appendix records four quantitative results that accompany the main text. Nothing in the body of the paper depends on them. Each fixes a constant, an exact order, or a ceiling that would otherwise be guessed, and each carries its own verification status, stated at the end of the item.

Conventions are those of Section 2. Total variation is half the ℓ^1 distance. The first and the last part of this appendix use sign outputs $A, B, C \in \{-1, +1\}$; for reals x, y, z with $|x| + |y| + |z| \leq 1$ let

$$\Pi(x, y, z)(\alpha, \beta, \gamma) = \begin{cases} \frac{1}{4}(1 + x\alpha + y\beta + z\gamma), & \alpha\beta\gamma = 1, \\ 0, & \alpha\beta\gamma = -1, \end{cases}$$

the parity-even three-bit law with observed means x, y, z . The middle two parts use zero-one outputs, as in the rest of the paper. In the second part the scenario is the four-cycle $\square$ with observations $A(X, W)$, $B(X, Y)$, $C(Y, Z)$, $D(Z, W)$, independent sources, arbitrary latent alphabets and stochastic responses; its order- n inflation has the $4n^2$ copied bits $A^{il}, B^{ij}, C^{jk}, D^{kl}$, and $\mathcal{C}_\square$ denotes its compatible set; $t_{\min}^{H, \square}(P) = \min\{n : P \notin \mathcal{I}_n^H(\square)\}$ is the first rejecting order there.

An orthogonal witness with quadratic decay

The family below improves the $\Omega(n^{-3})$ rate recorded for H_n in Section 12 along the family of Section 8. Its distance bound uses the parity rigidity of compatible laws rather than a Finner violation; every atomic Finner inequality holds for it.

Proposition C.1 (Quadratic lower bound from an orthogonal density). *Let $n \geq 2$, $\tau = 1/(5n^2)$ and $R_\tau = \Pi(\tau, \tau, -\tau)$. Then*

$$R_\tau \in \mathcal{I}_n^{\text{NW}}, \quad d_{\text{TV}}(R_\tau, \mathcal{C}_\Delta) \geq \frac{\tau}{10},$$

and consequently $H_n^{\text{NW}}(\Delta) = H_n \geq 1/(60n^2)$.

Proof. For a sign vector $S \in \{-1, +1\}^n$ and $0 \leq k \leq n$ put

$$h_k(S) = \binom{n}{k}^{-1} \sum_{|I|=k} \prod_{i \in I} S_i, \quad h_0 = 1,$$

the normalized elementary symmetric sign polynomial; on the weight classes of the cube these are the Krawtchouk polynomials [14]. Under the uniform law on $\{-1, +1\}^n$, character orthogonality gives

$$\mathbb{E}[h_k(S)h_\ell(S)] = \frac{\mathbf{1}[k = \ell]}{\binom{n}{k}}. \quad (38)$$

Take three independent uniform sign vectors $S, T, U \in \{-1, +1\}^n$ as reference measure, indexed by the three source families X, Z, Y of Section 2. For $r, s, t \geq 0$ with $r + s + t \leq n$ set

$$H_{rst} = h_{r+s}(S) h_{r+t}(T) h_{s+t}(U), \quad B_{rst} = \binom{n}{r+s} \binom{n}{r+t} \binom{n}{s+t},$$

and change the reference density to

$$W_\tau = \sum_{r+s+t \leq n} (-1)^t \tau^{r+s+t} B_{rst} H_{rst}. \quad (39)$$

Define every copied observation by

$$A^{ij} = S_i T_j, \quad B^{ik} = S_i U_k, \quad C^{jk} = T_j U_k, \quad (40)$$

and let Γ be the law of the resulting $3n^2$ signs under the reference measure reweighted by W_τ .

Normalization and positivity. Only the term $(r, s, t) = (0, 0, 0)$ has nonzero reference mean, so $\mathbb{E}W_\tau = 1$. Since $|H_{rst}| \leq 1$ and $B_{rst} \leq n^{2(r+s+t)}$,

$$|W_\tau - 1| \leq \sum_{d \geq 1} \binom{d+2}{2} (n^2 \tau)^d = (1 - \frac{1}{5})^{-3} - 1 = \frac{61}{64},$$

whence $W_\tau \geq 3/64 > 0$ pointwise. All weights are rational, so Γ is a rational probability table.

Symmetry. Independent permutations of the three index families permute the coordinates of S, T, U separately, leave each H_{rst} invariant, and transform (40) accordingly. Hence Γ is invariant under the group of Definition 2.1.

Diagonal law. Each copied triangle has $A^{ij}B^{ik}C^{jk} = 1$, so every site is parity-even and $1, A, B, C$ span the functions on its four outcomes. A basis character on the n diagonal sites uses A on r sites, B on s sites and C on t sites, with the three site sets disjoint, so $r + s + t \leq n$; by (40) its monomial uses exactly $r + s$ distinct S -indices, $r + t$ distinct T -indices and $s + t$ distinct U -indices. Averaging over independent index permutations replaces the monomial by H_{rst} . The

map $(r, s, t) \mapsto (r + s, r + t, s + t)$ is injective, so by (38) the H_{rst} are orthogonal with squared norms $1/B_{rst}$, and (39) gives

$$\mathbb{E}_{W_\tau} H_{rst} = (-1)^t \tau^{r+s+t}. \quad (41)$$

These are the corresponding moments of n independent triples of law R_τ . All 4^n basis characters agree, so the complete diagonal law is $R_\tau^{\otimes n}$ and $R_\tau \in \mathcal{I}_n^{\text{NW}}$.

Distance. Flipping the signs of A and B preserves compatibility and the parity ABC , and sends R_τ to a law with all three means equal to $-\tau$ and parity $+1$. Let $R \in \mathcal{C}_\Delta$ with $\delta = d_{\text{TV}}(R_\tau, R)$ and let R' be its flipped version, which is compatible with $d_{\text{TV}}(R'_\tau, R') = \delta$. Since $|\mathbb{E}_P f - \mathbb{E}_Q f| \leq 2d_{\text{TV}}(P, Q)$ for $|f| \leq 1$, the means of R' are at most $-\tau + 2\delta$ and its parity W satisfies $1 - W \leq 2\delta$. Lemma 6.6 gives $-\tau + 2\delta \geq -4(1 - W) \geq -8\delta$. Thus $\delta \geq \tau/10$ for every compatible R . $\square$

The witness has two limitations. First, it applies to the symmetric test only. The truncation $r + s + t \leq n$ in (39) is exactly what the diagonal conditions need, since a diagonal character occupies $r + s + t$ distinct sites, but it is not enough for the ancestral-independence prescriptions, which involve copied triangles spread over more indices. At $n = 2$ the family $\{A^{11}, B^{21}, C^{22}\}$ has pairwise disjoint parent sets, so Definition 2.2 prescribes the product of the three means, $\tau^2(-\tau) = -1/8000$ at $\tau = 1/20$; the corresponding moment of Γ is the coefficient of H_{111} , which (39) omits because $1 + 1 + 1 > 2$, so the witness returns 0. This is a statement about this witness, not a proof that R_τ lies outside $\mathcal{I}_2^{\text{AI}}$. Second, the bound is not the best available: Section 6 gives $H_n^{\text{AI}}(\Delta) = \Omega(1/n)$ by a witness of a different kind, superseding Proposition C.1.

Recorded from the research notes of 13 September 2026 archived with the repository ([research-handoffs/2026-09-13](#)). Status: analytic proof in the source, re-derived here; the order-two table was reconstructed by the packet's independent checker, which verified positivity with minimum density $7/16$, the 96 symmetry equations, the complete 64-equation diagonal law over the 4096-point domain, and rejection of a wrong-diagonal control, and which recorded the order-two ancestral-independence failure quoted above; not formalized.

A square family with quadratic rejection depth

The next item fixes the exact first rejecting order of a family in the four-cycle. It is the square counterpart of Proposition 10.1: there the rejecting order grows as a power of the distance along one direction, here it grows quadratically in a parameter of the target while the target itself stays at a fixed combinatorial complexity.

Fix $t \geq 2$, rational $0 < \varepsilon \leq 1/(2t^6)$, and write $h = 1 - \varepsilon$. The seed R_ε is the law on $ABCD \in \{0, 1\}^4$ with

$$R_\varepsilon(0001) = \varepsilon^2, \quad R_\varepsilon(0010) = R_\varepsilon(1100) = R_\varepsilon(1111) = \varepsilon h, \quad R_\varepsilon(1110) = h(1 - 2\varepsilon),$$

and zero elsewhere. Its one-variable marginals are $R_A(1) = R_B(1) = R_C(1) = h$ and $R_D(1) = \varepsilon$; every pair except AB is independent, so the two opposite-pair independences required of a square law hold exactly, while

$$R_\varepsilon(ABC \mid D = 1) = \varepsilon \delta_{000} + h \delta_{111}$$

is a three-way common-defect law. The target is $P_{t,\varepsilon} = F_t(R_\varepsilon)$, where F_t samples $U_q \sim R_\varepsilon$ independently over $q \in [t]^4$ and takes the coordinatewise conjunction of the t^2 roots on each line, $A = \prod_{j,k} U_{1jk1}^A$ and its three analogues. With

$$N = t^2, \quad p = h^N, \quad q = \varepsilon^N, \quad k = h^{2N-t},$$

the target has the compact description

$$P_{t,\varepsilon}(abc, 1) = qQ(abc), \quad P_{t,\varepsilon}(abc, 0) = S(abc) - qQ(abc), \quad (42)$$

where $S = P_{ABC} = K_{AB} \otimes \text{Bern}(p)$ with $K_{11} = k$, $K_{01} = K_{10} = p - k$, $K_{00} = 1 - 2p + k$, and where $Q = P_{ABC|D=1}$ is determined by binary moment inversion from

$$Q_A(1) = Q_B(1) = Q_C(1) = p, \quad Q_{AB}(11) = k, \quad Q_{AC}(11) = Q_{BC}(11) = \frac{p^2}{h}, \quad Q(111) = \frac{kp}{h}.$$

The witness that carries this target past order t replaces the independent cell roots by a rare selector supported on partial matchings. Its positivity is an instance of Shearer's measure on the line graph of a bipartite graph [21], with the occupied-bit convention reversed; the construction is not new here, and we record only the bound this application needs.

Lemma C.2 (Shearer's measure, positivity bound). *Give the edges of a finite bipartite graph nonnegative weights λ_e with $\sigma = \sum_e \lambda_e < 1$. For a partial matching M set*

$$\pi(M) = \sum_{\substack{L \supseteq M \\ L \text{ a matching}}} (-1)^{|L|-|M|} \prod_{e \in L} \lambda_e.$$

Then $\pi(M) \geq (1 - \sigma) \prod_{e \in M} \lambda_e \geq 0$, $\sum_M \pi(M) = 1$, and the random matching $\mathcal{D}$ with law π satisfies $\Pr[M \subseteq \mathcal{D}] = \prod_{e \in M} \lambda_e$, while any non-matching set has containment probability zero.

Proof. Factor $\prod_{e \in M} \lambda_e$ out of the sum. Among the edges compatible with M , let a_r be the total weight of the r -edge matchings, so $a_0 = 1$. Counting each $(r+1)$ -edge matching through each of its $r+1$ edges gives $(r+1)a_{r+1} \leq \sigma a_r$, so (a_r) is nonincreasing and nonnegative. Pairing successive terms of the finite alternating sum gives $\sum_r (-1)^r a_r \geq 1 - a_1 \geq 1 - \sigma > 0$, which is the stated bound. Normalization and the containment identity follow by exchanging the finite sums and using $\sum_{J \subseteq L} (-1)^{|L|-|J|} = \mathbf{1}[L = \emptyset]$, including when some weights vanish. $\square$

Proposition C.3 (Matching-selector witness). *For $t \geq 2$, rational $0 < \varepsilon \leq 1/(2t^6)$ and every $n \leq t^2 - t + 1$,*

$$P_{t,\varepsilon} \in \mathcal{I}_n^{\text{exp}}(\square) \subseteq \mathcal{I}_n^{\text{AI}}(\square) \subseteq \mathcal{I}_n^{\text{NW}}(\square).$$

Proof sketch. Put $b = N - n - t + 1 \geq 0$ and sample independently $G_i \sim \text{Bern}(h^{t-1})$, $T_{ijl} \sim \text{Bern}(h)$, $U_{il}^A, U_{ij}^B \sim \text{Bern}(h^b)$ and $C^{jk} \sim \text{Bern}(p)$. Set

$$A^{il} = G_i U_{il}^A \prod_j T_{ijl}, \quad B^{ij} = G_i U_{ij}^B \prod_l T_{ijl}.$$

Each such bit succeeds with probability p , and a pair A^{il}, B^{ij} shares G_i and exactly one T -bit, so it succeeds jointly with probability $h^{2N-t} = k$; hence every copied ABC triple has law S . For the selectors put $\kappa_{jl}(1 \mid T) = h^{N-n} \prod_i T_{ijl}$, and with $b_p(c) = p^c(1-p)^{1-c}$,

$$L_{kl}(T, C) = \prod_{j=1}^n \frac{\kappa_{jl}(C^{jk} \mid T)}{b_p(C^{jk})}, \quad \lambda_{kl} = q L_{kl},$$

then draw $\mathcal{D}$ from Lemma C.2 on the complete bipartite graph between the k - and l -indices, conditionally on (T, C) , and set $D^{kl} = \mathbf{1}[(k, l) \in \mathcal{D}]$. Positivity is uniform: writing $\alpha = 1 - p$,

one has $p \geq 1 - N\varepsilon > 1/2$ and $\alpha \geq \varepsilon$, so every ratio is at most $1/\alpha$ and, for every assignment of (T, C) ,

$$\sum_{k,l} \lambda_{kl} \leq n^2 q \alpha^{-n} \leq n^2 \varepsilon^{N-n} \leq N^2 \varepsilon \leq \frac{1}{2t^2} < 1,$$

using $N - n \geq t - 1 \geq 1$. The whole construction is therefore a finite rational probability space, and its pushforward is a nonnegative normalized table on all $4n^2$ copied bits.

One identity controls the witness. For a matching M let W_M retain the base variables but replace each selected C -row, that is each k with $(k, l) \in M$, by $C^{jk} = V_{jk}^C \prod_i T_{ijl}$ with private $V_{jk}^C \sim \text{Bern}(h^{N-n})$. Distinct selected edges carry distinct k -indices, so $\prod_{(k,l) \in M} L_{kl}$ is a normalized likelihood ratio even with T fixed, and Lemma C.2 gives, for every function f of the copied ABC observations,

$$\mathbb{E}_\Gamma \left[f \prod_{(k,l) \in M} D^{kl} \right] = q^{|M|} \mathbb{E}_{W_M} f, \quad (43)$$

while non-matching D -moments vanish. Independent permutations of the four index families permute the base variables, transform the likelihoods covariantly and permute the matchings, so the table is S_n^4 -invariant; (43) applied to one copied square reproduces (42), applied to the n diagonal squares gives the joint law $P_{t,\varepsilon}^{\otimes n}$, and applied to ancestrally disjoint injectable blocks gives the prescribed products of Definitions 2.2 and 2.3. $\square$

Rejection uses a strengthening of the first fan inequality of Theorem 9.1 that keeps two pair probabilities instead of discarding them.

Proposition C.4 (Pair-retaining fan inequality). *Let $n \geq 2$ and let P be a law on three bits with $a = P_A(0)$, $b = P_B(0)$, $c = P_C(0)$, $d = P_{AB}(00)$, $g = P_{AC}(00)$ and $z = P(000)$. If $P \in \mathcal{I}_n^{\text{NW}}$ then*

$$d + g + (n - 2)z \leq a + \binom{n}{2} bc. \quad (44)$$

Proof. For bits $b_1, \dots, b_n, c_1, \dots, c_n$ put $U = \sum_k b_k$, $V = \sum_k c_k$, $W = \sum_k b_k c_k$. The pointwise inequality is

$$UV - W \geq \frac{2}{n} ((n - 2)W + U + V - n). \quad (45)$$

To see it, multiply by n and collect terms on one side as $R = nUV - (3n - 4)W - 2U - 2V + 2n$. Since $W \leq \min(U, V)$ and its coefficient is negative, it suffices to treat $W = \min(U, V)$; assume $U \leq V$, so R becomes $nUV - (3n - 2)U - 2V + 2n$. For $U = 0$ this is $2(n - V) \geq 0$; for $U = 1$ it is $(n - 2)(V - 1) \geq 0$; for $U \geq 2$ it increases in V , so its minimum is at $V = U$, where it equals $n(U - 1)(U - 2) \geq 0$.

Now take the fan of the proof of Theorem 9.1: $\mathbf{A} = \mathbf{1}[A^{11} = 0]$, $\mathbf{B}_k = \mathbf{1}[B^{1k} = 0]$, $\mathbf{C}_k = \mathbf{1}[C^{1k} = 0]$, and set $b_k = \mathbf{B}_k$, $c_k = \mathbf{C}_k$ in (45). Definition 2.1 gives $\mathbb{E}\mathbf{A} = a$, $\mathbb{E}[\mathbf{A}\mathbf{B}_k] = d$, $\mathbb{E}[\mathbf{A}\mathbf{C}_k] = g$ and $\mathbb{E}[\mathbf{A}\mathbf{B}_k\mathbf{C}_k] = z$, since a permutation of the Y -indices carries each of these sets into the diagonal triangle Δ_{111} , and $\mathbb{E}[\mathbf{B}_k\mathbf{C}_\ell] = bc$ for $k \neq \ell$ by (35). Hence $\mathbb{E}(UV - W) = n(n - 1)bc$. Multiplying (45) by $\mathbf{A}$ and using $UV - W \geq 0$ gives

$$n(n - 1)bc \geq \frac{2}{n} (n(n - 2)z + nd + ng - na),$$

which rearranges to (44). $\square$

Since $d \geq z$ and $g \geq z$, (44) implies the first inequality of (33). We apply it to the square through a conditioning that costs no order.

Lemma C.5 (Same-order transfer). *Let $P \in \mathcal{I}_n^{\text{NW}}(\square)$. Then $P_{ABC|D=1} \in \mathcal{I}_n^{\text{NW}}(\triangle)$ whenever $P_D(1) > 0$, and in every case*

$$P_{ABD}(001) + P_{ACD}(001) + (n-2)P(0001) \leq P_{AD}(01) + \binom{n}{2}P_B(0)P_{CD}(01). \quad (46)$$

Proof. Condition an order- n square witness on $D^{11} = \dots = D^{nn} = 1$, an event of probability $P_D(1)^n > 0$, and retain the complete A, B, C arrays. Independent permutations of the X - and Y -indices, together with the simultaneous permutation of the Z - and W -indices, realize the three source symmetries of the triangle, and the conditional diagonal law is $(P_{ABC|D=1})^{\otimes n}$. Applying (44) to the conditional law, using the independence of B and D forced at order $n \geq 2$, and clearing the positive denominator gives (46). If $P_D(1) = 0$ every term of (46) containing $D = 1$ vanishes and the inequality is trivial. $\square$

Theorem C.6 (Exact first rejection). *For every $t \geq 2$ and rational $0 < \varepsilon \leq t^{-8}$,*

$$t_{\min}^{H, \square}(P_{t, \varepsilon}) = t^2 - t + 2, \quad H \in \{\text{NW}, \text{AI}, \text{exp}\},$$

the first rejection being witnessed by (46) with margin at least $\frac{1}{2}\varepsilon t^{2+1}$.

Proof sketch. Write $\alpha = 1 - p$, $\beta = 1 - 2p + k$, $\gamma = 1 - 2p + p^2/h$, so that the conditional law Q has $Q_A(0) = Q_B(0) = Q_C(0) = \alpha$, $Q_{AB}(00) = \beta$, $Q_{AC}(00) = Q_{BC}(00) = \gamma$, and moment inversion gives $z := Q(000) = \varepsilon + (\beta - \varepsilon)(1 - h^{N-1})$. Bernoulli's inequality gives $\alpha \leq N\varepsilon$; the AB pair has a representation with a shared $\text{Bern}(h^t)$ gate and independent private gates, so $\beta \geq 1 - h^t \geq t\varepsilon - \binom{t}{2}\varepsilon^2$, whence $\beta \geq \varepsilon$ and then $z \geq \varepsilon$; and $\gamma \geq z$ because the triple event is contained in the pair event. Let

$$\Delta_{t, \varepsilon}(n) = \beta + \gamma + (n-2)z - \alpha - \binom{n}{2}\alpha^2$$

be the violation of (44) by Q , so that $q \Delta_{t, \varepsilon}(n)$ is exactly the violation of (46) by $P_{t, \varepsilon}$. With $n_0 = N - t + 2$ and $K_t = \binom{t}{2} + \binom{n_0}{2}N^2$, the three bounds give $\Delta_{t, \varepsilon}(n_0) \geq \varepsilon(1 - K_t\varepsilon)$. Since $n_0 \leq N$ one has $K_t \leq N^4/2$, so $\varepsilon \leq N^{-4} = t^{-8}$ forces $K_t\varepsilon \leq 1/2$ and $\Delta_{t, \varepsilon}(n_0) \geq \varepsilon/2$. Order $n_0 - 1 = t^2 - t + 1$ is passed by Proposition C.3 and order n_0 is rejected by Lemma C.5; the inclusions (1) place the first rejection of all three hierarchies at the same integer. $\square$

For the original parameter $\varepsilon_t = 1/(2t^6)$ the same estimates give the bracket

$$t^2 - t + 2 \leq t_{\min}^{H, \square}(P_{t, \varepsilon_t}) \leq U_t, \quad U_t = \left\lfloor \frac{4t^2 + 1 - \sqrt{16t^3 - 8t^2 + 1}}{2} \right\rfloor + 1, \quad (47)$$

whose two sides agree at $t = 2$: there $N = 4$, $\varepsilon = 1/128$, $n_0 = 4$ and $K_2 = 97$, so $K_2\varepsilon = 97/128 < 1$ and $t_{\min}^{H, \square}(P_{2, 1/128}) = 4$ exactly, with conditional margin $\Delta_{2, 1/128}(4) \geq 31/16384$. Exact rational evaluation of $\Delta_{t, \varepsilon_t}$ improves (47) at small t , leaving $t_{\min}^{H, \square}(P_{3, \varepsilon_3}) \in \{8, 9\}$. The asymptotics of $t_{\min}^{H, \square}(P_{t, \varepsilon_t})/t^2$ are pinned only between 1 and 2, and no limit is claimed.

Recorded from the research notes of 13 September 2026 archived with the repository ([research-handoffs/2026-09-13](#)). Status: analytic proofs in the sources, with the matching-selector positivity and the ancestral-independence factorization read but not re-derived line by line here; exact finite checks passed (the pointwise inequality (45) on 814,380 count triples, and 18 exact small- ε rejection cases covering $t \leq 10$ at two values of ε each); not formalized.

Ceilings for two restricted construction classes

The two results below are limitations of ansatz families, not upper bounds on H_n . They say that the defect family of Section 8 and the wider class of independent-cell constructions cannot produce a better exponent than n^{-3} , whatever their parameters. Neither restricts arbitrary feasible tables.

Lemma C.7 (Compatible comparison curve). *For $0 \leq t \leq 1$, the law $Q(t^3/(1+t^3), 1-t^2)$ of (28) is compatible.*

Proof. Put $h = 1 - t + t^2$ and $p = t/h$. Both are valid probabilities, since $0 < h \leq 1$ and $h - t = (1 - t)^2 \geq 0$. Let each of the three sources carry an independent $\text{Bern}(p)$ bit, and let each node output 0 with probability h when both of its incident source bits are 1, and output 1 otherwise, with independent response coins. For the zero indicators Z_A, Z_B, Z_C this model has

$$\mathbb{E}Z_A = hp^2 = \frac{t^2}{h}, \quad \mathbb{E}Z_A Z_B = h^2 p^3 = \frac{t^3}{h}, \quad \mathbb{E}Z_A Z_B Z_C = h^3 p^3 = t^3,$$

together with the corresponding coordinate permutations. Since $1 + t^3 = (1 + t)h$, these are exactly the seven nonempty binary moments of $Q(t^3/(1+t^3), 1-t^2)$, and binary moments determine all eight atoms by inclusion and exclusion. $\square$

Proposition C.8 (Ceiling for the defect family). *Let*

$$M_n = \sup \left\{ d_{\text{TV}}(Q(\varepsilon, r), \mathcal{C}_\Delta) : 0 < \varepsilon < 1, 0 \leq r \leq (1 - \varepsilon)^{n-1} \right\},$$

the supremum of the distance over the whole parameter range for which Theorem 8.1 supplies an order- n witness. Then

$$M_n = \frac{4}{27n^3} + O(n^{-4}),$$

an asymptotically optimal sequence being $\varepsilon_n = 4/(9n^3)$ with $r_n = (1 - \varepsilon_n)^{n-1}$, whose distance to $\mathcal{C}_\Delta$ exceeds $1/(10n^3)$ for every $n \geq 2$.

Proof sketch. Write $q = 1 - r$, $m = \varepsilon + (1 - \varepsilon)q$ and $z = \varepsilon + (1 - \varepsilon)q^3$ for the common zero marginal and the atom at 000, and put $\varepsilon_c(q) = q^{3/2}/(1 + q^{3/2})$. Thinning the ones by a product of local channels maps $Q(\varepsilon, \varepsilon_c(q))$ to $Q(\varepsilon, r)$ for $\varepsilon \leq \varepsilon_c(q)$, so Lemma C.7 with $t = (\varepsilon/(1 - \varepsilon))^{1/3}$ shows $Q(\varepsilon, r) \in \mathcal{C}_\Delta$ in that range, and comparison at fixed r gives

$$d_{\text{TV}}(Q(\varepsilon, r), \mathcal{C}_\Delta) \leq [\varepsilon - \varepsilon_c(q)]_+(1 - q^3).$$

In the other direction, Finner [6] in the form $S(000) \leq \sqrt{S_A(0)S_B(0)S_C(0)}$ for compatible S , which is the classical specialization of the network inequality [19], applied to a nearest compatible law at distance δ gives $z - \delta \leq (m + \delta)^{3/2}$ and hence

$$d_{\text{TV}}(Q(\varepsilon, r), \mathcal{C}_\Delta) \geq \frac{[z - m^{3/2}]_+}{1 + \frac{3}{2}\sqrt{m + \varepsilon}}.$$

Local channels also reduce the two-parameter supremum exactly to $r = (1 - \varepsilon)^{n-1}$. At the critical scale $\varepsilon = x/n^3$ with x in a fixed bounded range one has $q = xn^{-2} + O(n^{-3})$, $m = xn^{-2} + O(n^{-4})$, $z = xn^{-3} + O(n^{-6})$ and $\varepsilon_c(q) = x^{3/2}n^{-3} + O(n^{-4})$, so both estimates give

$$d_{\text{TV}}(Q(x/n^3, (1 - x/n^3)^{n-1}), \mathcal{C}_\Delta) = \frac{[x - x^{3/2}]_+}{n^3} + O(n^{-4}),$$

maximized at $x = 4/9$ with value $4/27$. Scales outside this window are handled separately: for $\varepsilon \geq 1/8$ the comparison with δ_{000} gives an exponentially small bound, and for smaller ε the two displayed estimates close the gap. $\square$

Independent asymmetric private filtering of the three outputs and exclusive-or aggregation along the defect lines both reduce to local stochastic channels applied to the same family, so neither improves the exponent.

The second class fixes independent cell roots and symmetric line responses. Let U_{ijk} , $(i, j, k) \in [n]^3$, be independent with a common law μ on a finite alphabet, let each observed type $V \in \{A, B, C\}$ carry a permutation-invariant kernel g_V giving its probability of output one from the n roots on its line, and let the responses be independently randomized across copied observations. The resulting table is a full order- n witness, and with the profiles $f_V(s) = \mathbb{E}[g_V(s, U_2, \dots, U_n)]$ and $m_V = \sum_s \mu_s f_V(s)$ the target is $\sum_s \mu_s \otimes_V \text{Bern}(f_V(s))$. Write $\mathcal{L}_n^{(\kappa)}$ for the laws obtained this way from a root alphabet of size κ .

Lemma C.9 (Profile variance). *For every admissible line kernel, $\text{Var}_\mu(f_V) \leq m_V(1 - m_V)/n$.*

Proof. Write $f = f_V$, $m = m_V$, $h = f - m$, $v = \mathbb{E}h(U)^2$ and $T = \sum_{i=1}^n h(U_i)$, and let $g = g_V(U_1, \dots, U_n)$. Independence gives $\mathbb{E}T^2 = nv$, and symmetry of the kernel gives $\mathbb{E}[(g - m)T] = n\mathbb{E}[h(U_1)(g - m)] = nv$. Cauchy and Schwarz then give $(nv)^2 \leq \text{Var}(g)nv$. For $v > 0$ divide by nv and use $0 \leq g \leq 1$ to get $nv \leq \text{Var}(g) \leq m(1 - m)$; the case $v = 0$ is immediate. $\square$

Proposition C.10 (Independent-cell ceilings). *Let $P \in \mathcal{L}_n^{(\kappa)}$ for some finite κ . Then*

$$d_{\text{TV}}(P, \mathcal{C}_\Delta) \leq \frac{1}{2n},$$

and if P violates the Finner inequality $P(abc)^2 \leq P_A(a)P_B(b)P_C(c)$ at some atom, then

$$d_{\text{TV}}(P, \mathcal{C}_\Delta) \leq \frac{2}{n(n-1)^2} \leq \frac{8}{n^3}.$$

Proof sketch. For any three-bit law, replacing the pair (B, C) by an independent pair and keeping the conditional law of A produces a compatible law, so

$$d_{\text{TV}}(P, \mathcal{C}_\Delta) \leq 2 \min\{|\text{Cov}(A, B)|, |\text{Cov}(A, C)|, |\text{Cov}(B, C)|\},$$

the factor being the exact four-cell identity for a binary pair. Under the independent-cell hypothesis $\text{Cov}(V, W) = \mathbb{E}_\mu[(f_V - m_V)(f_W - m_W)]$, so Lemma C.9 and Cauchy and Schwarz give $|\text{Cov}(V, W)| \leq \sqrt{m_V(1 - m_V)m_W(1 - m_W)}/n \leq 1/(4n)$, which is the first bound. For the second, relabel the violating atom's events as ones and order the marginals $a \leq b \leq c$, and write $t_n(x) = x + (1 - x)/n$ and z for the atom probability. Lemma C.9 gives $\mathbb{E}f_V^2 \leq m_V t_n(m_V)$, so

$$z \leq \mathbb{E}f_A f_B \leq \sqrt{ab t_n(a) t_n(b)} \leq \sqrt{ab} t_n(c),$$

and $t_n(c) \leq \sqrt{c}$ whenever $c \geq (n - 1)^{-2}$, since with $y = \sqrt{c}$ one has $ny - 1 - (n - 1)y^2 = (1 - y)((n - 1)y - 1) \geq 0$. Hence $z^2 > abc$ forces $\max(a, b, c) < (n - 1)^{-2}$, and applying Lemma C.9 to the event indicators gives $|\text{Cov}(V, W)| \leq \sqrt{m_V m_W}/n < 1/(n(n - 1)^2)$. $\square$

For binary roots the exponent is exactly cubic, and the same holds for ternary roots with a large but finite constant:

$$\sup_{P \in \mathcal{L}_n^{(2)}} d_{\text{TV}}(P, \mathcal{C}_\Delta) = \Theta(n^{-3}), \quad \frac{1}{10n^3} < D_n^{(3)} := \sup_{P \in \mathcal{L}_n^{(3)}} d_{\text{TV}}(P, \mathcal{C}_\Delta) \leq \frac{10^{38}}{n^3}.$$

The lower bounds come from the family of Proposition C.8, which lies in both classes. The ternary upper bound needs two constructions rather than one estimate. When the product of the two rare root probabilities is small, two independent compatible binary resources approximate the categorical selector, and a count-overlap inequality makes their local response kernels valid up to a uniformly controlled clipping error. When that product is larger, a nested compatible resource reproduces the target exactly with signed local kernels, and clipping them gives a genuine model; the error must be weighted by the probability of the source configuration on which it occurs, since discarding that weight loses the cubic rate. The displayed constant is asymptotic bookkeeping and is not a useful numerical ceiling.

Recorded from the research notes of 13 September 2026 archived with the repository ([research-handoffs/2026-09-13](#)). Status: long analytic proofs in the sources, with Lemmas C.7 and C.9 re-derived here and the remaining estimates read but not re-derived; exact finite checks passed (the nested-resource algebraic identity on 24 parameter profiles, 192 exact moment equalities over 64 edge configurations each, and the fan substitutions of the defect family at $n = 2, 3, 4, 8$); not formalized; the all-parameter tail bounds behind the ternary constant are not covered by any finite check.

Parity extraction from a witness

The convergence bound of Theorem 6.16 extracts a compatible law from a feasible table by sampling source indices independently. Requiring the extracted law to come from a table in the support of the given witness loses nothing on the parity face, from order six on. At order two there is an exact obstruction.

For a deterministic table ω on the copied observations and index distributions u, v, w on $[n]$ let

$$q_\omega(u, v, w)(a, b, c) = \sum_{i,j,k} u_i v_j w_k \mathbf{1}[(A^{ij}, B^{ik}, C^{jk}) = (a, b, c)],$$

let $\mathcal{R}(\Gamma)$ be the set of all $q_\omega(u, v, w)$ with $\omega \in \text{supp } \Gamma$, and let $e_\Gamma(P) = \min_{Q \in \mathcal{R}(\Gamma)} d_{\text{TV}}(P, Q)$. Write $\mathcal{C}_{\text{par}} = \{\Pi(st, su, tu) : s, t, u \in [-1, 1]\}$ for the parity-perfect part of $\mathcal{C}_\Delta$; the parametrization is the parity rigidity used in Proposition C.1 and is established prior work [3].

Proposition C.11 (Finite catalogue). *Every law attainable by reweighting a given binary-triangle table is attainable using at most eight existing indices from each source, with no table entry changed. Consequently $\mathcal{R}(\Gamma)$ is a union of model images drawn from a catalogue of deterministic subtables whose size does not depend on n , crudely bounded by $\sum_{r,s,t \leq 8} 2^{rs+rt+st} \leq 2^{201}$.*

Proof. Fix two of the three index distributions. The output law is the convex combination $q = \sum_i u_i q_i$ of points of the seven-dimensional simplex of eight outcomes. If more than eight weights are positive, an affine dependence moves them without changing q until one weight vanishes; repeat, then compress the second and third source in turn. $\square$

Theorem C.12 (Identity transfer). *Let f be a polynomial of degree $d \leq n$ in the eight atom probabilities, and let Γ be an order- n symmetric witness for P . If $f(q_\omega(u, v, w)) = 0$ for every $\omega \in \text{supp } \Gamma$ and all u, v, w , then $f(P) = 0$.*

Proof. Homogenize f to degree d using $\sum_x p_x = 1$ and let F_f be its symmetric multilinear polarization, so $F_f(p, \dots, p) = f(p)$. For a fixed table the function $G_\omega(u, v, w) = F_f(q_\omega, \dots, q_\omega)$ is homogeneous of degree d in each of the three weight blocks and vanishes on the product of simplexes, hence, by homogeneity after normalizing positive weight vectors, on the product of positive cones; so it is the zero polynomial. Select d distinct indices in each source family and read off the coefficient of the corresponding square-free monomial: it is a sum of terms $F_f(\delta_{\Delta_1(\omega)}, \dots, \delta_{\Delta_d(\omega)})$ over copied triangles with pairwise distinct indices in every family, and it vanishes. Average over all ordered injections $[d] \hookrightarrow [n]$ in each family and then over $\omega \sim \Gamma$. By the symmetry and the degree- d diagonal condition of Definition 2.1 the selected triangles are jointly distributed as $P^{\otimes d}$, so multilinearity gives

$$0 = \mathbb{E} F_f(\delta_{\Delta_1}, \dots, \delta_{\Delta_d}) = F_f(P, \dots, P) = f(P). \quad \square$$

Theorem C.13 (Extraction on the parity face). *For every parity-perfect target P , every order- n symmetric witness Γ for P and every $n \geq 6$,*

$$e_\Gamma(P) = d_{\text{TV}}(P, \mathcal{C}_{\text{par}}).$$

More precisely, at least one of the following holds: a single supported table realizes every law of $\mathcal{C}_{\text{par}}$ by independent reweighting, or a single supported table realizes P exactly. At most two existing indices per source are needed, and no positivity of atoms or marginals is assumed.

Compatibility of the target alone does not give exact extraction at low order.

Proposition C.14 (Order-two obstruction). *Every parity-even three-bit law passes the order-two symmetric test. For the uniform parity law U , which is compatible, there is an order-two witness Γ with $e_\Gamma(U) = 1/2$.*

Proof. Draw an ordered pair $(x, y) \sim P^{\otimes 2}$ of parity-even triples. If $x = y$, let ω_{xy} be the constant table with that value everywhere. If $x \neq y$, the two triples differ at exactly two observed nodes, and those two nodes share a source; activate only that source, letting its index 1 produce the entries of x and its index 2 those of y at the two differing nodes, with the third node constant and all other source indices ignored. This defines the whole twelve-bit table. Its diagonal is (x, y) ; swapping the active source's two indices exchanges ω_{xy} with ω_{yx} , whose weights $P(x)P(y)$ agree; swapping inactive indices changes nothing. So the mixture is a normalized, nonnegative, symmetric table with diagonal law $P^{\otimes 2}$, which proves the first claim.

For U the four constant tables and the twelve oriented two-outcome tables each carry probability $1/16$. Every reweighting of a supported table is supported on at most two of the four parity-even outcomes, so for its support D one has $d_{\text{TV}}(U, Q) \geq U(D^c) \geq 1/2$, with equality for a nonconstant table whose active source is fair. Meanwhile $U = \Pi(0, 0, 0) \in \mathcal{C}_{\text{par}}$. $\square$

Theorem C.12 does not extend from identities to inequalities, and the failure is visible at degree two. Let n be even and take the table $A^{ij} = b_i$ with half the b_i equal to zero and half to one, and B, C constant. The polynomial $f(q) = (q_A(1) - \frac{1}{2})^2$ is nonnegative on every reweighting, while its degree-two polarization on distinct source indices is

$$\frac{1}{n(n-1)} \sum_{i \neq j} \left(b_i - \frac{1}{2}\right) \left(b_j - \frac{1}{2}\right) = -\frac{1}{4(n-1)} < 0,$$

because the centered sum vanishes while its sum of squares is $n/4$. Nonnegativity on all reweighted models therefore does not imply nonnegativity of the distinct-index expression that finite inflation tests. This identifies the loss in one route to a C/n distance bound; it is not a counterexample to such a bound. Proposition C.14 does show that any universal constant C with $e_\Gamma(P) \leq C/n$ for all n , P and Γ must satisfy $C \geq 1$.

Recorded from the research notes of 13 September 2026 archived with the repository ([research-handoffs/2026-09-13](#)). Status: analytic proofs in the source, with Theorem C.12 and Proposition C.14 checked by hand here and Theorem C.13 read but not re-derived; exact finite checks passed (the order-two witness over all 4096 assignments, 12,288 symmetry equations, the 64 diagonal equations, the classification of all 32 parity tables, 184 support-annihilator substitutions, the distinct-index polarization over 32 source assignments, and rejection of a normalized symmetric table with the wrong diagonal); not formalized. The order- $n \geq 6$ statement rests on the analytic proof alone.

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